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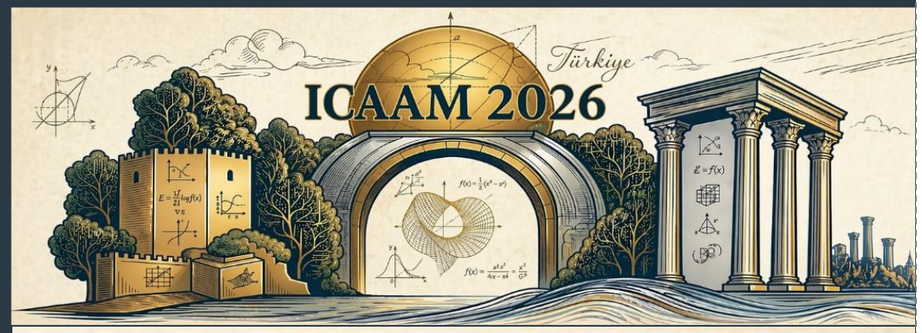
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Charyyar Ashyralyyev

and

Mahmud A. Sadybekov

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- Minisymposium MS1: Functional analysis in interdisciplinary applications
- Minisymposium MS2: Applied Statistics and Data Analysis
- Minisymposium MS3: Biomedical Signal Processing

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  Yildiz Technical University, Turkiye; Institute of Mathematics, Ministry of Science and Education of the Republic of Azerbaijan; Azerbaijan University of Architecture and Construction, Azerbaijan
  Title: On Applications of the Concept of $\otimes$-Basis in Approximation Theory and Partial Differential Equations
- Prof. Dr. Alexey Karapetyants
  Institute of Mathematics, Mechanic and Com- puter Sciences, Russia Southern Federal University, Russian Federation
  Title: Aggrandization of spaces of holomorphic functions
- Prof. Dr. Virginia Kiryakova
  Institute of Mathematics and Informatics, Bulgaria
  Title: Generalized fractional integrals related to Le Roy-, Feinman integrals- and Rathie I- generalized hypergeometric functions
- Prof. Dr. Ljubisa D.R. Kocinac
  University of Nis, Serbia
  Title: On sequential Karamata theory
- Prof. Dr. Alexander Meskhi
  Kutaisi International University, Georgia; TSU A. Razmadze Mathematical Institute, Georgia
  Title: Multilinear Rellich and Hardy inequalities
- Prof. Dr. Andrey Muravnik
  Peoples' Friendship University of Russia, Russia
  Title: Series expansions of solutions of initial-value problems for hyperbolic differential-difference equations: computational aspects
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  Peoples' Friendship University of Russia, Russia
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Opening speech

A. Arif Ergin[1]

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Dear participants!

On behalf of the Rectorate of Bahçeşehir University I welcome you all to the 8-th International Conference on Analysis and Applied Mathematics ICAAM-2026. The ICAAM conference aims to bring together mathematicians working in the area of analysis and applied mathematics to share new trends of applications of mathematics. The developments in the field of applied mathematics open new research areas in analysis and vice versa. That is why this conference provides a forum for researchers and scientists to communicate their recent developments and present their original results in various fields of analysis and applied mathematics. Previous ICAAM conferences were held in Türkiye (2012,2022, 2024), Kazakhstan (2014, 2016) and Northern Cyprus (2018, 2020). We are very happy to host you this time again in Türkiye. The proceedings from all previous ICAAM conferences held between 2012 and 2022 have been published in 6 volumes by the American Institute of Physics (AIP) Conference Proceedings and are indexed in the Scopus database. Selected manuscripts were published in the Special issues of various International Journals, such as Filomat, Boundary Value Problems, Numerical Functional Analysis and Optimization, International Journal of Applied Mathematics, Bulletin of the Karaganda University-Mathematics. The main organizers of ICAAM-2026 are:

- Bahçeşehir University, Türkiye
- Institute of Mathematics and Mathematical Modeling, Almaty, Kazakhstan
- Analysis PDE Center, Ghent University, Ghent, Belgium

It is a great pleasure for us to continue collaboration with these important Mathematical Centers of Kazakhstan and Belgium. We are particularly excited about our ongoing collaboration with these institutions through:

- Weekly "Analysis and Applied Mathematics" seminars, both in-person and online.
- The international journal "e-Journal of Analysis and Applied Mathematics".

The seminars provide a platform for researchers to share their latest findings and encourage international collaboration. Notably, 97 successful seminars have been held since their beginning in 2022. Recordings of almost all seminars are available on YouTube. A book of extended abstracts from these seminars and related talks was recently published by Springer Publishing center. This publication features contributions from researchers from ifferent countries. e-Journal of Analysis and Applied Mathematics, established by our teams in 2018, serves as a forum for the latest advancements in the field.

There are 227 mathematicians from 31 different countries participating in the ICAAM-2026 conference this year. It is a great pleasure to welcome our 8 Keynote Speakers from 5 different countries who are the best scientists in the world in the field of Analysis and Applied Mathematics. It is truly a big event, and I would like to thank the organizing committee as well as all the scientists and administrative staff who have contributed to this conference for their tremendous work to make this conference a success.

As Bahçeşehir University, it is a great pleasure for us to be an organizing side of the 8-th ICAAM conference. As a higher education institution dedicated to teaching, research, and service to our society, the mission of Bahçeşehir University is to educate the leading workforce of future who have an inquiring mind and a critical thinking ability; are sensitive to local and global issues; achieve international standards; contribute to scientific, technological, and cultural knowledge; are strong supporters of universal ideas and values. Bahçeşehir University is a leading university representing Turkish higher education in the world with its over 25 thousand students, 20 % of which are international, over 30 thousand alumni all over the world, over 1000 faculty members, over 500 staff, 5 campuses in Istanbul and thousands of national and international collaborations with universities and business as the flagship of BAU Global network. According to UK-based international higher education rating agency Times Higher Education (THE), Bahçeşehir University became the 5-th university in Turkey and rose to the 601-800 band worldwide in the "2023 World University Ranking".

Dear friends, I sincerely wish you very fruitful discussions over the next few days and I hope you will really enjoy your stay in Antalya. Thank you very much for your attention and as we say in Turkish "Kolay gelsin".

Welcome address

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Dear Esteemed Guests and Participants,

On behalf of the Organizing Committee of the Eighth International Conference on Analysis and Applied Mathematics (ICAAM 2026), I warmly welcome all our distinguished guests, invited speakers, participants, and colleagues to Antalya. It is a great pleasure and honor to have you with us for this important international scientific event. We sincerely appreciate the many mathematicians and researchers who have traveled from different countries and cities to participate in ICAAM 2026. Your presence, scientific contributions, and active participation are essential to the success of this conference. The ICAAM conference series provides an international platform for mathematicians and researchers working in analysis and applied mathematics to meet, exchange ideas, present recent results, and discuss emerging directions in research. The interaction between theoretical analysis and applied mathematics continues to generate new methods, applications, and opportunities for collaboration. ICAAM 2026 is intended to strengthen these interactions and to promote the exchange of recent advances and original contributions across a broad range of mathematical disciplines. The seven-day scientific program of ICAAM 2026 includes 7 plenary lectures and approximately 90 contributed presentations, covering a wide spectrum of topics in analysis and applied mathematics.

Highlights of ICAAM 2026

- 227 participants from 31 countries, representing a broad and diverse international mathematical community.
- Publication of an Abstract Book of ICAAM 2026 containing the abstracts of the scientific contributions.
- Selected full papers will be considered for publication in special issues or dedicated publication channels associated with Azerbaijan Journal of Mathematics, Journal of Mathematical Sciences (Series A), and the e-Journal of Analysis and Applied Mathematics, subject to the respective journals' editorial policies and peer-review procedures.
- Selected papers will also be considered for possible publication in Filomat. Following his lecture, Prof. Ljubiša D. Kočinac will provide information about the submission procedure and the requirements for consideration by the journal. Kočinac and I will jointly coordinate the selection of papers for possible consideration for publication in Filomat, in accordance with the journal's standard editorial and peer-review procedures.

We are also particularly pleased to welcome Prof. Virginia Kiryakova, one of the keynote speakers of ICAAM 2026 and Editor-in-Chief of Fractional Calculus and Applied Analysis (FCAA). ICAAM 2026 brings together many active researchers working in fractional calculus and its applications. We hope that, following the conference, selected high-quality contributions presented at ICAAM 2026 may be submitted for consideration in FCAA, subject, of course, to the journal's regular submission, editorial evaluation, and peer-review procedures.

Our Distinguished Speakers

Our heartfelt thanks and appreciation go to the distinguished plenary and keynote speakers of ICAAM 2026 for their valuable contributions and for sharing their expertise with our international mathematical community: **Virginia Kiryakova,** Institute of Mathematics and Informatics, Bulgarian Academy of Sciences, Bulgaria; **Ljubiša D. R. Kočinac,** University of Niš, Serbia; **Alexander L. Skubachevskii,** Peoples' Friendship University of Russia (RUDN University), Russia; **Andrey Muravnik,** Peoples' Friendship University of Russia (RUDN University), Russia; **Alexander Meskhi,** Kutaisi International University, Georgia, and A. Razmadze Mathematical Institute, Tbilisi State University, Georgia; **Bilal Bilalov,** Yıldız Technical University, Türkiye; Institute of Mathematics, Ministry of Science and Education of the Republic of Azerbaijan; Azerbaijan University of Architecture and Construction, Azerbaijan; **Nedyu Popivanov,** Sofia University, Bulgaria. We are especially grateful to all our speakers for their distinguished lectures, scientific leadership, and contributions to the academic program of ICAAM 2026.

Acknowledgments

I would like to express my sincere gratitude to the Scientific Committee of ICAAM 2026 for its invaluable support, guidance, and commitment to the conference. Special acknowledgment is also due to Tynysbek Kalmenov, Durvudkhan Suragan, Ravshan R. Ashurov,Mirsaid Aripov,Abdizhahan Sarsenbi,Abdullah S. Erdogan, Charyyar Ashyralyyev, Deniz Agirseven, Maksat Ashyraliyev, Yıldırım Özdemir, Özgür Yıldırım, Ramil Salimov, and Akmuhammet Ashyralyyev for their invaluable assistance in the organization and preparation of ICAAM 2026. We also gratefully acknowledge the support and sponsorship of the leadership and colleagues of Bahçeşehir University, Istanbul; the Institute of Mathematics and Mathematical Modeling, Almaty; and the Analysis and PDE Center, Ghent University. Their support has played an important role in making this international scientific gathering possible.

Finally, I would like to thank all participants for their enthusiasm, cooperation, and commitment to the advancement of analysis and applied mathematics. Conferences such as ICAAM are successful not only because of their scientific programs, but also because they create opportunities for new collaborations, friendships, and lasting academic connections. We believe that your participation will greatly enrich ICAAM 2026 and contribute to a stimulating and productive scientific atmosphere. We hope that your time in Antalya will be both scientifically rewarding and personally enjoyable. Thank you once again for joining us and for contributing to the success of ICAAM 2026. I wish you all a successful, fruitful, and memorable conference.

Warm regards,Prof. Dr. Allaberen Ashyralyev

On behalf of the Organizing Committee

ICAAM 2026

Bahçeşehir University, Istanbul, Türkiye

Generalized fractional integrals related to Le Roy-, Feinman integrals- and Rathie I-generalized hypergeometric functions

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Abstract: Till recently, the operators of Generalized Fractional Calculus (GFC) ([1]) with the Fox H- and Meijer G-functions as singular kernels have been considered as the most general case of integrations and differentiations of fractional orders and multi-orders. The H-function (as an extension of the G-function) is a generalized hypergeometric function including practically almost all Special Functions (SF) related to classical Calculus and to Fractional Calculus (FC). However, it happened that still there are important classes of SF that cannot be represented in terms of the H-functions, like the polylogarithms, Riemann Zeta-function, Feynman integrals, Le Roy function and their recent multi-index versions, appearing in models in statistical physics and probability, see [2]. That is why, the not popular I-function of Rathie has attracted our attention as a far extension of the H-function, since it encompasses also these mentioned classes of SF. Now, we introduce GFC operators with I-functions as singular kernels: analogues of the Erdélyi-Kober fractional integrals [3], and then generalized fractional integrals [4] extending these with H- and G-functions from [1]. It is proved that all the basic axioms of FC are satisfied, and important examples and relations to new classes of SF follow, some of them appearing as eigenfunctions for the new GFC operators.

Keywords: generalized fractional calculus, Erdélyi-Kober fractional integrals, special functions: I -, H-, G-, Mittag-Leffler and Le Roy type-functions

2020 Mathematics Subject Classification: 26A33, 33E20, 33E12, 30D20, 30D15, 34L10
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On sequential Karamata theory

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Abstract: We present some recent results on Karamata theory of regular variation of sequences of positive real numbers. The results concern the so-called index function and its relation to fixed point theory [1], and connections between sequential Karamata theory and statistical convergence [2].

Keywords: Regular variation, translational regular variation, index function, statistical convergence

2020 Mathematics Subject Classification: 26A12, 40A05, 40A35, 40G15

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Global solutions of the Cauchy problem for the Vlasov–Poisson system and plasma confinement

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Abstract: We consider the Vlasov–Poisson system with an external homogeneous magnetic field describing the kinetics of a two-component rarefied plasma in the three-dimensional case. For all initial density distribution functions with compact supports, we obtain sufficient conditions on the external magnetic field that provide the global existence of density distribution functions whose supports have arbitrarily small growth with respect to the space variables.

The work was supported by the Ministry of Science and Higher Education of the Russian Federation within the state assignement (project no. FSSF–20026–0013).

Keywords: Cauchy problem, Vlasov–Poisson system, plasma confinement

2020 Mathematics Subject Classification: 35Q83, 35M32

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Series expansions of solutions of initial-value problems for hyperbolic differential-difference equations: computational aspects

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Abstract: This report is devoted to the following prototype problem:

$$u_{tt}(x,t) - u_{xx}(x,t) = au(x+h,t),\ x \in (-\infty,\infty),\ t > 0, \tag{1}$$

$$u(x,0) = \varphi(x),\ x \in (-\infty,\infty), \tag{2}$$

$$u_t(x,0) = \psi(x),\ x \in (-\infty,\infty), \tag{3}$$

with arbitrary real parameters a and h.

Its solution $\sum\limits_{j=0}^{\infty} a^j \left[u_{1,j}(x,t) + u_{2,j}(x,t)\right]$ found in [1], where the series terms are defined by explicit recurrent integral relations, is investigated from the computational viewpoint.

Here, we provide direct relations to compute $u_{1,j}(x,t)$ and $u_{2,j}(x,t)$. Further, using the piecewise-linear interpolation of the initial-value functions, we show that the total error of the proposed numerical procedure is equal to the error of the proposed piecewise-linear approximation.

The work of the first author was supported by the Ministry of Science and Higher Education of the Russian Federation (project number FSSF-2026-0013).

Keywords: Hyperbolic equations, differential-difference operators, solution expansions, Picard approximations

2020 Mathematics Subject Classification: 35L10, 35R10

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1.6**A. Muravnik, O. Yaremko, N. Yaremko, Series expansions of solutions of initial-value problems for hyperbolic differential-difference equations: computational aspects**

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Pohozhaev identities and applications for mixed type equations

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Joint work with Daniela Lupo (Italy), Kevin Payne (USA, Italy), Lubomir Dechevski (Norway) and Evgeny Moiseev (Russia)

Abstract: Beginning from the seminal Pohodzaev paper (1965), the Pohodzaev Identities, first developed for the Dirichlet problem for elliptic quasilinear equations, have been extended to study many cases for different PDE classes of equations, even for some fractional PDE's. One of reasons is that, for example, from the Pohodzaev identities in the case of supercritical power follows very fast the nonexistence of nontrivial solutions in so called "starshaped domains". Also, in the opposite case of subcritical power (especially for elliptic equations) there exist some nontrivial solutions, look for example the remarkable results of Brezis-Nirenberg (1983) (for the history of the area see the very new book [1]).

The present talk will focus on the extension of Pohodzaev approach in the last 25 years to the study its applications for mixed elliptic-hyperbolic quasilinear equations. This study began from the series of results of Lupo - Payne, where using some symmetries in the linear homogeneous case, the dilation invariance law is strongly related to certain Pohodzaev type identities that give examples of critical exponent phenomena. Let mention here some BVP, in the linear case with Tricomi equation, Gellerstedt equation, Grushin equation, or in the last few years with the operator $L := y^{m_1}\partial_{xx}^2 + x^{m_2}\partial_{yy}^2$, in some appropriated "starshaped domain", connecting with the linear case statements.

In the present talk we will fix and discuss two possible interesting questions:

(1) What happen with "existence of nontrivial solutions" in this area? First of all, as a big difference with the elliptic case, we must look in this situation not in the class of classical smooth solutions, but in some special functional weight class of "generalized solutions". And the first question is to find such class of "generalized solutions", connecting the cases of existence and nonexistence in both respectively super and sub power situations (for example see our paper [2] in the 2D case of Goursat problem for a degenerated hyperbolic equation).

(2) This is the reason why in the present talk we will give some examples of nonexistence of nontrivial solutions in supercritical and critical cases not only for the classical smooth solutions, but in the class of generalized solutions also. Some open problems will be formulated and discussed.

(3) Also, to fix how different the situation sometimes is, we explain a very old classical Guderley-Morawetz problem for mixed-type hyperbolic-elliptic equations on the plane that models transonic flows in fluid dynamics. It has been studied by Morawetz, Lax and Phillips and going later to some multidimensional BVP, namely Protter-Morawetz problems, where the situation is quite different than on the plane. In the last case using the Pohodzaev identities we proved nonexistence results (in the supercritical and critical cases) but for classical solutions only. Let mention also that in the linear situation, the Protter-Morawetz problem is strongly overdetermined for the classical solvability. Alternatively, the notion of generalized solutions that may have singularities was introduced. Looking for some appropriated class of generalized singular solutions, we found (see [3]) that even for the usual wave equation in the Protter-Morawetz domain (the Protter statement is including this case also) we constructed an infinity smooth right-hand side function, for which the uniquely determined generalized solution has an exponential growth at the only one point of singularity. The singularity is isolated at the vertex of the boundary characteristic light cone and does not propagate along the cone.

Keywords: Pohozhaev identities, power nonlinearity, a semi-Fredholm operator, critical exponent, supercritical cases, nonexistence of nontrivial solutions

2020 Mathematics Subject Classification: 35B33, 35L05, 35M12

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Contents

General Section

A class of polynomial differential systems with an unique algebraic limit cycle

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Abstract: In this report we consider a multi-parameter family of polynomial differential systems of the form

$$(1)\quad \begin{cases} x' = \beta^4 x + \left(4\beta^3\lambda x^3 + bx^2y + 4\beta^3\lambda xy^2 + ay^3\right) + \\ 6\beta^2 x\left(\lambda x^2 + \lambda y^2\right)^2 + 4\beta x\left(\lambda x^2 + \lambda y^2\right)^3 + x\left(\lambda x^2 + \lambda y^2\right)^4, \\ \\ y' = \beta^4 y + \left(-bx^3 + 4\beta^3\lambda x^2y - axy^2 + 4\beta^3\lambda y^3\right) + \\ 6\beta^2 y\left(\lambda x^2 + \lambda y^2\right)^2 + 4\beta y\left(\lambda x^2 + \lambda y^2\right)^3 + y\left(\lambda x^2 + \lambda y^2\right)^4, \end{cases}$$

where a, b, β, λ are real constants. We prove that these systems are integrable and determine the explicit expression of their first integral. Moreover, we explicitly determine an algebraic limit cycle of the system, concrete examples are presented to illustrate the applicability of our results.

Keywords: Planar polynomial differential system, First integral, Algebraic limit cycle.

2020 Mathematics Subject Classification: 34A05, 34C05, 34CO7, 34C25.

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Modeling Tumor-Immune Dynamics: Synergistic Combination Treatment and the Role of Dynamic PD-L1 Regulation

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Abstract: The efficacy of cancer immunotherapies, particularly immune checkpoint inhibitors (ICIs), is often hindered by adaptive immune resistance mediated by the PD-1/PD-L1 pathway. Combination therapies, such as pairing an ICI with a potent immunostimulant, represent a promising strategy to overcome this challenge. Foundational models of single-agent and combination (Avelumab/NHS-muIL12) immunotherapies, while capturing key dynamics, fail to reproduce complex preclinical data like non-monotonic tumor responses.

The main focus of this talk is the introduction of a refined model that addresses this gap by treating PD-L1 expression as a dynamic variable. This variable accounts for basal production, activation by immunostimulants, and suppression by anti-PD-L1 drugs. We will show how this refined model, with a single set of parameters, successfully reproduces the experimental data across all six treatment arms including monotherapies and combinations. This approach provides a robust, mechanistic explanation for both dose-dependent treatment failure (driven by adaptive PD-L1 upregulation) and therapeutic synergy. We will discuss the systematic development of the model and simulation results, alongside a rigorous mathematical analysis of the generic and full models focusing on the existence, uniqueness, and local asymptotic stability of tumor-free and tumorous equilibria [1].

Keywords: cancer immunotherapy; differential equations; PD-L1; adaptive immune response; mathematical modeling; immunostimulant.

2020 Mathematics Subject Classification: 92C50, 92C60, 34D23, 34C60

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Mathematical modeling of heat transfer in hollow building blocks with internal partitions

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Abstract: The work examines mathematical modeling of heat transfer in hollow building blocks with internal partitions. The installation of internal partitions suppresses large-scale convective flows and reduces radiative transfer, resulting in a reduction in overall heat flux by tens of percent.

The authors' own numerical results obtained using computational fluid dynamics methods for various block configurations, including those with vertical and inclined partitions and low-emission coatings, will be presented. The influence of geometric parameters (height, orientation, number of partitions) on the flow structure, temperature fields and effective thermal conductivity is shown. Furthermore, the work presents a critical review of current research in this area, including experimental and numerical studies. Special attention is given to a comparative analysis of various methods for increasing the thermal resistance of blocks and identifying the most promising configurations [1-4]. The results can be used in the design of energy-efficient wall materials.

This work was supported by the Russian Science Foundation (Project No. 25-79-10293).

Keywords: Mathematical modeling, hollow building blocks, boundary conditions, convection, radiation, conduction

2020 Mathematics Subject Classification: 76D05, 80A19, 80A21

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A new approach to finding global extrema of multivariable objective functions

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Abstract: This report is devoted to a deterministic method for the global optimization of a multimodal objective function in many variables, based on a special auxiliary function.

The auxiliary function is constructed via an integral transformation of the objective function $F(x)$ and is a function of a single variable $\alpha \in \mathbb{R}$:

$$g_m(F,\alpha) = \int_E \left(|F(x)-\alpha| - (F(x)-\alpha)\right)^m dx, \qquad m \in \mathbb{N},\ m > 1.$$

The function $g_m(F,\alpha)$ possesses several important properties, including continuity, differentiability, convexity, monotonicity, and others. These properties, together with its dependence on only one variable, make the auxiliary function an effective and unique tool for solving global optimization problems.

In this work, based on the properties of the function $g_m(F,\alpha)$ previously established by the authors [1-2], necessary and sufficient conditions for the existence of a global minimum of the function $F(x)$ are obtained. It is shown that the value of the global minimum of the objective function coincides with the upper bound of the zeros of the auxiliary function. Thus, the problem of finding the global minimum of the original objective function is reduced to the problem of finding the largest zero of the auxiliary function.

In addition to various multimodal functions of many variables, the objective function may also represent loss functions arising in machine learning problems, as well as functions describing economic, physical, and other processes. The main requirement imposed on the objective function is continuity.

Using the proposed method, the problem of minimizing the loss function of a nonlinear regression model and recovering its parameters is solved. A complete algorithm for finding the global minimum of the loss function is proposed, and its numerical implementation is carried out with recovery of the model parameters to a prescribed accuracy.

Keywords: Global optimization, nonlinear regression, loss function, auxiliary function, deterministic methods.

2020 Mathematics Subject Classification: 90C26, 62F30, 65K05 **References:**

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Development an algorithm of equilibrium growth of Kazakhstan's financial and insurance activities

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Abstract: This report is devoted to study the problems of hidden imbalances in transactions of platform industry, internal and external ecosystem agents. Based on historical data from the OECD for 1995–2018, a methodology, computational models and algorithms for determine the equilibrium state of transactions of Kazakhstan's financial and insurance activity agents have been developed. The results are presented in the following steps.

Step 1. Let we have the supply function of the equilibrium transactions of the platform industry produce/selling agent is:

$$z'(t) = \text{Intercept}' + \text{Slope}' \times z'(t), \qquad t = \overline{1995; 1995 + (T-1)},$$
$$\text{(1)}\quad \text{R-square} \qquad\qquad T = 1, 2, \ldots, 24,$$
$$\text{(Std. Error)} \qquad \text{(Std. Error)}$$

and we have the demand function of the equilibrium transactions of the platform industry purchase/buying agent is:

$$z(t) = \text{Intercept} + \text{Slope} \times z(t), \qquad t = \overline{1995; 1995 + (T-1)},$$
$$\text{(2)}\quad \text{R-square} \qquad\qquad T = 1, 2, \ldots, 24,$$
$$\text{(Std. Error)} \qquad \text{(Std. Error)}$$

Then exists $\exists! t^* = \overline{1995; 1995 + (T-1)}$, $T = \overline{1; 24}$ such that: $z'(t^*) = z(t^*)$.

Step 2. Let we have the supply function of the equilibrium transactions of the internal ecosystem payment agent is:

$$p(t) = \text{Intercept}' + \text{Slope}' \times p(t), \qquad t = \overline{1995; 1995 + (T-1)},$$
$$\text{(3)}\quad \text{R-square} \qquad\qquad T = 1, 2, \ldots, 24,$$
$$\text{(Std. Error)} \qquad \text{(Std. Error)}$$

and we have the demand function of the equilibrium transactions of the internal ecosystem final demand agent is:

$$f(t) = \text{Intercept} + \text{Slope} \times f(t), \qquad t \in \overline{1995; 1995 + (T-1)},$$
$$\text{(4)}\quad \text{R-square} \qquad\qquad T = 1, 2, \ldots, 24,$$
$$\text{(Std. Error)} \qquad \text{(Std. Error)}$$

Then exists $\exists! t^* = \overline{1995; 1995 + (T-1)}$, $T = \overline{1; 24}$ such that: $p(t^*) = f(t^*)$.

Step 5. Let we have the supply function of the equilibrium transactions of the external ecosystem exports agent is:

$$e(t) = \text{Intercept}' + \text{Slope}' \times e(t), \qquad t = \overline{1995; 1995 + (T-1)},$$
$$\text{(5)} \quad \text{R-square} \qquad\qquad T = 1, 2, \ldots, 24,$$
$$\text{(Std. Error)} \qquad \text{(Std. Error)}$$

and we have the demand function of the equilibrium transactions of the external ecosystem imports agent is:

$$m(t) = \text{Intercept} + \text{Slope} \times m(t), \qquad t = \overline{1995; 1995 + (T-1)},$$
$$\text{(6)} \quad \text{R-square} \qquad\qquad T = 1, 2, \ldots, 24,$$
$$\text{(Std. Error)} \qquad \text{(Std. Error)}$$

Then exists $\exists! t^* = \overline{1995; 1995 + (T-1)}$, $T = \overline{1; 24}$ such that: $e(t^*) = m(t^*)$.

Throughout this note we mainly use techniques from our works [1].

Keywords: computational models; equilibrium; algorithm; development; financial and insurance activities; Kazakhstan.

Acknowledgments. This research was funded by the Committee of Science of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant No. AP26196337).

2020 Mathematics Subject Classification: 68M01, 68R01, 68U35

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Very weak well-posedness of parabolic equations with singular coefficients and inhomogeneous Dirichlet data

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Abstract: We study the very weak well-posedness of the initial–boundary value problem for the parabolic equation

$$\begin{cases} \partial_t u(t,x) - \nabla \cdot \big(A(t,x)\nabla u(t,x)\big) + b(t,x)\cdot \nabla u(t,x) + V(x)u(t,x) = f(t,x), \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad (t,x) \in (0,T)\times\Omega, \\ u(0,x) = u_0(x), \quad x \in \Omega, \\ u(t,x) = g(t,x), \quad (t,x) \in (0,T)\times\partial\Omega, \end{cases}$$

where the diffusion matrix A, drift term b, potential V, source term f, initial datum u_0, and boundary datum g may have singularities of distributional type. In the regular case, we first establish existence, uniqueness, and a priori estimates for weak solutions under non-homogeneous Dirichlet boundary conditions. We then introduce a very weak solution framework based on regularisation of the coefficients and data. Under suitable distributional ellipticity of A and non-negativity of V, we prove existence and uniqueness of very weak solutions. Finally, we show consistency with the classical theory: when the coefficients and data are regular, representatives of the very weak solution converge to the unique weak solution in the natural energy norm.

Keywords: parabolic equation, divergence form, shift term, energy methods, Galerkin approximation, singular coefficients, regularisation, very weak solution

2020 Mathematics Subject Classification: 35K20, 35D30, 35K67

Solution of a nonlocal problem for the heat equation using the potential method

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Abstract: One nonlocal problem for the heat equation is considered

(1) $$L(u) \equiv u_t - u_{xx} = 0 \quad in\ domain \quad \Omega = \{(x,t) : 0 < x < l, \quad 0 < t < T\}$$

with initial conditions

(2) $$u(x,0) = f(x), \quad 0 \le x \le l$$

boundary conditions of the form

(3) $$\begin{cases} a_1 u(0,t) + a_2 u(l,t) = \varphi(t), \\ b_1 u_x(0,t) + b_2 u_x(l,t) = \psi(t), \end{cases} \qquad 0 \le t \le T$$

the coefficients $a_i, b_i (i = 1, 2)-$ real numbers.

Using the potential method, problem (1)-(3) is reduced to Volterra integral equations. The solution of the Volterra integral equation was found and the solvability of the stated problem (1)-(3) was investigated.

Acknowledgments: This research is funded by the Science Committee of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant No. BR31714735)

Keywords: heat equation, nonlocal problems, potential method, integral equations, solvability conditions

2020 Mathematics Subject Classification: 35A09, 35L20, 35M13

Solution of forward and inverse problems for the convection-diffusion-reaction equation using neural networks

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Abstract: This paper considers forward and inverse problems for a two-dimensional convection–diffusion–reaction equation describing the propagation of harmful aerosol substances in the atmosphere. The mathematical model takes into account diffusion processes, convective transport by airflow, chemical reactions, and localized pollution sources represented by delta functions. The Physics-Informed Neural Networks (PINN) approach is applied for solving the considered problems.

The computational domain is defined as

$$\Omega = (0,1) \times (0,1) \times (0,T), \tag{1}$$

and the governing equation is given by

$$\frac{\partial \varphi_q}{\partial t} + \nabla \cdot (\mathbf{w}\varphi_q) = \Delta \varphi_q + \alpha_q \varphi_q + \beta_q + f_q(x,y), \tag{2}$$

where $\mathbf{w} = (u, \vartheta)$ is the airflow velocity vector, $\varphi_q(x,y,t)$ denotes the aerosol concentration, α_q and β_q are reaction coefficients, and $f_q(x,y)$ describes pollution sources.

The pollution sources are represented as a superposition of delta functions:

$$f_q = \sum_{j=1}^{m} Q_j \delta(\mathbf{r} - \mathbf{r}_j), \tag{3}$$

where Q_j is the source intensity and $\mathbf{r}_j$ denotes the source location.

The initial concentration distribution is defined by

$$\varphi_0(x,y) = \frac{Q_j}{2\sqrt{\mu\sigma}} \exp\left(-\sqrt{\frac{(x-x_j)^2 + (y-y_j)^2}{\mu}} \right). \tag{4}$$

In the forward problem, the concentration distribution is determined from known initial and boundary conditions. In the inverse problem, unknown model parameters such as source intensities, diffusion coefficients, or reaction parameters are reconstructed using observational concentration data.

Within the PINN framework, the solution is approximated by a neural network $\varphi_\theta(x, y, t)$, where θ denotes trainable parameters. The loss functional is defined as

$$\mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_{IC} + \mathcal{L}_{BC} + \mathcal{L}_{data}. \tag{5}$$

Here $\mathcal{L}_{PDE}$ enforces the governing equation, $\mathcal{L}_{IC}$ corresponds to the initial condition, $\mathcal{L}_{BC}$ represents the boundary conditions, and $\mathcal{L}_{data}$ minimizes the discrepancy between observational data and neural network predictions.

Numerical experiments demonstrate that the PINN method provides stable and accurate approximations without constructing computational meshes. The proposed approach effectively solves both forward and inverse problems and can be applied to atmospheric pollution modeling and other multidimensional problems of mathematical physics.

Keywords: convection–diffusion equation, PINN, neural networks, forward problem, inverse problem, atmospheric pollution, deep learning.

2020 Mathematics Subject Classification: 35K57, 35R30, 65M99, 68T07.

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I-lacunary statistical convergence in partial metric spaces

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Abstract:

In this paper, we make a new framework for statistical convergence by using ideals and introduce new notions, namely, I-statistical convergence and I-lacunary statistical convergence in Partial Metric spaces and mainly investigate their relationship and also make some observations about these classes

Keywords: Ideal, filter, I-statistical convergence, I-lacunary statistical convergence, partial metric space.

2020 Mathematics Subject Classification: 40A05, 40D25

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Parametric nonlocal initial-boundary value problems for a degenerate hyperbolic equation

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Abstract: Let $\Omega = \{(x,y) : \ 0 < x < 1, \ 0 < y < T\}$ be a rectangular domain. Consider the following degenerate hyperbolic equation:

$$Lu \equiv y^m u_{xx} - u_{yy} - b^2 y^m u = 0, \tag{1}$$

where $m > 0$, $b \geq 0$, and $T > 0$ are given real parameters.

In this work, we study two problems for equation (1) with the initial conditions and different parameter-dependent nonlocal boundary conditions.

Problem P_α**.** *Find a function* $u(x,y) \in C^1\big(\overline{\Omega}\big) \cap C^2(\Omega)$, *which is a solution of equation* (1) *in the domain* Ω, *satisfies the classical initial conditions*

$$u(x,0) = \tau(x), \quad u_y(x,0) = \nu(x), \quad 0 \leq x \leq 1, \tag{2}$$

and the following nonlocal boundary conditions:

$$u_x(0,y) = \alpha\, u_x(1,y), \quad 0 \leq y \leq T, \tag{3}$$

$$u(0,y) = u(1,y), \quad 0 \leq y \leq T, \tag{4}$$

where α *is a given real parameter, and* $\tau(x)$ *and* $\nu(x)$ *are sufficiently smooth given functions.*

Problem P_β**.** *Find a function* $u(x,y) \in C^1\big(\overline{\Omega}\big) \cap C^2(\Omega)$, *which is a solution of equation* (1) *in the domain* Ω, *satisfies the classical initial conditions* (2), *and the following nonlocal boundary conditions:*

$$u_x(0,y) = u_x(1,y), \quad 0 \leq y \leq T, \tag{5}$$

$$u(0,y) = \beta\, u(1,y), \quad 0 \leq y \leq T, \tag{6}$$

where β *is a given real parameter, and* $\tau(x)$ *and* $\nu(x)$ *are sufficiently smooth given functions.*

Using the spectral analysis method, the main results on the existence and uniqueness of solutions to Problems P_α and P_β have been obtained, and the corresponding theorems have been proved.

The key feature of the problem under study is that the system of eigenfunctions of the spectral problem (which arises when applying the Fourier method) does not form an unconditional basis. Therefore, direct application of the Fourier method is impossible. To overcome this, an auxiliary basis is constructed using eigenfunctions.

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Keywords: degenerate hyperbolic equation, initial condition, nonlocal boundary conditions, regular boundary conditions, root functions, Riesz basis,

2020 Mathematics Subject Classification: 35L20

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The mixed initial-boundary value problem for the heat equation with a discontinuous coefficient under general conjugation conditions

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Abstract: An initial-boundary value problem for the heat equation with a discontinuous coefficient is considered:

$$\frac{\partial u}{\partial t} = a_i^2 \frac{\partial^2 u}{\partial x^2}, \tag{7}$$

in the domain $\Omega = \Omega_1 \cup \Omega_2$, $\Omega_1 = \{(x,t) : l_0 < x < l_1, 0 < t < T\}$, $\Omega_2 = \{(x,t) : l_1 < x < l_2, 0 < t < T\}$, $(i = 1, 2)$, with initial conditions

$$u(x,0) = \varphi(x), \quad l_0 \le x \le l_2, \tag{8}$$

boundary conditions of the form

$$u(l_0,t) = u_x(l_2,t) = 0, \quad 0 \le t \le T, \tag{9}$$

and conjugation conditions [1]

(10)
$$\begin{cases} a_{11}u_x(l_1-0,t) + a_{12}u_x(l_1+0,t) + a_{13}u(l_1-0,t) + a_{14}u(l_1+0,t) = 0, \\ a_{21}u_x(l_1-0,t) + a_{22}u_x(l_1+0,t) + a_{23}u(l_1-0,t) + a_{24}u(l_1+0,t) = 0. \end{cases}$$

The theorem on the existence and uniqueness of a classical solution to the problem (1)-(4) is proved [2].

Keywords: heat equation, discontinuous coefficients, eigenvalues, eigenfunctions, method of separation of variables

2020 Mathematics Subject Classification: 35K10, 35K20, 35P10

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Absolute stable difference scheme for numerical solving source identification elliptic problem with nonlocal condition

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Abstract Source identification problems arise in many areas of science and engineering, where unknown internal sources must be reconstructed from indirect observations. Such inverse problems are encountered in heat transfer, diffusion processes, geophysics, medical imaging, and environmental modeling ([1]-[3]).

In the paper [1], the well-posedness of the source identification problem for an abstract elliptic equation with nonlocal boundary conditions in a Banach space was established. The problem is formulated as

$$\begin{cases} -v''(t) - Av(t) = p + f(t), 0 < t < 1 \ , \\ v(0) = v(1), \\ v'(0) = v'(1) \end{cases} \tag{1}$$

where A is a positive operator, the parameter $p \in E$ and the function $v : [0, 1] \to E$ are unknown, while $f : [0, 1] \to E$ is given.

Although the continuous problem has been shown to be well-posed, practical applications require reliable numerical methods. This motivates the development of stable and convergent difference schemes that can efficiently approximate both the unknown source parameter and the solution.

This work is devoted to construct and analyze a finite difference scheme for the source identification problem, establish stability estimates that guarantee the well-posedness of the discrete problem, and demonstrate the accuracy and efficiency of the proposed method through numerical experiments.

This research was funded by the Science Committee of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant No. AP23487362).

Keywords: difference scheme, elliptic equation, inverse problem, source identification problem, stability estimate.

2020 Mathematics Subject Classification: 39A14, 65N06, 65J22

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Theoretical analysis of variational formulations and finite element approximations for the Navier–Stokes equations

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Abstract: This report is devoted to the variational formulation and finite element approximation of the nonstationary Navier–Stokes equations:

$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p = f, \qquad \nabla \cdot u = 0. \tag{1}$$

The problem is supplemented with the initial condition

$$u(x, 0) = u_0(x) \tag{2}$$

and homogeneous Dirichlet boundary conditions.

Let $V \subset H^1(\Omega)^d$ and $Q \subset L^2(\Omega)/\mathbb{R}$ be the velocity and pressure spaces, respectively. The bilinear and nonlinear forms are defined as follows: $a(u, v)$ is the diffusion term, $b(v, p)$ is the pressure term, and $c(w; u, v)$ is the nonlinear convective term.

The weak formulation consists in finding $(u, p) \in V \times Q$ such that:

$$(\partial_t u, v) + c(u; u, v) + a(u, v) + b(v, p) = (f, v), \tag{3}$$

$$b(u, q) = 0, \qquad \forall (v, q) \in V \times Q. \tag{4}$$

Finite-dimensional approximations satisfying the discrete Ladyzhenskaya–Babuška–Brezzi (LBB) condition are considered. Under suitable regularity assumptions, the finite element approximation satisfies the error estimate:

$$\|u(t^n) - u_h^n\|_{L^2} \leq C(h^2 + \Delta t). \tag{5}$$

The results confirm the convergence of the finite element method for incompressible flow problems.

Keywords: Navier–Stokes equations, variational formulation, finite element method.

2020 Mathematics Subject Classification: 35Q30, 65M60, 76D05.
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Geometric criteria for controllability in arbitrary Banach spaces

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Abstract: We present geometric criteria ensuring controllability properties for dynamical systems evolving in arbitrary (possibly infinite-dimensional) Banach spaces. Let $(X, \|\cdot\|)$ be a real Banach space and consider the set-valued evolution

$$x'(t) \in F(t, x(t)) \subset X,\ t \in [t_0, T],\ x(t_0) = x_0, \tag{1}$$

where $F(t, x)$ is a nonempty subset of X. The reachable set at time t is

$$\mathcal{R}(t; x_0) = \{x(t) \in X :\ x(\cdot) \text{ solves the dynamics with } x(t_0) = x_0\}. \tag{2}$$

Controllability is reformulated as a geometric property of these reachable sets. For a closed target $S \subset X$, exact controllability requires $\mathcal{R}(T; x_0) \cap S \neq \emptyset$, while approximate controllability requires $\overline{\mathcal{R}(T; x_0)} \cap S \neq \emptyset$; approximate null-controllability corresponds to $0 \in \overline{\mathcal{R}(T; x_0)}$.

The approach is based on set-valued analysis and viability-type arguments on suitable constraint sets, together with separation theorems and tangent (Bouligand/contingent) cones. For a closed set $K \subset X$ and $x \in K$, the contingent cone is

$$T_K(x) = \{v \in X :\ \exists\, h_n \downarrow 0^+,\ \exists\, v_n \to v,\ x + h_n v_n \in K\}. \tag{3}$$

A tangency (viability) condition of the form $F(t, x) \cap T_K(x) \neq \emptyset$ keeps trajectories in K. Encoding the control objective through an epigraph set $\mathcal{K} = \{(x, r) \in X \times \mathbb{R} : r \geq \phi(x)\}$, where ϕ is a lower semicontinuous distance-to-target functional (e.g. $\phi(x) = \|x\|$ or $\phi(x) = \operatorname{dist}(x, S)$), near-viability of $\mathcal{K}$ under an augmented dynamics yields quantitative decrease estimates of $\phi(x(t))$ and hence approximate steering to the target. Finally, in the dual space X^*, approximate controllability is shown to be equivalent to the nonexistence of a separating functional between the target and the reachable sets, providing the infinite-dimensional analogue of finite-dimensional geometric separation arguments.

Throughout this note we mainly use techniques from set-valued analysis and viability theory [1, 2], control theory [3], and semigroup methods [4].

Keywords: Banach spaces, controllability, reachable set, viability, contingent cone, set-valued dynamics, separation

2020 Mathematics Subject Classification: 93B05, 49J53, 34A60, 47D06

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Pathwise Fourier continuation for Bermudan and American options: a sinh-accelerated NUFFT–RQMC framework

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Abstract: We develop a pathwise Fourier-continuation framework for Bermudan and American option pricing. The method combines randomized quasi-Monte Carlo path simulation, sinh-accelerated Fourier inversion, and type-3 nonuniform FFT evaluation so that, at each exercise date, continuation values are computed from the one-step characteristic function directly at the simulated states. This replaces the regression fit in least-squares Monte Carlo by a transform-based conditional expectation while retaining a simulation architecture. The framework is formulated for one-factor exponential models with explicit one-step characteristic functions; the continuously monitored barrier treatment considered here is specific to geometric Brownian motion, where a Brownian-bridge survival correction is applied in log space without changing the Fourier continuation step.

Numerical experiments recover standard GBM American-put benchmarks accurately, show competitive bias and RMSE relative to polynomial and Laguerre-basis LSMC on representative cases, and demonstrate the continuation machinery under a pure-jump CGMY model. For continuously monitored up-and-out puts under GBM, the barrier extension gives prices in close agreement with published benchmark values.

Keywords: American options; Bermudan options; Fourier continuation; nonuniform fast Fourier transform; randomized quasi-Monte Carlo; sinh acceleration; barrier options; CGMY model.

JEL classification: C63; G12; G13.

Heat equation with a singular potential in domains with non-smooth boundaries

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Abstract. This paper investigates the behavior of nonnegative generalized solutions to the first initial-boundary value problem for a linear heat equation with a singular potential in a cylindrical domain $Q = \Omega \times (0, T)$, where $\Omega \subset R^n$ $(n \geq 3)$ is a neighborhood of the edge containing the origin and $T > 0$. In a neighborhood of the origin, the domain Ω is locally diffeomorphic to a a dihedral angle of opening β_0 $(0 < \beta_0 \leq 2\pi)$. A sharp condition on the potential is found that ensures the existence and/or nonexistence of nonnegative solutions to this problem.

Keywords: heat equation, singular potential, generalized solution, nonnegative solution, behavior of solutions, sharp condition, existence and nonexistence.
2020-Mathematics Subject Classification: 35K05, 35K20, 35K67

Let Ω be a bounded domain in R^n $(n \geq 3)$, whose boundary $\partial\Omega$ consists of two smooth $(n-1)-$dimensional surfaces Γ_1 and Γ_2 intersecting along an $(n-2)-$dimensional manifold (edge) Γ_0, such that the origin lies on the edge: $0 \in \Gamma_0$. It is assumed that in a neighborhood of the origin, the domain Ω is locally diffeomorphic to a dihedral angle of opening β_0 $(0 < \beta_0 \leq 2\pi)$. We consider the first initial-boundary value problem in the cylinder $Q = \Omega \times (0, T)$:

$$u_t - \Delta u - V(x)u = 0, \quad (x, t) \in \Omega \times (0, T), \tag{1}$$

$$u(x, t) = 0, \quad x \in \partial\Omega, \ \ t \in (0, T); \quad u(x, 0) = u_0(x), \quad x \in \Omega. \tag{2}$$

It is assumed that $0 \leq V \leq L^1_{loc}(\Omega)$ and $0 \leq u_0 \leq L^2(\Omega)$.

The initial condition holds in the sense of

$$ess \lim_{t \to +0} \int\limits_{\Omega} u(x, t)\eta(x)dx = \int\limits_{\Omega} u_0(x)\eta(x)dx$$

for any function $\eta \in D(\Omega) = C_0^\infty(\Omega)$.

Definition. Let $Q = \Omega \times (0, T)$. A function $u(x, t)$ is called a *nonnegative weak (generalized) solution* to the problem (1)-(3) in the sense of distributions if $u \in L^1_{loc}(Q)$, $Vu \in L^1_{loc}(Q)$, and for any function $\psi \in C^\infty(\overline{Q})$ such that $\psi(x, T) = 0$ and $\psi|_{\partial\Omega \times (0,T)} = 0$, the following integral identity holds:

$$\iint\limits_{Q} u(-\psi_t - \Delta\psi - V(x)\psi)dxdt = \int\limits_{\Omega} u_0(x)\psi(x, 0)dx.$$

In a neighborhood of the point $0 \in \Gamma_0$, we introduce local spherical coordinates: $r = |x| \in (0, \delta)$, $(\delta > 0)$ and $\omega \in \Omega_\omega \subset S^{n-1}$, where Ω_ω is the spherical cross-section of the dihedral angle on the unit sphere. The Laplace operator takes the form:

$$\Delta = \frac{\partial^2}{\partial r^2} + \frac{n-1}{r} \cdot \frac{\partial}{\partial t} + \frac{1}{r^2}\Delta_\omega,$$

where Δ_ω is the Laplace-Beltrami operator on S^{n-1}.

Consider the Dirichlet eigenvalue problem on the spherical segment Ω_ω :

$$-\Delta_\omega Y(\omega) = \lambda Y(\omega), \quad \omega \in \Omega_\omega; \qquad Y(\omega) = 0, \quad \omega \in \partial\Omega_\omega.$$

Let λ_1 be the principal (first) eigenvalue of this problem and let $Y_1(\omega)$ be the eigenfunction corresponding to the principal eigenvalue λ_1.

Let $\varphi(x) = |x|^{-\alpha} Y_1(\omega)$. Let us find the function $V_0(x)$ satisfying the equation $\Delta\varphi + V_0(x)\varphi = 0$ in $D'(\Omega)$. Since $-\Delta\varphi = [\alpha(n-2-\alpha) + \lambda_1]|x|^{-\alpha-2}Y_1(\omega)$, we obtain: $-\Delta\varphi/\varphi = c|x|^{-2}$, where $c = \alpha(n-2-\alpha)+\lambda_1$. From the last equality, it follows that the number α is defined by the formula

$$\alpha = \frac{n-2}{2} - \sqrt{\frac{(n-2)^2}{4} + \lambda_1 - c} = \frac{n-2}{2} - \sqrt{C_H - c}.$$

It is clear that when $0 \le c \le C_H = \frac{(n-2)^2}{4} + \lambda_1$, the number α is a real and $\Delta\varphi \in L^1_{loc}(\Omega)$.

We set: $V_0(x) = c|x|^{-2}$, $x \in \Omega$.

Note that the constant C_H is the best Hardy constant for non-smooth domains and is not achievable by functions in the Sobolev space.

The main result of this paper is the following theorem:

Theorem. 1. If $0 \le c \le C_H$ and $V(x) \le V_0(x)$ in the domain Ω, then problem (1),(2) has a nonnegative generalized solution for any nonnegative initial function $u_0 \in L^2(\Omega)$.

2. If $c > C_H$ and $V(x) \ge V_0(x)$ in the domain Ω, then for $u_0(x) > 0$ problem (1),(2) has not nonnegative generalized solutions.

Note that in the case of a conical domain with vertex at the origin, problem (1),(2) was studied in [1].

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Method of generalized jacobi elliptic functions for solving some classes of nonlinear elliptic systems of second-order equations on the plane

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Abstract: In this paper, generalized Jacobi elliptic functions $dn\,\omega(z)-$ delta-amplitude are applied to the construction of particular solutions of certain classes of elliptic systems of partial differential equations of second order on the complex plane C with Cauchy–Riemann operators $2\partial_{\overline{z}} = \partial_x + i\partial_y,\ 2\partial_z = \partial_x - i\partial_y$, and the Bitsadze operator $4\partial^2_{\overline{z}\,\overline{z}} = 4\partial_{\overline{z}}(\partial_{\overline{z}}) = \partial^2_{xx} - \partial^2_{yy} + 2i\partial^2_{xy}$, where the variables $z = x + iy,\ \overline{z} = x - iy$ are independent.

The function $\omega(z)$ is a quasi-periodic homeomorphism of the Beltrami equation

$$\omega_{\overline{z}} - q\omega_z = 0, \quad |\,q\,| \neq 1,$$

satisfying the condition

$$\omega(0) = 0, \quad \omega(z + h_j) = \omega(z) + \tau_j, \quad j = 1, 2,$$

where $\mathrm{Im}(h_2/h_1) \neq 0,\ \mathrm{Im}(\tau_2/\tau_1) \neq 0$.

The function $\omega(z)$ quasiconformally maps any parallelogram Ω with vertices $z_0,\ z_0 + h_1,\ z_0 + h_1 + h_2,\ z_0 + h_2$ onto a quadrilateral Ω' with vertices $\omega(z_0),\ \omega(z_0)+\tau_1,\ \omega(z_0)+\tau_1+\tau_2,\ \omega(z_0)+\tau_2$, and for $|\,q\,| < 1$ the orientation of the boundaries of Ω and Ω' is preserved.

Keywords: solution of equations, modulus of elliptic function, Jacobi elliptic functions, analytic function.

2020 Mathematics Subject Classification: 5A0,9 35B36

Heat transfer model with involution: analytical approaches and FDM analysis

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Abstract: In this study, a heat transfer model on a metal rod of length L, folded in half from its exact center ($t = 0$) into a U-shape, is investigated theoretically and numerically. While differential equations with involutions have been extensively studied by various researchers in the literature [2-5], the symmetric interaction resulting from the geometric structure of the rod and the proximity of its arms is modeled using a first-order differential equation with involution. Two different analytical methods frequently encountered in the literature, namely the separation into even-odd components and the order-elevation method, are evaluated for the solution of the model; both approaches are demonstrated to yield consistent general solutions in integral form [1]. For the numerical analysis of the equation, a Finite Difference Method (FDM) discretization based on the centered difference approximation for derivative expressions is formulated. Finally, under a contraction condition defined on the coefficients of the system, the existence, uniqueness, and uniform convergence of the sequence of successive approximations to the analytical solution are theoretically proven within the framework of the Banach fixed-point theorem.

Keywords: Heat transfer model, finite difference method, Banach fixed point theorem

2020 Mathematics Subject Classification: 34K06, 34A30, 34K10

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Historic behaviour for a Lotka–Volterra operator on the simplex

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Abstract: We study the dynamics of a four-dimensional Lotka–Volterra type operator $\mathcal{K}_r$ on the 3-simplex S^3, arising from a discrete-time Kolmogorov system. We first give a complete description of the fixed points: the unique fixed point in the interior is the symmetric centre $c = (1/4, 1/4, 1/4, 1/4)$, which is repelling, while every point on the opposite edges Γ_{13} and Γ_{24} is a (non-hyperbolic) fixed point. A Lyapunov-type function is used to show that, for any initial state in the interior of S^3 distinct from c, the ω-limit set is contained in the boundary of the simplex. By analysing the restrictions of $\mathcal{K}_r$ to the two-dimensional faces, we refine this picture and prove that every ω-limit set coming from the interior is in fact contained in the union of one-dimensional edges, with convergence to vertices excluded.

Our main results concern the interior dynamics. We describe the asymptotic ordering of coordinates and construct a set $X \subset \operatorname{int} S^3$ of positive Lebesgue measure such that the orbits of all $x \in X$ follow a cyclic itinerary of order regions near the four non-fixed edges $\Gamma_{12}, \Gamma_{23}, \Gamma_{34}$ and Γ_{14}. This intertwined motion produces long alternating blocks in which different pairs of coordinates dominate and implies that the averages of $\mathcal{K}_r^m$ fail to converge for every $x \in X$. In particular, $\mathcal{K}_r$ is non-ergodic and exhibits historic behaviour on a set of positive Lebesgue measure, providing a higher-dimensional example of rich historic dynamics in the class of Lotka–Volterra operators

Keywords: Lotka–Volterra operator, stochastic operator, Kolmogorov system, historic behaviour, non-ergodicity

2020 Mathematics Subject Classification: 37C75, 47H25, 37N25

Study of Gaussian primes using quantum mechanics

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Abstract: We introduce the concept of the prime state over Gaussian integers, defined as a uniform quantum superposition of all Gaussian primes within a bounded norm. By mapping Gaussian integers to a Hilbert space basis, we define Hamiltonians, norm operators, Hecke-type observables, and entanglement measures acting on the prime state. Time evolution generates interference patters revealing correlations among Gaussian primes. We analyze the spectrum using random-matrix analogies and discuss quantum primality oracles. Connections to the Prime Number Theorem, Gaussian prime number estimation, Chebyshev-type bounds, and prime-counting functions are provided.

Keywords: Gaussian Primes, Quantum superposition, Prime number theorem

2020 Mathematics Subject Classification: 11A05, 11A04, 11R11

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30 years of applied mathematics for radar signature computation

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Abstract: Prediction of radar signatures of targets and terrain using cumputer techniques has been of interest since early 1970s. This field of expertise requires a good understanding of advanced mathematics as well as how it can be applied to physical phenomena. In this presentation, I will review certain key mathematical theories and their applications, that have been game-changers in the area of radar signature analysis.

The first such application has made use of the Whittaker transform (1902), prolate spheroidal functions (1965), and spatial bandwidth of scattered fields (1987). The result was the invention of the Plane-Wave Time-Domain (PWTD) method [1] which reduced the computational complexity of solving time-domain integral equations from O(NtNs2) to O(NtNslogNs). Application of the PWTD method to both acoustic and electromagnetic problems received wide attention with 2000+ citations and awards.

The second application is the use of Radon Transform (1917) interpretation to evaluate highly oscillatory physical optics integrals exactly [2]. Prior to this application, these integrals used to be evaluated with approximate techniques such as steepest descent path, stationary phase, or saddle point integrations. This made it possible to calculate the radar cross section (RCS) of large targets on a laptop computer instead of super computers.

The third application is the use of matrix pencil method (1980) to extract infinite bandwidth resolution from bandlimited radar echoes [3]. This application resulted in big improvement in target classification and identification especially in synthetic aperture radar (SAR) images. Combination of the knowledge acquired from this application with surface equivalence theorem (1901; Huygens) resulted in near-field measurement of RCS [4].

Keywords: Plane-wave time-domain (PWTD), Radon Transform for phycal optics, Matrix pencil method, radar cross-section (RCS), Near-field RCS measurementt

2020 Mathematics Subject Classification: 45B05, 65D05, 65D18, 65D30, 78M16, 78M35

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Probabilistic representation and Monte Carlo solution of initial–boundary value problems for the advection–diffusion equation in air pollution modelling

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Abstract: We study the initial–boundary value problem (IBVP) for the advection–diffusion–reaction equation

$$\frac{\partial c}{\partial t} + \mathbf{u}\cdot\nabla c = \nabla\cdot(K\nabla c) - \lambda c, \qquad \mathbf{x}\in\Omega,\ 0<t\le T, \tag{1}$$

$$c(\mathbf{x},0) = c_0(\mathbf{x}), \qquad \mathcal{B}[c] = g \ \text{ on } \ \partial\Omega, \tag{2}$$

which governs the spatial distribution of a pollutant concentration c transported by a wind field $\mathbf{u}$, dispersed by a turbulent diffusivity K, and lost at the first-order rate λ. The boundary operator $\mathcal{B}$ ranges over the physically relevant Dirichlet (prescribed background or inflow), Neumann (impermeable, reflecting surface) and Robin (dry deposition, $-K\,\partial_n c = v_d\,c$) conditions.

We present the probabilistic, Feynman–Kac representation of the solution. Equation (1) is the Kolmogorov forward (Fokker–Planck) equation of the Itô diffusion $d\mathbf{X}_t = \mathbf{u}\,dt + \sqrt{2K}\,d\mathbf{W}_t$, so the concentration coincides, up to the released mass, with the sub-probability density of diffusing pollutant parcels killed at rate λ. The pointwise value of the solution admits the stochastic representation

$$c(\mathbf{x},t) = \mathbb{E}\Big[e^{-\lambda(t\wedge\tau)}\big(\mathbf{1}_{\{\tau>t\}}\,c_0(\mathbf{Y}_t) + \mathbf{1}_{\{\tau\le t\}}\,g(\mathbf{Y}_\tau,\,t-\tau)\big)\Big], \tag{3}$$

where $\mathbf{Y}$ is the adjoint diffusion $d\mathbf{Y}_s = -\mathbf{u}\,ds + \sqrt{2K}\,d\mathbf{W}_s$ started at $\mathbf{Y}_0 = \mathbf{x}$, and τ is its first hitting time of the boundary $\partial\Omega$. In this representation every boundary condition becomes a behaviour of the path: Dirichlet corresponds to absorption, Neumann to reflection, and Robin to partial absorption (an elastic boundary). As a consequence, we show that the classical Ogata–Banks solution of the semi-infinite inflow problem is exactly a first-passage probability, $c(x,t)/c_0 = \mathbb{P}(\tau\le t)$.

On this basis we develop and compare two Monte Carlo solvers: a backward Feynman–Kac estimator for pointwise concentrations, equipped with a Brownian-bridge correction for sub-step boundary crossings, and a forward random-walk particle method for the whole field, in which the dry-deposition (Robin) boundary is realised by absorbing a particle upon contact with probability $v_d\sqrt{\pi\,\Delta t/K}$. Well-posedness is addressed through the parabolic maximum principle and an energy estimate yielding uniqueness and continuous dependence on the data. The estimators are validated against the analytic fundamental solution, the Ogata–Banks formula, and an implicit (Crank–Nicolson) finite-difference solution, in one, two, and three spatial dimensions.

For a diagonal diffusivity the multidimensional diffusion factorises into independent one-dimensional processes, which renders the method mesh-free and free of numerical diffusion, the ground reflection, capping inversion, and deposition acting on the vertical coordinate alone. An interactive R-Shiny web application has been developed that lets the user vary the wind, diffusivity, source, and boundary parameters and visualise the resulting concentration fields in real time.

Keywords: advection–diffusion equation, initial–boundary value problem, Feynman–Kac formula, Monte Carlo method, stochastic differential equation, first-passage time, Robin boundary condition, atmospheric dispersion

2020 Mathematics Subject Classification: 35K20, 60H30, 65C05

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The ITD approach in quasilinearization of integro-differential equations

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Abstract: Integro-differential equations frequently serve as mathematical frameworks across a variety of disciplines. Investigating Volterra integro-differential equations alongside their real-world uses has emerged as a prominent research domain. Within this context, let us consider the initial value problem (IVP) formulated below:

$$\frac{dx}{dt} = g(t, x(t)) + \int_{t_0}^{t} \varphi(t, s, x(s))ds, \qquad x(t_0) = x_0, \tag{1}$$

with $t \in J_1 = [t_0, t_0 + T], t \geq 0, T > 0$, assuming appropriate conditions on g and φ.

This report focuses on employing the method of quasilinearization incorporating the initial time difference (ITD) technique to analyze a Volterra integro-differential equation (1), thereby yielding monotone sequences of functions that converge uniformly and quadratically to the unique solution of (1).

Keywords: Quasilinearization method, initial value problem, integro-differential equations, lower and upper solutions

2020 Mathematics Subject Classification: 34A12, 45J05

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Weak solvability of Navier–Stokes–Voigt thermal model

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Abstract: We consider the following initial–boundary value problem

$$\frac{\partial v}{\partial t}+\sum_{i=1}^{n} v_i\frac{\partial v}{\partial x_i}-\text{Div }(\nu(I_2(v))\mathcal{E}(v))-\varkappa\frac{\partial \Delta v}{\partial t}+\nabla\theta+\nabla p=f;$$

$$\frac{\partial \theta}{\partial t}+\sum_{i=1}^{n} v_i\frac{\partial \theta}{\partial x_i}-\chi\Delta\,\theta=\nu(I_2(v))\mathcal{E}(v):\mathcal{E}(v)+\varkappa\frac{\partial \mathcal{E}(v)}{\partial t}:\mathcal{E}(v)+g;$$

$$\text{div } v=0;\quad v|_{t=0}=v_0;\quad v|_{[0,T]\times\partial\Omega}=0;\quad \theta|_{t=0}=\theta_0\Omega;\quad \theta|_{[0,T]\times\partial\Omega}=0.$$

Here, v is un unknown velocity, p is un unknown pressure, $f=f(t,x)$ is the external force, $\nu>0$ is the fluid viscosity, $\varkappa>0$ is the retardation time, $\mathcal{E}(v)=(\mathcal{E}_{ij}(v)),\quad \mathcal{E}_{ij}(v)=\frac{1}{2}\big(\frac{\partial v_i}{\partial x_j}+\frac{\partial v_j}{\partial x_i}\big),\quad i,j=\overline{1,n}$, is the strain-rate tensor, the function $I_2(v)$ is defined by the relation $I_2(v)^2=\mathcal{E}(v):\mathcal{E}(v)$.

This initial–boundary value problem describes the flow of a non–linearly elastically retarded Voigt fluid (for example, the motion of a viscous fluid with polymer components) see [1].

Theorem. Let $\Omega\subset\mathbb{R}^n$, $n=2,3$, be bounded domain with smooth boundary. Let $f\in L_2(0,T;V^*)$, $v_0\in V$, $g\in L_1(0,T;H_p^{-2(1-1/p)}(\Omega))$ and $\theta_0\in W_p^{1-2/p}(\Omega)$. The viscosity function ν have to be continuously differentiable and satisfy the inequalities (ν_1) $0<C_1\le\nu(s)\le C_2<\infty$; (ν_2) $-s\nu'(s)\le\nu(s)$ for $\nu'(s)<0$; (ν_3) $|s\nu'(s)|\le C_3<\infty$. Also let $1<p<4/3$ for $n=2$ and $1<p<10/9$ for $n=3$. Then the initial-boundary value problem has weak solution $v\in W=\{v: v\in C(0,T;V),\ v'\in L_2(0,T;V)\}$.

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Keywords: Navier–Stokes–Voigt system of equations, weak solution, thermodynamic model, existence theorem

2020 Mathematics Subject Classification: 35Q35, 76A05 **References:**

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Finite propagation and space localization properties of cross-diffusion systems with time-dependent coefficients and variable density

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Abstract: We investigate the Cauchy problem for a class of degenerate nonlinear cross-diffusion systems with time-dependent diffusion coefficients, variable density, and absorption terms. In the domain

$$Q = \{(t,x) : t > 0,\ x \in \mathbb{R}^N\}$$

investigated properties of the solutions of a nonlinear diffusion-reaction system with variable density:

$$(1) \qquad \begin{cases} |x|^r \dfrac{\partial u}{\partial t} = \operatorname{div}\big(\bar{k}_1(t)|x|^\alpha v^{m_1-1}|\nabla u|^{p-2}\nabla u\big) - |x|^r \gamma_1(t) u, \\ |x|^r \dfrac{\partial v}{\partial t} = \operatorname{div}\big(\bar{k}_2(t)|x|^\beta u^{m_2-1}|\nabla v|^{p-2}\nabla v\big) - |x|^r \gamma_2(t) v. \end{cases}$$

subject to the initial conditions

$$(2) \qquad u(0,x) = u_0(x) \geq 0, \qquad v(0,x) = v_0(x) \geq 0, \qquad x \in \mathbb{R}^N,$$

here $k \geq 1$, $n, p, m_i, k_i(t) > 0$, $i = 1,2$, are given positive constants, characterizing the nonlinear media, $\nabla(\cdot) = \operatorname{grad}(\cdot)$, -functions $u_0(x)$, $v_0(x) \geq 0$, $x \in \mathbb{R}^N$, and $0 < k_i(t), \gamma_i(t) \in C(0,\infty)$, $i = 1,2$.
The model describes nonlinear processes arising in porous media, the theory of gas filtration, nonlinear heat conduction, and reaction–diffusion cross-systems with variable density and absorption. Under suitable assumptions on the nonlinear diffusion exponents, absorption coefficients, and initial data, we analyze the qualitative properties of nonnegative weak solutions. By introducing an appropriate nonlinear self-similar transformation, the original evolutionary system is reduced to a family of nonlinear degenerate self-similar differential equations. In the present work, based on the self-similar analysis of solutions to problem (1),(2) and the comparison theorem, we investigate the qualitative properties of solutions to the problem under consideration; these include the finite velocity of disturbance propagation and spatial localization under the influence of absorption and time-dependent coefficients of the cross-system [1–5]. An estimate of the solutions and the free boundary of the problem is obtained for the case of slow diffusion. Furthermore, the asymptotics of the self-similar solution are investigated.

It is shown that the coefficient of the leading term of the solutions asymptotic satisfies a certain system of nonlinear algebraic equations for both slow and fast diffusion [2–4]. Critical values of the numerical parameters of the nonlinear medium, at which the behavior of the solution changes and takes on an exponential form, are established. Critical and singular cases are also considered. Finally, the asymptotics of self-similar solutions for weak and quenching solutions are established, and a numerical analysis of the solution is discussed.

Keywords: Degenerate parabolic system; cross-diffusion; weak solutions; finite-speed propagation; free boundary; space localization; self-similar solutions; asymptotic and numerical analysis.

2020 Mathematics Subject Classification. 35K65, 35K67, 35B40, 35B44, 35R35.

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On some inequalities of two-parameter noncommutative martingales

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Abstract: This report is devoted to extending the main results of [1] on two-parameter noncommutative martingales from the finite von Neumann algebra setting to the semi-finite setting. We prove convergence theorems for L_p-bounded martingales ($1 < p < \infty$). Furthermore, we establish the Burkholder–Gundy inequality for general two-parameter martingales and Davis's inequality for strong martingales within this broader semi-finite context. Our approach relies crucially on the powerful interpolation machinery developed in [2], specifically the complex and real interpolation formulae between BMO, $\mathcal{H}_p$, h_p, and their conditioned counterparts. This interpolation framework allows us to transfer the known finite-trace estimates systematically to the semi-finite setting. The results obtained unify and extend the classical L_p-theory for two-parameter martingales, providing essential tools for further developments in noncommutative harmonic analysis and quantum probability. Throughout this note, we mainly use techniques from the works [3–5].

Keywords: von Neumann algebra; two-parameter martingales; Burkholder–Gundy inequality; Davis's inequality; strong martingales

2020 Mathematics Subject Classification: 46L52; 46L53; 60G42

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Optimal control problems with nonlocal conditions for systems of integro-differential equations

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Abstract: This paper considers an optimal control problem in which the system state is determined by a system of integro-differential equations with nonlocal boundary conditions. Admissible controls are selected from the class of bounded and measurable functions with values in an open set. A formula for the second-order increment of the functional is derived. Based on control variations, the necessary optimality conditions for singular controls in the classical sense are obtained.

Similar problems have been considered in [1].

Keywords: nonlocal boundary conditions, optimal control, necessary optimality conditions, second-order functional increment

2020 Mathematics Subject Classification: 34B10, 34A37, 34H05

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Boundedness of convoultion pperators in generalized local Morrey spaces

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Abstract: This paper investigates norm estimates for convolution operators in generalized local Morrey-type spaces. Sufficient conditions for boundedness are established, and the corresponding associated (dual) spaces are characterized.

We prove Young-O'Neil-type inequalities in this setting, extending classical results to a more general framework. The obtained inequalities provide new estimates for convolution operators with kernels from weak Lebesgue spaces.

These results generalize known theorems for Lebesgue, Lorentz, and Morrey spaces, offering refined criteria for the boundedness of integral operators. Applications include potential operators and singular integrals in Morrey-type spaces.

Throughout this note we mainly use techniques from [1].

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Keywords: Morry space, Generalised Local Morrey Space, convolution operator, Young-O'Neil's inequality, Riesz potential

2020 Mathematics Subject Classification: 47A05, 47B06

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On the class of super-singular integro-differential equation

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Abstract: In this work on $\Gamma = \{x : a \leq x \leq b\}$ we investigate a class of first-order integro-differential equation with super-singular kernel

$$D_x^\beta y + A(x)y - \int_x^b \frac{B(x,t)}{(b-t)^\beta} y(t)\,dt = f(x), \tag{1}$$

where $\beta > 1$, $D_x^\beta = (b-x)^\beta \dfrac{d}{dx}$ is a special differential operator, $A(x)$, $f(x)$ are given continues functions at Γ, $B(x,t)$ is given continues function at rectangular $R = \{(x,t) : a < x < b,\ a < t < b\}$ and $y(x)$ is unknown function.

Since in equation (1) $\beta > 1$, therefore the kernel of equation has singularity of degree greater then 1. On the other hand it is well known that the kernel of type $\frac{B(x,t)}{(b-t)^\beta}$ upon $|B(x,t)| \leq K$ is a divergent kernel, because the improper integral

$$\int_x^b \frac{B(x,t)}{(b-t)^\beta}\,dt$$

diverges.

On the other hand, even though the kernel is divergent, our investigation shows that there exist some classes of functions in which the equation (1) has solutions.

In this work we show that in the class of such functions denoted by $y(x) \in C_{\beta-1}[a,b]$, which at point $x = b$ tends to zero and its behavior determined from the asymptotic formula

$$y(x) = o\left[(b-x)^\delta\right], \qquad \delta > \beta - 1, \qquad \text{when } x \to b.$$

the equation (1) solvable and its solution in general contains two arbitrary constants.

There also exist some cases in which the general solution of investigated equation contains one arbitrary constant or it has only the unique solution.

Keywords: Integro-differential equations, singular coefficients, Hölder class functions, degenerate kernel.

2020 Mathematics Subject Classification: 35J05, 35J08, 35J25

Analytical solutions for fractional stochastic equations via Martingale processes

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Abstract: The purpose of this paper is to put forward some analytical solutions to solve the fractional stochastic equation (FSDE)

$$\begin{cases} {}^{C}D_t^{\alpha} X(t) = a(X(t), t)\, dt^{\alpha} + \sigma(X(t), t)\, dW_t, & t \in [0, T], \\ X(0) = X_0, \end{cases}$$

where ${}^{C}D_t^{\alpha}$ is a Caputo fractional derivative of order $\alpha,\ 0 < \alpha \leq 1$, $W(t)$ is a Wiener process and triple $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space under some conditions and special relations between drift and volatility.

If functions a and σ are globally Lipschitz in $\mathbb{R}$ and satisfy the linear growth condition, then FSDE has a unique t-continuous solution adapted to the filtration $\mathcal{F}_t,\ t \geq 0$ generated by $W(t)$ and

$$E\left[\int_0^T |X(s)|^2\, ds\right] < \infty.$$

In our paper, we find a couple of analytical solutions of some set of fractional stochastic equations was indicated via the martingale process from a stochastic process. Converting fractional stochastic differential equations to ordinary ones as another suitable method was posed.

Keywords: Fractional order, martingale processes, drift term, diffusion, volatility term.

2020 Mathematics Subject Classification: 35J05, 35J08, 35J25

On the conditions for the existence of solutions of a difference inclusions

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Abstract: Let X be a complex Banach space and A be a closed linear relation on X, i.e., $A \subset X \times X$ is a closed linear subspace.

In the Banach space $l_p = l_p(\mathbb{Z}, X), 1 \leqslant p \leqslant \infty$, of two-sided sequences $x : \mathbb{Z} \to X$, we consider the difference inclusion

$$x(n) \in Ax(n-1) + f(n), n \in \mathbb{Z}, \tag{1}$$

here $f \in l_p, Ax(n-1) = \{y \in X : (x(n-1), y) \in A\}$.

The study of inclusions (1) is based on the closed relation $\mathcal{A} = \overline{A}S(-1)$ on l_p, where $\overline{A} = \{(x, y) \in l_p \times l_p : (x(n), y(n)) \in A\}$ and $(S(-1)x)(n) = x(n-1), n \in \mathbb{Z}, x \in l_p$. The spectrum of $\sigma(\mathcal{A})$ is invariant under rotation around zero in the field $\mathbb{C}$, and if $1 \notin \sigma(\mathcal{A})$, then the representation $\sigma(\mathcal{A}) = \sigma_{int} \cup \sigma_{out}$, is valid, where $\sigma_{int}(\mathcal{A}) = \{\lambda \in \sigma(\mathcal{A}) : |\lambda| < 1\}$ and $\sigma_{out}(\mathcal{A}) = \{\lambda \in \sigma(\mathcal{A}) : |\lambda| > 1\}$.

Theorem 1. The difference inclusion (1) has a unique solution $x \in l_p, 1 \leqslant p \leqslant \infty$ if and only if the spectrum $\sigma(A)$ of A satisfies the condition

$$\sigma(A) \cap \mathbb{T} = \varnothing, \tag{2}$$

where $\mathbb{T}$ is the unit circle in $\mathbb{C}$.

According to condition (2), the Riesz projector decomposes the space $X = X_0 \oplus X_1$, where $X_0 = ImP, KerP = X_1$, and for the restrictions $A|X_0 = A_0$ and $A|X_1 = A_1$ the equalities $\sigma(A_0) = \sigma_{int}(A)$ and $\sigma(A_1) = \sigma_{out}(A)$ hold. Based on this, any solution of difference inclusion (1) can be represented as $x(n) = \sum\limits_{m \in \mathbb{Z}} G(-m)f(m)$, where $G : \mathbb{Z} \to X, G(k) = A_0^k P$ for $k \geqslant 0, G(k) = -A_1^k(I - P)$ for $k \leqslant -1$.

Throughout this note we mainly use techniques from work [1].

Keywords: Banach space, linear relation, difference inclusion, spectrum

2020 Mathematics Subject Classification: 39A70, 39A20

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Numerical solution of the multidimensional identification elliptic telegraph problem with Dirichlet condition

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Abstract: In the present paper, the theorem on stability estimates for the solution of source identification problem (SIP) of the multidimensional elliptic-telegraph differential equation with Dirichlet's conditions is presented. Furthermore, numerical results for multidimensional of identification elliptic-telegraph problem are given.

Keywords: Source identification problem, elliptic-telegraph differential equation, difference scheme

2020 Mathematics Subject Classification: 65N06, 35M10, 35R30.

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Comparative analysis of different classes of adomian polynomials for solving nonlinear volterra integral equation of the second kind

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Abstract: In this research, the Standard Laplace Adomian decomposition Method (SLADM), Modified Laplace Adomian Decomposition Method (MLADM) and new modification in Standard Laplace Adomian Decomposition Method called the Two-Step Laplace Adomian Decomposition Method (TSLADM) for solving nonlinear Volterra integral equation of the second kind was considered. The analytical and numerical solutions of Nonlinear Volterra integral equation of second kind were solved for the classes of the Adomian polynomials. The results obtained are compared with exact solution and it was shown that these three methods are capable to achieve a good agreements with exact solution. This research also demonstrated that new modification in standard Laplace Adomian decomposition method (TSLADM) is reliable and efficient than standard Laplace Adomian decomposition method. However, the comparison between the exact solution and numerical solution obtained using SLADM, MLADM and TSLADM was established. Numerical solutions was carried out and the results shows that the TSLADM is less time consuming when compared to modified Laplace Adomian decomposition method and standard Laplace Adomian decomposition method.

Throughout this note we mainly use techniques from the works [1].

Keywords: Adomian Decomposition Method , nonlinear volterra integral equation , polynomial, numerical solutions.

2020 Mathematics Subject Classification: 35J05, 35J08, 35J25

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Some results for analytic functions related with modified Sigmoid function

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Abstract: In this study, we explore the analytical properties and applications of a Modified Sigmoid Function within the framework of geometric function theory, with a focus on the Schwarz Lemma. The classical sigmoid function, widely used in machine learning, is adjusted to better conform to the unit disk and its boundary behavior, enabling a rigorous examination using tools from complex analysis. We demonstrate that under appropriate normalization, the modified sigmoid satisfies the conditions of the Schwarz Lemma, and we investigate the implications of this conformity on the function's growth, distortion, and fixed-point behavior. Furthermore, we discuss how such modifications influence stability and convergence in iterative schemes, potentially offering refined activation functions for neural networks with provable analytical constraints. These results establish a bridge between classical function theory and modern computational methods, highlighting the role of complex analysis in evaluating and enhancing non-linear mappings in applied contexts.

Keywords: Modified Sigmoid Function, Analytic function, Schwarz lemma

2020 Mathematics Subject Classification: 30C80 **References:**

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Inverse boundary value problem for a second-order hyperbolic integro-differential equation with a periodic condition and an integral condition of the first kind

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Abstract: An inverse boundary value problem for a second-order hyperbolic integro-differential equation with a periodic condition and an integral condition of the first kind is investigated.

In the domain $D_T = \{(x,t) : 0 < x < 1, 0 < t < T\}$ we consider an inverse boundary value problem

$$u_{tt} - u_{xx} = \int_0^t a(\tau)u(x,t-\tau)d\tau + f(x,t) \ \ (x,t) \in D_T, \tag{1}$$

with the initial conditions

$$u(x,0) = \varphi(x), \ u_t(x,0) = \psi(x) \ \ (0 \le x \le 1), \tag{2}$$

the periodic boundary condition

$$u(0,t) = u(1,t) \ \ (0 \le t \le T), \tag{3}$$

the integral condition of the first kind

$$\int_0^1 u(x,t)dx = 0 \ \ (0 \le t \le T), \tag{4}$$

and the additional condition

$$\int_0^1 \omega(x)u(x,t)dx = h(t) \ \ (0 \le t \le T), \tag{5}$$

here, $\varphi(x), \psi(x), f(x,t), h(t), \omega(x) \neq const$ are the given functions, and $u(x,t)$ and $a(t)$ are the unknown functions.

First, the original problem has been reduced to an equivalent problem for which the existence and uniqueness theorem has been proved. Then, using these results, the existence and uniqueness of a classical solution to the original problem are established.

Our results develop the results of [1].

Keywords: inverse problem, second-order hyperbolic integro-differential equation, existence, uniqueness, classical solution.

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2020 Mathematics Subject Classification: 35R30, 35A09.

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Solvability of a nonlocal problem for a spatially loaded hyperbolic system of differential equations

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Abstract: This paper investigates a nonlocal problem for a second-order hyperbolic system of loaded differential equations with a spatial loading term depending on the value of the solution at a fixed spatial point. To establish the solvability of the problem, we propose a parametrization-based approach. The considered problem is formulated in the domain $\Omega = [0,T] \times [0,X]$ as follows:

(1)
$$\frac{\partial^2 u}{\partial x \partial t} = A(t,x)\frac{\partial u}{\partial x} + B(t,x)\frac{\partial u}{\partial t} + C(t,x)u(t,x) + C_0(t,x)u\,(t,x_0) + f(t,x),$$

(2)
$$P(x)\frac{\partial u(0,x)}{\partial x} + S(x)\frac{\partial u(T,x)}{\partial x} = \varphi(x), \quad x \in [0,X],$$

(3)
$$u(t,0) = \psi(t), \quad t \in [0,T].$$

To investigate problem (1)-(3), we employ a parametrization-based approach. The proposed approach introduces auxiliary unknown functions and employs the parametrization method. As a result, the original nonlocal problem is transformed into an equivalent family of parameter-dependent Goursat and Cauchy problems.

The proposed approach extends the parametrization method developed in [1, 2] to a class of nonlocal hyperbolic systems with spatial loading.

Keywords: nonlocal problem, hyperbolic loaded differential system, parametrization method, Goursat problem, Cauchy problem, fundamental matrix, iterative method, solvability

2020 Mathematics Subject Classification: 35L15; 34A12; 34B08; 45A05; 39B05

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Spectral properties of a second-order differential operator with the spectral parameter in the boundary conditions

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Abstract: We consider the spectral problem for the second-order differential operator

$$L(y) \equiv -y''(x) = \lambda y(x),\ 0 < x < 1, \tag{1}$$

subject to the boundary conditions

$$y(0) = 0, \tag{2}$$

$$\lambda Y_1(y) + Y_0(y) = 0, \tag{3}$$

where

$$Y_i(y) = a_{i1}y'(0) + a_{i2}y'(1) + a_{i3}y(0) + a_{i4}y(1) \tag{4}$$

are linear boundary forms.

The spectral properties of boundary value problems for equation (1) without the spectral parameter in the boundary conditions have been extensively investigated in the literature [1]. However, introducing the spectral parameter into the boundary conditions substantially changes the structure of the characteristic equation and the spectral properties of the problem, which requires a separate investigation.

In the abstract, we study the spectral properties of the boundary value problem (1)–(3), where the spectral parameter enters one of the boundary conditions linearly. A classification of the problem into regular and irregular cases is established in terms of the coefficients of the boundary condition (3). The asymptotic behaviour of the eigenvalues and the properties of the corresponding eigenfunctions are investigated.

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Keywords: Ordinary differential operator, nonlocal boundary conditions, spectral parameter in the boundary conditions, spectral properties, regular boundary conditions, irregular boundary conditions

2020 Mathematics Subject Classification: 34L05,34L10, 34L15 **References:**

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Deformation of the complex structure in $A(z)$-geometry

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Abstract: This paper introduces $A(z)$-geometry through the space of $A(z)$-analytic functions and its reproducing kernel K_A. The corresponding lemniscates and metric describe the complex structure deformed by the coefficient $A(z)$. Unlike quasiconformal geometry, this framework is based on reproducing-kernel and integral structures.

It is well established that solutions to the Beltrami equation

$$\bar{D}_A = \frac{\partial f}{\partial \bar{z}} - A(z)\frac{\partial f}{\partial z} = 0 \tag{1}$$

are closely connected with the theory of quasiconformal mappings. In the general setting, the coefficient $A(z)$ is assumed to be measurable in the domain $D \subset \mathbb{C}$ and to satisfy the ellipticity condition

$$|A(z)| \leq c$$

almost everywhere in D, where $c < 1$. Functions satisfying equation (1) are referred to as $A(z)$-analytic functions.

Krantz [1] develops the classical foundations of geometric function theory, while Sadullaev and Zhabborov [2] extend them to $A(z)$-analytic functions. These works provide the basis for $A(z)$-geometry and its associated kernel, lemniscates, and metric.

As an illustrative example, consider a multiply connected domain, such as the annulus. In this setting, the Bergman kernel may vanish at points away from the diagonal, a classical phenomenon demonstrating that the finer analytical properties of the kernel are strongly influenced by the topology of the underlying domain:

$$K_A(z,\zeta) = \frac{1}{\pi}\left[\sum_{n=0}^{\infty}\frac{n+1}{1-r^{2n+2}}\left(z\bar{\zeta}\right)^n + \sum_{n=0}^{\infty}\frac{(n+1)r^{2n+2}}{1-r^{2n+2}}\left(\bar{z}\zeta\right)^{n-2}\right].$$

Thus, in contrast to the case of the unit disk, the kernel associated with an annulus may possess zeros outside the diagonal set $z = \zeta$.

Classical Bergman geometry relies on the holomorphic structure of the domain and on invariance under biholomorphic transformations. In the setting of equation (1), however, the underlying complex structure is modified by the prescribed field, with the result that the conventional geometric objects no longer retain their invariance.

Within this framework, a novel geometric structure emerges in a natural manner. We refer to it as $A(z)$-geometry, since it is determined by the space of $A(z)$-analytic functions together with its associated reproducing kernel K_A.

Fix a point $\zeta \in D$ and introduce the function

$$\psi_\zeta(z) = \int\limits_{\gamma(\zeta,z)} A(\tau)d\tau,$$

where $\gamma(\zeta, z)$ denotes an arbitrary curve in D joining ζ to z. If D is convex, the function ψ_ζ is uniquely defined.

Definition 2.1. *For $r > 0$, the set*

$$L_r(\zeta) = \{z \in D : \big|\psi_\zeta(z)\big| < r\}$$

is called the $A(z)$-lemniscate with center at ζ.

Whereas ellipses serve as the fundamental geometric objects in quasiconformal geometry, the natural geometric elements of $A(z)$-geometry are lemniscates. Their form is governed by the integral structure associated with the coefficient field $A(z)$.

The reproducing kernel $K_A(z, \zeta))$ provides a natural basis for defining a metric that serves as an analogue of the classical Bergman metric.

Definition 2.2. ***($A(z)$-metric).*** *The $A(z)$-metric is defined by*

$$ds^2_A(z) = \partial_z\partial_{\bar{z}} \log K_A(z, z)|dz|^2.$$

This metric has the following properties:

- *it is preserved by those isometries of the space $\mathcal{H}^2_A(D)$ that are generated by $A(z)$-compatible mappings, whenever such mappings exist;*
- *it remains invariant under $A(z)$-automorphisms;*
- *it characterizes the intrinsic geometric structure of $A(z)$-lemniscates;*
- *in the particular case $A(z) \equiv 0$, it reduces to the classical Bergman metric.*

Consequently, the $A(z)$-metric may be regarded as a natural geometric invariant associated with the deformed complex structure.

The essential distinction between $A(z)$-geometry and quasiconformal geometry can be described as follows:

- quasiconformal mappings preserve the moduli of curve families;
- $A(z)$-automorphisms preserve the level sets of the $A(z)$-potential together with the corresponding lemniscatic structure, although they are not necessarily required to preserve extremal length.

Hence, $A(z)$-geometry should not be viewed as a particular subclass of quasiconformal geometry. Rather, it constitutes a distinct geometric framework determined by the kernel of a first-order elliptic differential operator.

The orthogonal decomposition

$$L^2(D) = \ker \bar{D}_A \oplus \operatorname{Ran} \bar{D}^*_A.$$

also admits a natural geometric interpretation. More precisely,

- the subspace $\ker \bar{D}_A$ represents the $A(z)$-analytic, and therefore geometrically intrinsic, component of a function;
- the complementary non-analytic component is described through the fundamental solution and does not contribute directly to the $A(z)$-geometric structure.

Accordingly, $A(z)$-geometry can be interpreted as the geometry induced by the orthogonal projection onto the solution space of the deformed Cauchy–Riemann equation.

In the limiting case $A(z) \equiv 0$, all of the constructions introduced above, namely, $A(z)$-lemniscates, the $A(z)$-metric and the $A(z)$-kernel reduce to their classical counterparts in Bergman theory. This limiting correspondence supports both the consistency and the natural character of the proposed geometric framework.

Remark 2.3. ***(Relation to quasiconformal geometry).*** *In general, the $A(z)$-geometry generated by the reproducing kernel K_A differs from the quasiconformal geometry developed by Ahlfors. Its structure is governed not by conformal moduli or extremal-length invariants, but by the reproducing-kernel Hilbert space of $A(z)$-analytic solutions and by the integral characteristics of the coefficient field $A(z)$. In the particular case $A(z) \equiv 0$, this geometric framework reduces to the classical Bergman geometry.*

Keywords: $A(z)$-analytic functions, the Bergman kernel, reproducing kernel, lemniscates, deformed complex structure, $A(z)$-geometry.

2020 Mathematics Subject Classification: 30C40, 30C62, 30H20

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An operator-symbol construction of sixth-order difference schemes for a fifth-order differential equation

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Abstract: Taylor's decomposition on four points for a third-order differential equation, introduced by Ashyralyev and Arjmand [1], is reformulated in operator-symbol form. This formulation is extended to the fifth-order differential equation

$$y^{(5)}(t) + a(t)y(t) = f(t), \qquad 0 < t < T.$$

A six-point decomposition is obtained and used to construct formally sixth-order difference schemes for an initial-value problem, a boundary-value problem, and a nonlocal boundary-value problem. The operator-symbol formulation provides a unified derivation of the common interior scheme and the corresponding one-sided closure formulas. Numerical examples exhibit the predicted sixth-order rate.

Keywords: operator-symbol method, six-point decomposition, fifth-order differential equation, sixth-order scheme, nonlocal boundary-value problem

2020 Mathematics Subject Classification: 65L05, 65L10, 65L12

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Existence and uniqueness for a variable-order fractional $p(\cdot)$-Laplacian equation with a causal delay term

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Abstract: This paper investigates the existence and uniqueness of weak solutions to a nonlinear parabolic problem involving the fractional $p(\cdot)$-Laplacian operator of variable order, with a source term containing a causal deviating argument. The analysis is carried out via a semi-discretization approach, where each time step is reformulated as the minimization of a strictly convex and coercive functional on a variable-order, variable-exponent Sobolev space. This formulation guarantees the existence, uniqueness, and stability of the discrete solutions, with estimates independent of the time step, and yields the corresponding weak solution of the continuous problem. The theoretical findings are illustrated through a concrete example, demonstrating the applicability of the proposed framework.

Keywords: Fractional $p(\cdot)$-Laplacian; variable order; deviating argument; semi-discretization method; weak solution.

2020 Mathematics Subject Classification: 35K55; 35R11; 35R10.

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Solution of a nonlocal problem with boundary conditions for a homogeneous ordinary differential equation with continuously varying second-order derivatives

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Abstract: In recent years, fractional mathematical modelling using nonlocal fractional derivatives has become a reliable tool and a new resource for describing the dynamics of complex systems with memory effects and heritable characteristics. Such systems are found in a variety of fields, including physics, fluid dynamics, materials science, signal processing, engineering, chemistry, biology, medicine, finance, social sciences, economics, and ecology.

In this paper, a new method for constructing an analytical solution to a nonlocal boundary value problem for a homogeneous ordinary differential equation with continuously varying second-order derivatives is proposed. To do this, an invariant expression that allows for algebraization of the problem under consideration is found. The resulting characteristic equation, corresponding to the equation under study, is transcendental, and its roots are found using the Lambert operator.

Keywords: Differential equations of variable order, invariant solution, transcendental characteristic equation, W-Lambert transform

2020 Mathematics Subject Classification: 34A08, 34B05, 34B60

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Study of a class boundary value problem for ordinary differential equations whose order varies continuously

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Abstract: Fractional and multifractional differential equations are encountered and widely used in certain areas of physics, such as chemistry, astrophysics, biology, medicine, and so on. The study of flow dynamics observed in inhomogeneous and irregular regions also leads to the solution of differential equations of variable order.

Of particular interest is the study of differential equations whose order varies continuously.

The main goal of this paper is to find an invariant expression that allows for the algebraization of differential equations whose order varies continuously, similar to Euler's method for solving ordinary equations with constant coefficients, and to use it to construct an analytical solution to the problem under consideration

Keywords: multifractional differential equation, differential equations of variable order, invariant solution, W Lambert transform

2020 Mathematics Subject Classification: 34A08, 34B05, 34B60

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New examples of regular, but not strongly regular, boundary value problems for the operator of fourfold differentiation on an interval

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Abstract: We consider the problem

$$y^{(IV)}(x) = \lambda y(x) \tag{1}$$

on the interval $(0,1)$ with two-point boundary conditions

$$B_\nu(y) = a_{\nu 1}y'''(0) + a_{\nu 2}y'''(1) + a_{\nu 3}y''(0) + a_{\nu 4}y''(1) + a_{\nu 5}y'(0) + a_{\nu 6}y'(1) + a_{\nu 7}y(0) + a_{\nu 8}y(1) = 0, \tag{2}$$

where the $B_\nu(y)$, $(\nu = 1,2,3,4)$ are linearly independent forms with arbitrary complex-valued coefficients. We assume that boundary conditions are normalized. The function $y(x)$ is an arbitrary complex-valued function of the class $W_2^4(0,1)$.

We can call boundary conditions normalized, if they can be written in the form

$$B_\nu(y) \equiv B_{\nu 0}(y) + B_{\nu 1}(y) = 0, \tag{3}$$

where

$$B_{\nu 0}(y) = \alpha_\nu y^{k_\nu}(0) + \sum_{j=0}^{k_\nu - 1} \alpha_{\nu j} y^j(0), \tag{4}$$

$$B_{\nu 1}(y) = \beta_\nu y^{k_\nu}(1) + \sum_{j=0}^{k_\nu - 1} \beta_{\nu j} y^j(1), \tag{5}$$

$$3 \geq k_1 \geq k_2 \geq k_3 \geq k_4 \geq 0, \qquad k_\nu > k_{\nu+2},$$

and for each index ν at least one of the numbers α_ν, β_ν are nonzero.

As an example, we give the following boundary conditions here:

Orders (k_1, k_2, k_3, k_4)	Leading part (example)	Strengthened irregularity
$(3, 3, 2, 1)$	$y^{(3)}(0) = 0, \quad y^{(3)}(1) = 0,$ $y''(0) = 0, \quad r\,y'(0) + y'(1) = 0$	$r = \pm\sqrt{2}$
$(3, 3, 1, 0)$	$y^{(3)}(0) = 0, \quad y^{(3)}(1) = 0,$ $y'(0) = 0, \quad r\,y(0) + y(1) = 0$	$r = \pm\sqrt{2}$
$(3, 3, 2, 0)$	$y^{(3)}(0) = 0, \quad y^{(3)}(1) = 0,$ $y''(0) = 0, \quad r\,y(0) + y(1) = 0$	$r = \pm 1$
$(3, 2, 2, 1)$	$y^{(3)}(0) = 0, \quad y''(1) = 0,$ $y''(0) = 0, \quad r\,y'(0) + y'(1) = 0$	$r = \pm 1$
$(3, 2, 2, 0)$	$y^{(3)}(0) = 0, \quad y''(1) = 0,$ $y''(0) = 0, \quad r\,y(0) + y(1) = 0$	$r = \pm\sqrt{2}$
$(3, 2, 1, 0)$	$y^{(3)}(0) = 0, \quad y''(1) = 0,$ $y'(0) = 0, \quad r\,y(0) + y(1) = 0$	$r = \pm 2$
$(3, 1, 1, 0)$	$y^{(3)}(0) = 0, \quad y'(1) = 0,$ $y'(0) = 0, \quad r\,y(0) + y(1) = 0$	$r = \pm\sqrt{2}$
$(2, 2, 1, 0)$	$y''(0) = 0, \quad y''(1) = 0,$ $y'(0) = 0, \quad r\,y(0) + y(1) = 0$	$r = \pm\sqrt{2}$
$(2, 1, 1, 0)$	$y''(0) = 0, \quad y'(1) = 0,$ $y'(0) = 0, \quad r\,y(0) + y(1) = 0$	$r = \pm 1$

These studies continue what was started in [1].

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Keywords: operator of fourfold differentiation, nonlocal boundary conditions, regular boundary conditions, not strongly regular boundary conditions, asymptotic distribution of eigenvalues

2020 Mathematics Subject Classification: 34L10, 34L20

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On weak singularity of Prabhakar kernels

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Abstract: We consider the 3-parametric generalization of the Mittag-Leffler type function

$$E^{\gamma}_{\alpha,\beta}(z)=\sum_{k=0}^{\infty}\frac{(\gamma)_k z^k}{k!\,\Gamma(\alpha k+\beta)},\qquad z,\alpha,\beta,\gamma\in\mathbb{C},\quad \operatorname{Re}\alpha>0, \tag{1}$$

where $(\gamma)_k$ is the Pochhammer symbol, i.e.

$$(\gamma)_k=\frac{\Gamma(\gamma+k)}{\Gamma(\gamma)},$$

and $\Gamma(\cdot)$ is Euler's gamma function. In this case the Prabhakar kernel [1] has the following form

$$e^{\gamma}_{\alpha,\beta}(\lambda;t)=t^{\beta-1}E^{\gamma}_{\alpha,\beta}(\lambda t^{\alpha}), \tag{2}$$
$$t\in\mathbb{R}^{+},\quad \lambda,\alpha,\beta,\gamma\in\mathbb{C},\quad \operatorname{Re}\alpha>0,\ \operatorname{Re}\beta>0.$$

The regularity properties of the function (1) in terms of the kernel (2) have been obtained.

Keywords: Mittag-Leffler function, Prabhakar kernel, Pochhammer symbol, weak singularity, fractional calculus, regularity properties

2020 Mathematics Subject Classification: 33E12, 26A33, 45D05

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On biparty graphs with concept lattices

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Abstract: This paper studies bipartite graphs as a natural representation of formal contexts and the concept lattices that arise from them within Formal Concept Analysis (FCA). In this paper we recall the basic apparatus of FCA, discuss the structural properties of concept lattices with a particularly simple, flat partial order, and consider practical applications of such lattices, including data analysis, clustering, and the identification of relations between objects and attributes. Using a concrete bipartite graph, we build the corresponding formal context table, extract the formal concepts, and construct the resulting star-shaped, modular concept lattice. The results are illustrated with examples from access-control systems, e-commerce, and security monitoring.

Keywords: bipartite graphs, formal concept analysis, formal context, formal concepts, concept lattice, modular lattice.

2020 Mathematics Subject Classification: 05C76, 06B15, 68R10

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Numerical analysis of a nonlocal problem for a loaded parabolic equation

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Abstract: This paper considers a nonlocal boundary value problem for a loaded parabolic equation defined on a closed rectangular domain. Such equations arise in mathematical models of diffusion, heat and mass transfer, environmental processes, and other systems involving nonlocal sources or interactions. The method of lines is applied to the spatial variable, reducing the original partial differential equation to a two-point boundary value problem for a system of loaded ordinary differential equations. The resulting problem is solved using the Dzhumabaev parameterization method. This approach introduces auxiliary parameters and transforms the boundary value problem into a family of initial value problems supplemented by a system of algebraic equations. Based on this transformation, a computational algorithm for approximating the solution of the original nonlocal problem is constructed. The accuracy and effectiveness of the proposed algorithm are evaluated through representative computational experiments. A comparison between the numerical and exact solutions confirms the reliability and applicability of the developed approach.

The analysis presented in this paper is primarily based on the techniques developed in our previous studies [1, 2].

Keywords: loaded parabolic equation, nonlocal boundary condition, method of lines, parameterization method, numerical algorithm

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2020 Mathematics Subject Classification: 35K10, 34K10, 35R10, 65M22

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Investigating and solving eigenvalue problems including linear and nonlinear difference equations using discrete additive and multiplicative differential equations

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Abstract: In this paper, we design and study eigenvalue problems involving linear and nonlinear difference equations with parameters. For such problems, first; we study simple cases of first-order equations including parameters and find their eigenvalues and eigenfunctions.

Then, for the general case of a second-order discrete multiplicative equations ,we will calculate their eigenvalues and eigenfunctions. eigenvalue problems involving nonlinear difference equations is one of the unsolved problems of this theory, therefore we try to generalize the Storm-Liouville theory to discrete multiplicative calculus and non-Newtonian analysis.

Finally, we apply the above method to boundary value problems including nonlinear difference equations with parameters, and illustarte the distribution eigenvalues and eigenfunctions in some graphics.

Keywords: Discrete multiplicative and additive derivative, Eigenvalue problem, Sturm-Liouvill problem, Non-linear difference equations

2020 Mathematics Subject Classification: 45J05, 34A08

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On a time nonlocal problem for a polyharmonic equation with involution

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Abstract: Let $\Omega \subset R^n$ be an arbitrary n-dimensional domain with a sufficiently smooth boundary $\partial\Omega$. Let us assume that the domain Ω is invariant with respect to the transformation $Sx = -x$. Denote by $Q_T = \{(t,x) : 0 < t < T, x \in \Omega\}, \Gamma$ the lateral boundary of the cylinder Q_T. Consider the following problem in Q_T:

$$u_t(t,x) + a_0(-\Delta)^m u(t,x) + a_1(-\Delta)^m u(t,-x) = 0, (t,x) \in Q_T, \tag{1}$$

$$u(0,x) + \gamma u(T,x) = \varphi(x), \tag{2}$$

$$\Delta^k u(t,x) = 0, k = 0, 1, ..., m-1, (t,x) \in \Gamma, \tag{3}$$

where γ is a real number, $\varphi(x)$ is a given function.

Direct and inverse problems for equation (1) in the case $m = 1$ and $\gamma = 0$ have been studied by numerous authors (see, e.g., [1, 2]). In these papers, the well-posedness of problem (1)–(3) in the case $a_0 > |a_1|$ was proven.

In the present paper, we find all possible values of the parameters a_0, a_1 and γ for which problem (1)–(3) becomes well-posed.

Keywords: involution, nonlocal problem, polyharmonic operator, Navier problem

2020 Mathematics Subject Classification: 35K35, 35K65, 35R30

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An inverse problem for time fractional pseudo-parabolic equation

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Abstract: In this work, we will discuss the solvability of an inverse problem for the following time fractional pseudo-parabolic equation :

$$\mathcal{D}_t^{\alpha}\big(u(x,t)+\beta\mathcal{L}u(x,t)\big)+\mathcal{L}u(x,t)=f(x,t), \qquad x\in\Omega, \quad t\in(0,T), \tag{1}$$

supplemented with the initial condition

$$u(x,0)=a(x), \qquad x\in\Omega, \tag{2}$$

the nonhomogeneous boundary condition encoded by

$$\mathcal{B}u(x,t)=p(t)h(x) \qquad x\in\partial\Omega, \quad t\in[0,T], \tag{3}$$

and the overdetermination condition

$$\mathcal{P}u(t)=e(t). \qquad t\in[0,T]. \tag{4}$$

Let $\mathcal{H}$ be a separable Hilbert space. The inverse problem (1)-(4) consists of finding the pair of functions :

- $p:[0,T]\to\mathbb{R}$ and
- $u:Q_T:=\Omega\times(0,T)\mapsto\mathbb{R}$,

which to be determined by the given data: functions $a(x), f(x,t), h(x), e(t)$ and a positive constant β. Here,$\mathcal{L}$ is a second order linear differential operator,$\mathcal{B}$ is a linear boundary (trace) operator $\mathcal{B}:\mathcal{H}^{\gamma_b}\to\mathcal{H}_{\partial\Omega}$, $\mathcal{P}$ is linear bounded functional and $\mathcal{D}_t^{\alpha}$ denotes the Caputo fractional derivative of order $\alpha\in(0,1)$.

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Keywords: Inverse problem,pseudo- parabolic equation, Caputo fractional derivative, boundary condition, existence, uniqueness, stability.

2020 Mathematics Subject Classification: 35R30, 35A01, 35A02

On a generalization of the Robin problem for a nonlocal analogue of the Poisson equation

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Abstract: Let S be an orthogonal transformation satisfying the condition $S^l = E, l \in N$, where E denotes the identity transformation. In the unit ball Ω, we consider the following boundary value problem:

$$\sum_{j=0}^{l-1} a_j(-\Delta u)(S^j x) = f(x), x \in \Omega, \tag{1}$$

$$\sum_{j=0}^{l-1} b_j \frac{\partial u}{\partial \nu}(S^j x) + \sum_{j=0}^{l-1} c_j u(S^j x) = g(x), x \in \partial\Omega \tag{2}$$

where a_j, b_j, c_j are given real constants, while f(x) and g(x) are prescribed functions. We note that, in [1], the main boundary value problems for equation (1) with the classical Dirichlet, Neumann, and Robin boundary conditions were previously investigated. The method used to investigate problem (1)–(2) is based on the decomposition of the function space into a direct sum of invariant subspaces by means of a system of orthogonal projections. This decomposition makes it possible to reduce the original nonlocal problem to a collection of classical local Dirichlet, Neumann, and Robin boundary value problems for the Poisson equation. Conditions ensuring the existence and uniqueness of a solution are obtained, together with necessary and sufficient solvability conditions in the case where a Neumann-type problem arises. An integral representation of the solution is constructed in terms of the Green functions corresponding to the associated classical boundary value problems. Furthermore, the corresponding spectral problem is investigated. It is shown that its spectral characteristics are determined by the spectra of the associated Dirichlet, Neumann, and Robin problems posed in the corresponding invariant subspaces.

Keywords: involution, nonlocal boundary value problem, Poisson equation with involution, projector, Dirichlet problem, Neumann problem, Robin problem, Green function, spectral problem, eigenvalues, eigenfunctions

2020 Mathematics Subject Classification: 35J05, 35J25, 35P05

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On a method of transferring of non-local boundary value problem for ordinary differential equations of the first order to the Cauchy problem

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Abstract: In this article a new method for transforming a nonlocal boundary value problem for a first-order ordinary differential equation into a Cauchy problem with an initial condition defined at an arbitrary point in the domain is proposed. A fundamental solution of the considered equation is established, with help which yields two necessary conditions. Using those conditions, the value of the solution at point c, i.e., $y(c)$, is found, that allowing the transformation of the boundary value problem into a Cauchy problem. The proposed method provides to easy find as an analytical as well numerical solution of the considered problem.

Keywords: Ordinary differential equation, nonlocal boundary value problem, main relationship, necessary conditions.

2020 Mathematics Subject Classification: 34A08, 34B05

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Optimal parameter identification of a discrete gas-lift process

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Abstract: The motion of gas in the annular space and the gas fluid in the lift of a gas-lift well are described by the following system of nonlinear diferential equations, the right side of which includes a small parameter:

$$\begin{cases} \dot{Q}_i = -\dfrac{2a_i\rho_i F_i Q_i^2}{c_i^2\rho_i^2 F_i^2\mu - Q_i^2}, & Q(0) = u, \\ \dot{P}_i = \dfrac{2a_i c_i^2\rho_i^2 F_i^2 Q_i^2}{c_i^2\rho_i^2 F_i^2\mu - Q_i^2}, & P(0) = P_0, \qquad i = 1, 2. \end{cases} \tag{1}$$

where the parameter a_i is found from the expression

$$2a_i = \frac{g_i}{\omega_i} + \frac{\lambda_i\omega_i}{2D_i}, \qquad i = 1, 2. \tag{2}$$

Here, g_1 and g_2 denote the gas release rates, ω_1 and ω_2 are the mixture velocities, λ_1 and λ_2 are the hydraulic resistance coefficients, D_1 and D_2 are the effective diameters, F_1 and F_2 are the cross-sectional areas, Q_i represents the mass flow rates of gas and gas-liquid mixture, and μ is a small parameter.

İnvestigates the determination of the hydraulic resistance coefficient (HRC) in a discrete gas-lift process. An asymptotic approach was applied by dividing the well into two sections, with the reciprocal of the well depth used as a small parameter. The HRC parameters were identified using the least squares method with Tikhonov regularization to ensure solution uniqueness. The obtained results are consistent with those from other methods and are close to the exact solution.

Throughout this note we mainly use techniques from our work [1].

Keywords: Gas-lift process, parameter identification, asymptotic method, nonlinear difference equations.

2020 Mathematics Subject Classification: 93B30, 65L12, 49J20

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The oscillatory of one class of the high-order Sturm-Liouville difference equation

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Abstract: In this work, we consider the problem of oscillatory of the higher-order Sturm-Liouville difference equation

$$(-1)^n \Delta^n (\rho_k \Delta^n y_k) - v_k y_{k+n} = 0, \tag{3}$$

where $n \in \mathbb{N}$, $n > 1$, $\Delta y_k = y_{k+1} - y_k$, $\rho = \{\rho_k\}_{k=0}^{\infty}$ is a sequence of positive numbers, $v = \{v_k\}_{k=0}^{\infty}$ is a sequence of non-negative numbers, and $\mathbb{N}$ is a set of natural numbers.

The oscillatory properties of the equation (3) are determined by the behavior of its solutions in a neighborhood of infinity. However, since the behavior of these solutions depends on the behavior of the equation coefficients, the determination of the oscillatory properties of equation (3) in terms of the coefficients ρ, v is important. We work directly with the variation principle [1]. We establish conditions equivalent to the fulfillment of the variational principle, on the basis of which the oscillatory conditions of the equation (3) are determined.

Our main result reads as follows:

Let

$$B_n(N) = \max_{0 \le t \le n-1} \left(\frac{1}{t!(n-t-1)!} \right)^2 \sup_{i \ge N} B_{n,i}(t),$$

$$B_{n,i}(t) = \sum_{k=i}^{\infty} \left[(k-i)^{(t)} \right]^2 u_k \sum_{j=N}^{i} \rho_j^{-1} \left[(i-j)^{(n-t-1)} \right]^2.$$

Theorem 1. If $\limsup_{N \to \infty} B_n(N) > 1$, then the equation (3) is oscillatory.

Keywords: Sturm-Liouville difference equation, discrete Sobolev space, oscillatory, variational principle

2020 Mathematics Subject Classification: 34C10, 39A70, 26D15

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Nonlinear integro-differential model of pest population dynamics with age structure for plant protection tasks

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Abstract: This paper investigates a nonlinear mathematical model of an agroecosystem consisting of three trophic levels (plants, harmful insects, and beneficial insects) that takes into account the age structure of insect populations. The model is described by the system of integro-differential equations with partial derivatives (1). Necessary and sufficient conditions for the existence of solutions to the plant protection problem are established.

Consider the agroecosystem "plant – harmful insects – beneficial insects" with age structure:

$$(1)\quad \begin{cases} \dfrac{dN_0}{dt} = Q - \alpha_0 N_0 N_1, \\ \dfrac{dN_1}{dt} = k_0\alpha_0 N_1 - V_1(N_1)\widetilde{N}_2 - m_1 N_1, \\ \dfrac{\partial N_2}{\partial t} + \dfrac{\partial N_2}{\partial a} = k_1 V_1(N_1)N_2 - V_2(N_2)\widetilde{N}_3 - m_2 N_2, \quad N_2|_{t=0} = N_2^0(a), \\ \dfrac{\partial N_3}{\partial t} + \dfrac{\partial N_3}{\partial a} = k_2 V_2(N_2)N_3 - \varepsilon N_3^2 - m_3 N_3, \quad N_3|_{t=0} = N_3^0(a), \\ N_2(0,t) = \displaystyle\int_0^\infty B_2(\xi,t,N_1)N_2(\xi,t)\,d\xi, \\ N_3(0,t) = \displaystyle\int_0^\infty B_3(\xi,t,\widetilde{N}_2)N_3(\xi,t)\,d\xi, \end{cases}$$

where $N_0(t)$ is external resource, $N_1(t)$ is plant biomass, $N_i(a,t)$ $(i=2,3)$ are age-specific densities of harmful and beneficial insects; $V_i(\cdot)$ are trophic functions; $B_i(\cdot)$ are birth rates; $\widetilde{N}_i(t) = \int_{\alpha_i}^{\beta_i} N_i(a,t)\,da$.

Keywords: mathematical model, agroecosystem, plant protection problem, age structure, insect population, trophic function, integro-differential equations, pest dynamics

2020 Mathematics Subject Classification: 35R09, 92D25

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An efficient high-order numerical method for the transverse vibration of a damped beam

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Abstract: In this paper, a high-order and unconditionally stable difference method is proposed for the numerical solution of the one-dimensional damped beam vibration equation (Euler-Bernoulli beam equation with damping).
We apply a fourth-order compact finite difference approximation for discretizing the fourth-order spatial derivative of this equation and a fifth-order Padé approximation for the resulting system of ordinary differential equations. It is shown through analysis that the proposed scheme is unconditionally stable.
This new method is easy to implement, produces very accurate results and requires short CPU time. Several numerical examples are included to demonstrate the validity and applicability of the technique

Keywords: Padé approximation, compact finite difference scheme, unconditionally stable, damped Euler-Bernoulli beam equation, fourth-order PDE, high accuracy.

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A generalized averages based variational joint image segmentation and registration model

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Abstract: Image segmentation and registration are tightly coupled tasks, yet many variational models struggle when object and background intensities vary or when multiple structures are present. We propose a generalized averages based variational joint segmentation-registration (**GAV-JSR**) model that replaces Chan-Vese averages with generalized averages parameterized by γ, and embeds them in a non-rigid registration framework regularized by linear curvature. This yields an energy functional that (i) remains backward-compatible with prior **NJSR** [2] formulations when $\gamma = 1$, (ii) better handles high- and low-intensity backgrounds without pre-segmentation, and (iii) produces smooth, realistic deformations suitable for substantial affine changes. We validate the method on clinical (brain MRI, hand X-ray) and synthetic datasets, demonstrating consistent improvements over **GV-JSR** [**?**] and **NJSR** [2] baselines in both registration error and segmentation quality for example, error drops from 69.1% to 30.85% on one white backgroud multi-object case while maintaining clear object boundaries with across multiple structures along with positive F-measure value shown by suggested model. These results indicate that integrating generalized averages with curvature-based regularization provides a robust and accurate solution for monomodal joint segmentation and registration.

Keywords: segmentation, registration, generalized averages, medical images, synthetic images

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Mathematical modeling of cotton agrocenosis with age structure and arbitrary trophic functions

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Abstract: This paper considers mathematical modeling of the process of protecting cotton agrocenosis taking into account age structure and with arbitrary trophic functions.

Assume that the state of the model agrocenosis is described by the following system of differential equations [1]:

$$\begin{cases} l\dfrac{dN_0}{dt} = Q - \alpha_0 N_0 N_1, \\ \dfrac{dN_1}{dt} = k_0\alpha_0 N_0 N_1 - V_1(N_1)\sum\limits_{i=1}^{3}\tilde{N}_{i+1} - m_1 N_1, \\ \dfrac{\partial N_{i-1}}{\partial t} + \dfrac{\partial N_{i-1}}{\partial a} = k_{i-2}V_1(N_1)N_{i-1} - V_{i-1}(N_{i-1})\tilde{N}_{i+2} + \delta N_{i+2} - m_{i-1}N_{i-1}, \quad i = \overline{3,5}, \\ \dfrac{\partial N_{i+2}}{\partial t} + \dfrac{\partial N_{i+2}}{\partial a} = k_{i+1}V_{i-1}(N_{i-1})N_{i+2} - \epsilon\tilde{N}_{i+2}^2 - m_{i+2}N_{i+2}, \quad i = \overline{3,5}, \\ N_i|_{t=0} = N_i^0, \quad i = \overline{0,7}, \\ N_i(0,t) = \displaystyle\int_0^\infty B_i(\xi)N_i(\xi,t)\,d\xi, \quad i = \overline{2,7} \\ \tilde{N}_i = \displaystyle\int_{\alpha_i}^{\beta_i} N_i(a,t)\,da, \quad i = \overline{2,7} \end{cases}$$

where $N_0(t)$ is the mass of the external resource at time t, $N_1(t)$ is the biomass of cotton, Q is the rate of external resource supply, $N_i(a,t)$ is the biomass (or abundance) of harmful and beneficial insects of age a at time t, $B_i(\cdot)$ are the birth rates of harmful and beneficial insects, $V_i(\cdot)$ is the trophic function, $\alpha_0, m_i, k_i, \varepsilon$ are biological parameters of the populations in the agrocenosis; $t \in [0,\tau]$, $\tau = \text{const} < \infty$, and a is age, $a \in [0,\infty)$.

Keywords: mathematical modeling, agrocenosis, age structure, trophic functions, integral equations, biological control

2020 Mathematics Subject Classification: 35R09, 92D25

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Solving a transport problem with fuzzy tariffs on the example of cargo transportation at the Rogun HPP

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Abstract: This report is devoted to solving the transport problem with fuzzy tariffs on the example of cargo transportation at the Rogun HPP. A homogeneous cargo is available at m supply points A_i with volumes a_i and must be delivered to n demand points B_j with volumes b_j. The cost c_{ij} of transporting a unit from i to j is minimized:

$$(1) \qquad Z = \sum_{i=1}^{m}\sum_{j=1}^{n} c_{ij}x_{ij} \to \min, \quad \sum_{j=1}^{n} x_{ij} = a_i, \quad \sum_{i=1}^{m} x_{ij} = b_j, \quad x_{ij} \geq 0.$$

For fuzzy tariffs $\tilde{c}_{ij} = (a_{ij}, m_{ij}, b_{ij})$ of triangular type, we use defuzzification via level sets. The deterministic equivalent is:

$$(2) \quad s(\tilde{c}_{ij}) = \sum_{p=0}^{k} w_p L_p, \quad L_p = \frac{1}{2}\{(m_{ij} - a_{ij}(1-\alpha_p)) + (m_{ij} + b_{ij}(1-\alpha_p))\}.$$

The problem is solved by the potential method. For the Rogun HPP case: two suppliers (18,951 and 41,439 tons) deliver cement to four warehouses (17,019, 4,996, 36,444, 1,931 tons). Using level sets $\alpha = 0.05, 0.1, 0.2, 0.3$ with weights $W = (0.05, 0.1, 0.15, 0.3, 0.4)$, the optimal plan yields total cost $Z = 6,208,772$ somoni.

Keywords: transport problem, northwest corner method, potential method, fuzzy tariffs, Fuzzy set, optimal plan, triangular membership function

2020 Mathematics Subject Classification: 90B06, 90C05, 90C70

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On the strong solvability of the Laplace equation with a non-classical boundary condition

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Abstract: This paper is devoted to the question of strong solvability of the following nonlocal boundary value problem for the Laplace equation in the domain $D = (0,1) \times (0,h)$:

$$\frac{\partial^2 u(x,y)}{\partial x^2} + \frac{\partial^2 u(x,y)}{\partial y^2} = 0, \qquad (x,y) \in D, \tag{1}$$

$$u(x,0) = \varphi(x), \qquad u(x,h) = \psi(x), \qquad 0 \le x \le 1, \tag{2}$$

$$u(0,y) = u(1,y), \qquad \frac{\partial u}{\partial x}(0,y) = \beta \frac{\partial u}{\partial x}(1,y) + \alpha u(1,y), \qquad 0 \le y \le h, \tag{3}$$

where $\alpha, \beta \in \mathbb{R}$, $\beta \neq -1$, $\varphi(x)$ and $\psi(x)$ are given functions. The solution to problem (1)-(3) is sought in a weighted Sobolev space $W^2_{p,\omega}(D)$ generated with a mixed norm

$$\|u\|_{L_{p,\omega}(D)} := \left(\int_0^h \int_0^1 \omega(x) |u(x,y)|^p \, dx \right)^{1/p} dy,$$

where the weight function $\omega(x)$ belongs to the Muckenhoupt class A_p. The boundary conditions of the corresponding spectral problem are regular, but not strongly regular. For $\beta \neq \pm 1$ and $\alpha \neq 0$, the spectral problem has only eigenfunctions that do not form a basis in $L_{p,\omega}[0,1]$, and direct application of the Fourier method is impossible. From the linear combinations of these eigenfunctions, a new system is constructed, which already forms a basis in $L_{p,\omega}[0,1]$, and this system is used to solve the equation under consideration by the Fourier method. In this work we use the approach from [1].

Keywords: Laplace equation, nonlocal boundary value problem, spectral problem, regular boundary conditions, eigenvalues, eigenfunctions, basis

2020 Mathematics Subject Classification: 35J05, 35J08, 46E35, 46M05

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Positive invariance of an admissible region in a discrete predator–prey model

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Abstract: Discrete predator–prey models are widely used to describe population processes [1,2]. The construction of positively invariant regions is an important step in the analysis of discrete predator–prey systems [3]. Consider the following model with rational prey growth:

$$\begin{cases} x_{n+1} = \dfrac{\alpha x_n}{1+\beta x_n+\gamma x_n^2} - a x_n y_n, \\ y_{n+1} = y_n + b x_n y_n - d y_n, \end{cases} \tag{1}$$

where $x_n, y_n \geq 0$ denote the prey and predator populations, respectively, and $\alpha > 0$, $\beta \geq 0$, $\gamma > 0$, $a > 0$, $b > 0$, $d > 0$.

Here α characterizes the reproductive potential of the prey, β, γ describe intraspecific self-regulation, a describes the predator's impact on the prey, b accounts for predator growth due to prey consumption, and d denotes the natural mortality of the predator.

Set

$$F(x) = \frac{\alpha x}{1+\beta x+\gamma x^2}.$$

The entire nonnegative quadrant $\mathbb{R}^2_+$ is not automatically invariant: if

$$y_n > \frac{\alpha}{a(1+\beta x_n+\gamma x_n^2)},$$

then $x_{n+1} < 0$. Therefore, consider the biologically admissible region

$$D_K = \left\{ (x,y) \in \mathbb{R}^2_+ : \ 0 \leq x \leq K, \quad 0 \leq y \leq \frac{\alpha}{a(1+\beta x+\gamma x^2)} \right\}.$$

Theorem 1.

Let $\alpha > 0$, $\beta \geq 0$, $\gamma > 0$, $a > 0$, $b > 0$, $0 < d \leq 1$, and

$$K \geq \frac{\alpha}{\beta + 2\sqrt{\gamma}}. \tag{2}$$

Assume that for all $x \in [0, K]$ and $z \in [0, 1]$,

$$(H_D) \qquad z(1-d+bx)\Big[1+\beta F(x)(1-z)+\gamma F^2(x)(1-z)^2\Big] \leq 1+\beta x+\gamma x^2.$$

Then D_K is a positively invariant region for system (1):

$$(x_0, y_0) \in D_K \quad \Longrightarrow \quad (x_n, y_n) \in D_K \qquad \text{for all } n \geq 0.$$

Since $\max_{x \geq 0} F(x) = \alpha/(\beta + 2\sqrt{\gamma})$, condition (2) controls the prey component, whereas condition (H_D) controls the predator component relative to the upper boundary of D_K. Together, these conditions guarantee positive invariance of the biologically admissible region and provide a basis for further investigation of fixed points, local stability, and bifurcation regimes of the system.

Keywords: discrete predator–prey model, positive invariance, invariant region, population dynamics, rational growth

2020 Mathematics Subject Classification: 39A60, 92D25

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Basicity of the sine and cosine systems in weighted Sobolev spaces

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Abstract: In this work, we present a method for establishing basicity in weighted Sobolev spaces. Under suitable conditions on the weight function, and assuming that the systems $\{\sin \lambda_n x\}_{n\in\mathbb{N}}$ and $\{\cos \lambda_n x\}_{n\in\mathbb{N}}$ form Schauder bases in $L_{p,\rho}(0,\pi)$, we establish sufficient conditions under which the cosine system $\{\cos \lambda_n x\}_{n\geq 0}$ forms a Schauder basis in the weighted Sobolev space $W^1_{p,\rho}(0,\pi)$.

This work is supported by the Azerbaijan Science Foundation-Grant no. AEF-MGC-2024-2(50)-16/02/1-M-02

Keywords: Schauder basis, weighted Lebesgue space, weighted Sobolev space, biorthogonal system, sine system, cosine system

2020 Mathematics Subject Classification: 46B15, 46E35, 42C15

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On the integral representation of regular solutions to the heat equation and boundary conditions

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Abstract: Let $\Omega \subset \mathbb{R}^n$ be a bounded domain with a smooth boundary, $Q = \Omega \times \{t > 0\}$.

Definition 2.4. *The operator L is called a coercively solvable operator of the heat equation*

$$Lu(x,t) = \left(\frac{\partial}{\partial t} - \Delta_x\right) u(x,t) = f(x,t), \quad (x,t) \in Q \tag{1}$$

if for any $f \in L_2(Q)$ there exists a unique solution $u \in W_2^{2,1}(Q)$ of equation (1) *satisfying the inequality*

$$\|u\|_{W_2^{2,1}(Q)} \leq c\,\|f\|_{L_2(Q)}. \tag{2}$$

We seek a class of regular solutions of equation (1) in the following form

$$\begin{aligned} u(x,t) = &\int_0^t \int_\Omega \varepsilon(x,t,\xi,\tau) f(\xi,\tau) d\xi d\tau \\ &+ \int_\Omega \varepsilon(x,t,\xi,0)\varphi(\xi) d\xi + \int_0^t \int_{\partial\Omega} \frac{\partial \varepsilon(x,t,\xi,\tau)}{\partial n_\xi} \nu(\xi,\tau) dS_\xi d\tau, \end{aligned} \tag{3}$$

where

$$\varepsilon(x,t) := \frac{\Theta(t)}{(2\sqrt{\pi t})^n} e^{-\frac{|x|^2}{4t}} \tag{4}$$

is the fundamental solution of the heat equation.

Next, using the Riesz representation theorem for linear functionals, we derive a unified integral representation of the solutions and find the boundary conditions. For this purpose, we apply the method developed by the authors in [1], [2].

Keywords: heat equation, fundamental solution, boundary condition.

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2020 Mathematics Subject Classification: 35K05, 35K20

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Maximal F-regularity of an abstract Cauchy problem with sectorial operators

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Abstract: We consider the abstract Cauchy problem

$$\frac{du(t)}{dt} + Au(t) = f(t), \quad t > 0,$$
$$u(0) = 0,$$

where A is a closed linear operator in a Banach space X with dense domain $D(A)$, and f belongs to a Bochner–Luxemburg space $F(\mathbb{R}_+; X)$. The space $F(\mathbb{R}_+)$ is assumed to be rearrangement invariant, with Boyd indices α_F and β_F satisfying

$$0 < \alpha_F \leq \beta_F < 1.$$

We define maximal F-regularity by requiring that, for every $f \in F(\mathbb{R}_+; X)$, the corresponding solution satisfies

$$\frac{du}{dt} \in F(\mathbb{R}_+; X), \qquad Au \in F(\mathbb{R}_+; X).$$

The main result is obtained by transferring the classical maximal L_p-regularity result to the rearrangement invariant setting. For this purpose, we use R_F-bounded families of operators and the relation between R_p-boundedness and R_F-boundedness. If X is a UMD space and A is sectorial with spectral angle less than $\pi/2$, then the existence of maximal L_p-regularity for one admissible exponent implies maximal F-regularity. More precisely, it is sufficient that one of the standard equivalent L_p-regularity conditions holds, including R_p-boundedness of the corresponding resolvent family or R_p-sectoriality of A. Thus, the result applies simultaneously to a broad class of rearrangement invariant spaces, including Lebesgue, grand Lebesgue, Orlicz, Lorentz and Marcinkiewicz spaces. The obtained framework also covers Bochner–Luxemburg spaces and provides a convenient setting for further applications to elliptic operators in symmetric function spaces.

Keywords: abstract Cauchy problem, maximal regularity, rearrangement invariant spaces, Boyd indices, sectorial operators, UMD spaces

2020 Mathematics Subject Classification: 35J40, 35K50, 42B15

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About an explicit solutions of a second-order differential equation of Riemann-type with many singular points

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Abstract: The paper is devoted to the investigation of the second order differential equation of the Riemann-type with three singular points

$$(1)\quad \frac{d^2w}{dz^2}+\left(\frac{1-r_1^{(1)}+r_2^{(1)}}{z-a_1}+\frac{1-r_1^{(2)}+r_2^{(2)}}{z-a_2}+\frac{1-r_1^{(3)}+r_2^{(3)}}{z-a_3}\right)\frac{dw}{dz}+$$
$$+\left(\frac{r_1^{(1)}r_2^{(1)}(a_1-a_2)(a_1-a_3)}{z-a_1}+\frac{r_1^{(2)}r_2^{(2)}(a_2-a_1)(a_2-a_3)}{z-a_2}+\right.$$
$$\left.+\frac{r_1^{(3)}r_2^{(3)}(a_3-a_1)(a_3-a_2)}{z-a_3}\right)\frac{1}{(z-a_1)(z-a_2)(z-a_3)}w=0,$$

where $r_i^{(j)}(i=1,2;j=1,2,3)$–are the roots of the indicial equations corresponding to the singular points a_1,a_2,a_3 and these roots satisfy the Fuchs condition

$$(2)\qquad r_1^{(1)}+r_2^{(1)}+r_1^{(2)}+r_2^{(2)}+r_1^{(3)}+r_2^{(3)}=1.$$

and many singular points

$$(3)\quad w''+\left(\sum_{r=1}^{n}\frac{1-\alpha_r-\beta_r}{z-a_r}\right)w'+\left(\sum_{r=1}^{n}\frac{\alpha_r\beta_r}{(z-a_r)^2}+\sum_{r=1}^{n}\frac{E_r}{z-a_r}\right)w=0,$$

where α_r,β_r are the roots of the indicial equations corresponding to singular points a_r, and the following conditions are satisfied

$$(4)\quad \sum_{r=1}^{n}(\alpha_r+\beta_r)=n-2,\sum_{r=1}^{n}E_r=0,\sum_{r=1}^{n}(a_rE_r+\alpha_r\beta_r)=0,$$
$$\sum_{r=1}^{n}(a_r^2E_r+2a_r\alpha_r\beta_r)=0.$$

In the work, a three-pointed differential operator and n-pointed differential operator are introduced, and with the help of these operators, second-order operator-differential equations constructed. Further, it is shown the equivalence of the constructed operator-differential equations to the second-order Riemann equations with three and n singular points, but only with the distinction that the Fuchs condition is replaced by a Fuchs-type condition with zero right-hand side. It is also obtained the explicit solutions of the second order operator-differential equation with three and many singular points.

The classical results about these type equations was obtained in works [1] – [5].

Keywords: differential equation, Riemann-type differential equation, singular point, characteristic equation

2020 Mathematics Subject Classification: 44A15, 45J05, 45E99, 45D99

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On the solvability of a problem for a nonlinear pseudoparabolic equation

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Let $\Omega = [0,1] \times [0,T]$ be a rectangle, and let $a_i(x,t) \in C(\Omega), \quad i = \overline{0,3}$. We consider the differential operator L defined by the equality:

$$Lu = u_{xtt} - a_0 u_{xt} - a_1 u_x - a_2 u_t - a_3 u.$$

We study the following nonlinear boundary value problem in the domain Ω:

$$Lu = f(t,u), \;\; u(0,t) = 0, \;\; u(x,0) = u(x,T), \;\; u_t(x,0) = u_t(x,T). \tag{1}$$

where the function $f(t,u) : [0,T] \times \mathbb{R}^1 \to \mathbb{R}^1$. In what follows, we assume that f is continuous on $[0,T] \times \mathbb{R}^1$ and satisfies the Lipschitz condition with respect to u; that is, there exists a constant $L > 0$ such that

$$|f(t,u) - f(t,u_1)| \le L\,|u - u_1|\,. \tag{2}$$

for any $t \in [0,T]$ and $u, u_1 \in \mathbb{R}^1$.

We further assume that problem (1) has a unique classical solution, provided that the Lipschitz constant L is sufficiently small.

For this purpose, we make essential use of a result derived from work [1], which provides a uniform estimate for the solution of problem (1) and all its derivatives in the linear case.

In this work, the following theorem is proved:

Theorem. Assume that f is continuous on $[0,T] \times \mathbb{R}^1$ and satisfies the Lipschitz condition (2) on u and

$$LK < 1,$$

then problem (1) has a unique classical solution $u(x,t) \in C_3(\Omega)$. Where the constant K is the coefficient bounding u in the linear case in [1]. $C_3(\Omega)$ is the space of functions u such that u, u_t, u_x, u_{xt}, and u_{xtt} are continuous on Ω and equipped with the norm

$$\|u\|_3 = \max_{(x,t)\in\Omega} \big\{ |u(x,t)|,\, |u_t(x,t)|,\, |u_x(x,t)|,\, |u_{xt}(x,t)|,\, |u_{xtt}(x,t)| \big\}.$$

Keywords: pseudoparabolic equation, uniform convergence, small parameter, perturbation.

2020 Mathematics Subject Classification: 35A09, 35A02

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A coupled dynamical system for the social facilitation of eating: stability and transient network effects

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Abstract: The social facilitation of eating, defined as the increased food consumption observed when individuals are in the company of others has been represented in models as a scaling factor. This approach does not capture the dynamic effects of companions and the relationships in their network. We develop a coupled system of k nonlinear differential equations for the food intake $E_i(t)$ of each agent, parameterized by the agent's satiety $E_{\max,i}$, eating rate θ_i, and meal duration r_i:

$$\dot{E}_i = \frac{\theta_i g_i(t)}{E_{\max,i}^2 \tau (1 + g_i(t))} \left(E_{\max,i}\tau + g_i(t) E_i\right) \left(E_{\max,i}\tau - E_i\right), \tag{1}$$

with $g_i(t) = (F_i(t) r_i)^{0.45}$. Social facilitation acts as a state variable $F_i(t)$, which decays at rate λ toward a level determined by the unconsumed capacity of companions within a familiarity network $S_{ij} \in [0, 1]$ and the coupling strength $\alpha \geq 0$:

$$\dot{F}_i = \lambda\Big(f(n)\Big[1 + \frac{\alpha}{k-1} \sum_{j \neq i} S_{ij}\Big(1 - \frac{E_j}{E_{\max,j}}\Big)\Big] - F_i\Big). \tag{2}$$

The equation simplifies to an individual agent model when $k = 1$ and remains bounded across all parameter values, as intake growth diminishes to zero whenever $E_i \to E_{\max,i}\tau$, irrespective of F_i. This term ensures the Jacobian of the coupled system adopts a block lower-triangular form at the satiety equilibrium of the group: at that point, facilitation cannot feed back into intake, so the spectrum remains negative for any α and any network S, thus precluding a Hopf bifurcation. Applying nonlinear simulation revealed a transient rise and decline in social facilitation whose magnitude depended on the network position rather than on group size alone. Consequently, the model represents heterogeneous social effects that a scalar parameter for group size would not be capable of capturing.

Keywords: social facilitation of eating, coupled nonlinear ODE systems, multi-agent dynamical models, local stability analysis, Jacobian, boundedness, mathematical biology

2020 Mathematics Subject Classification: 34D20, 92B05, 37N25

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Age-structured Bayesian SIR modelling with reversible-jump MCMC: a comparative study of two social contact matrix projections

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Abstract: We develop a fully Bayesian age-structured discrete-time stochastic SIR model in which the transmission rate $\beta_i(t)$ and recovery probability $\gamma_i(t)$ for age group i are piecewise constant, with the number and location of changepoints treated as unknown. For n_a age groups with populations N_i and a contact matrix $C = (C_{ij})$, the force of infection is

$$\lambda_i(t) = \beta_i(t) \sum_{j=1}^{n_a} C_{ij} \frac{I_j(t-1)}{N_j}, \qquad i = 1, \ldots, n_a. \tag{1}$$

Estimation is carried out through reversible-jump Markov chain Monte Carlo (RJMCMC) coupled with a Product Partition Model (Barry and Hartigan [1]), extending the single-population changepoint framework of Gu and Yin [2] to the age-structured, contact-matrix-coupled setting. We apply the model to Indonesia's 2022 Omicron BA.2 wave using two independently generated contact matrices, the Bayesian hierarchical projection of Prem et al. [3] and the Poisson generalised additive model of Tierney et al. [4], under both a two-group and a three-group age stratification – four configurations in total. All four RJMCMC runs independently converge on a single structural changepoint within a two-to-three day window, without any prior specification of its number or location. We further establish an identifiability result for this class of models: if two contact matrices satisfy $\sum_j C_{ij}^{(2)} = \rho \sum_j C_{ij}^{(1)}$ for a row-wise scalar ρ, then the two parametrisations produce the same likelihood whenever $\beta_i^{(2)} = \beta_i^{(1)}/\rho$, so that only the product $\beta_i(t) \times \sum_j C_{ij}$ is estimable from aggregate incidence data. This is confirmed empirically: the product is conserved across the two independently generated contact matrices to within 1.3–1.5% in every model configuration, despite the matrices themselves differing substantially in total contact rate. The result gives a precise account of what is, and is not, identifiable in contact-matrix-coupled compartmental models.

Keywords: age-structured epidemic model, reversible-jump MCMC, Product Partition Model, changepoint detection, social contact matrix, parameter identifiability

2020 Mathematics Subject Classification: 62F15, 92D30, 65C40

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Existence of radially symmetric nonnegative solutions for a class of nonpositone elliptic problems in a ball using shooting method

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Abstract: In this paper, we investigate the existence of nonnegative radial solutions for a semipositone semilinear elliptic boundary value problem of the form $-\Delta u = \lambda f(u)$ in Ω, with $u = 0$ on $\partial\Omega$, where Ω is the unit ball in $\mathbb{R}^n$, $n \geq 2$, and $\lambda > 0$ is a parameter. The nonlinearity satisfies the semipositone condition $f(0) < 0$, which creates analytical difficulties since the maximum principle cannot be directly applied near the origin.
Assuming that $f \in C(\mathbb{R}^+, \mathbb{R})$ has exactly one positive zero, is convex and satisfies suitable superlinearity and growth conditions, we analyze the structure of radial solutions. Using radial symmetry, the problem is reduced to an ordinary differential equation. We apply the shooting method and study the qualitative behavior of the associated initial value problem.

Keywords: Semipositone problem, Radial solutions, Elliptic equation, Shooting method.

2020 Mathematics Subject Classification: 35J66, 35B18.

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Noetherness of oblique derivative problems in weighted spaces of harmonic functions

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Abstract: We study the Noether property of an oblique derivative problem for the Laplace equation in the unit disk within weighted spaces of harmonic functions. Let $D = \{z \in \mathbb{C} : |z| < 1\}$ and $\Gamma = \partial D$. For a Muckenhoupt weight $\rho \in A_p(\Gamma)$, $1 < p < \infty$, we consider the weighted Lebesgue space $L_{p;\rho}(\Gamma)$ and the corresponding harmonic space $h^{(1)}_{p;\rho}$. The boundary value problem is

$$\Delta_{r,\varphi} u = 0, \qquad (r, \varphi) \in D, \tag{1}$$

$$\gamma^+ \left(\cos\varphi \frac{\partial u}{\partial r} + \sin\varphi \frac{\partial u}{\partial \varphi} \right) = f(\varphi), \qquad \varphi \in [-\pi, \pi), \tag{2}$$

where $f \in L_{p;\rho}(\Gamma)$ and

$$\Delta_{r,\varphi} u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r}\frac{\partial u}{\partial r} + \frac{1}{r^2}\frac{\partial^2 u}{\partial \varphi^2}.$$

Here $h^{(1)}_{p;\rho}$ consists of harmonic functions whose radial and angular derivatives belong to $h_{p;\rho}$. The main result states that, for every $\rho \in A_p(\Gamma)$, the above oblique derivative problem is Noetherian in $h^{(1)}_{p;\rho}$. Moreover, its index is equal to -2. Thus, the result establishes the Fredholm-type solvability structure of the problem in weighted harmonic spaces.

Keywords: harmonic functions, oblique derivative problem, weighted spaces, Muckenhoupt weights, Noether operator, index.

2020 Mathematics Subject Classification: 35J05, 35J08, 46E35, 46M05.
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Pre-compactness of sets in the discrete Morrey-type spaces

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Abstract: Let $m \in \mathbb{Z}$, $N \in \omega := \mathbb{N} \cup \{0\}$, and write $S_{m,N} := \{m - N, \dots, m, \dots, m + N\}$. Then $|S_{m,N}| = 2N + 1$ — the cardinality of $S_{m,N}$. (It is independent of m, but we write it in the subscript to keep track of the set.) Let $\mathbb{K}$ be $\mathbb{R}$ or $\mathbb{C}$ and $1 \le p \le q < \infty$. We denote by $\ell^p_q = \ell^p_q(\mathbb{Z})$ the set of sequences $x = (x_k)_{k\in\mathbb{Z}}$ taking values in $\mathbb{K}$ such that $\|x\|_{\ell^p_q} := \sup_{m\in\mathbb{Z},\, N\in\omega} |S_{m,N}|^{\frac{1}{q}-\frac{1}{p}} \left(\sum_{k\in S_{m,N}} |x_k|^p\right)^{\frac{1}{p}} < \infty$. Clearly ℓ^p_q is a vector space, which we shall call a *discrete Morrey space.* We remark that when $p = q$, we have $\ell^p_p = \ell^p$, the space of p-summable sequences with integer indices. For a sequence x to be in ℓ^p_q, x has to have some decay, but not as fast as those in ℓ^p. In general, for $p < q$, ℓ^p_q is a larger space than ℓ^p, as we shall see below. In the following theorem we establish conditions for the pre-compactness of sets in discrete Morrey space.

Theorem 2.5. *Let $1 \le p \le q < \infty$ and let $\ell^p_q = \ell^p_q(\mathbb{Z})$ be the discrete Morrey space. A subset $\mathcal{F} \subset \ell^p_q$ is relatively compact (precompact) in ℓ^p_q if and only if the following conditions are satisfied:*

(i) *(Boundedness) $\mathcal{F}$ is bounded in ℓ^p_q, i.e., $\sup_{x\in\mathcal{F}} \|x\|_{\ell^p_q} < \infty$.*

(ii) *(Vanishing at infinity uniformly in $\mathcal{F}$) For every $\varepsilon > 0$ there exists $R > 0$ such that*

$$\sup_{x\in\mathcal{F}} \sup_{m\in\mathbb{Z},\, N\in\omega S_{m,N}\cap\{k\in\mathbb{Z}:|k|>R\}\neq\varnothing} |S_{m,N}|^{\frac{1}{q}-\frac{1}{p}} \left(\sum_{k\in S_{m,N}} |x_k|^p\right)^{\frac{1}{p}} < \varepsilon.$$

(iii) *(Tightness on finite windows) For every fixed finite interval $I \subset \mathbb{Z}$, the set of restrictions $\{\, (x_k)_{k\in I} : x \in \mathcal{F} \,\}$ is relatively compact in $\mathbb{K}^I$.*

In particular, if $\mathcal{F}$ is bounded in ℓ^p_q and satisfies the uniform vanishing at infinity condition (ii)*, then $\mathcal{F}$ is relatively compact in ℓ^p_q.*

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On the lattices generated varieties

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Abstract: The main result is the determination of an equational basis for the variety of lattices generated by a finite non-modular and non-semidistributive lattice. It is established that the resulting quasivariety is a variety defined by three identities. Structural constraints are determined for minimal non-trivial covers of join-irreducible elements in lattices satisfying the weak modularity identity, and the possible closures of individual join-irreducible elements are described.

Keywords: variety, finite lattice, quasivariety, non-modular lattice, non-semidistributive lattice.

2020 Mathematics Subject Classification: 06B10, 06B20, 06B05

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A characterization: fixed points and weak compactness in $W^{1,1}((a,b)\cup(c,d),\mu)$

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Abstract: Let $\Omega = (a,b)\cup(c,d)$ be the union of disjoint open intervals in $\mathbb{R}$ and (Ω,Σ,μ) be a finite measure space. We examine the Sobolev space $W^{1,1}(\Omega,\mu)$ equipped with its standard norm and establish a precise characterization of weak compactness for closed, bounded and convex subsets of $W^{1,1}(\Omega,\mu)$. It is formulated in terms of the existence of fixed points for specific classes of affine, uniformly Lipschitz mappings.

Keywords: Weak Compactness, Sobolev Space, Lipschitzian mapping

2020 Mathematics Subject Classification: 47H10, 46B50, 46E35

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Boundedness and integral inequalities for Sonin-type fractional integral operators on Morrey spaces

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Abstract: This work develops a functional-analytic framework for Sonin-type fractional integral operators acting on Morrey spaces. Let k be a locally integrable kernel admitting an associated kernel ℓ such that the pair (k, ℓ) satisfies the Sonin convolution relation. The corresponding generalized fractional integral operator is defined by

$$(\mathcal{S}_k f)(x) = \int_0^x k(x-t) f(t)\, dt. \tag{3}$$

We investigate the mapping properties of $\mathcal{S}_k$ in Morrey-type spaces, with particular emphasis on the interaction between the nonlocal structure of the kernel and the local scale-sensitive behavior encoded by the Morrey norm. Under appropriate size and integrability assumptions on the Sonin kernel, sufficient conditions are established for the boundedness of $\mathcal{S}_k$ between suitable Morrey spaces. The resulting estimates are subsequently employed to derive a family of integral inequalities associated with Sonin-type operators. The proposed approach is formulated at the kernel level and therefore does not depend on a particular power-law representation. Consequently, suitable choices of the kernel recover important fractional integral structures as special cases and provide corresponding reductions of the obtained estimates. This establishes a unified connection between Sonin-type fractional calculus, operator-theoretic estimates, and scale-sensitive function spaces. The results provide an analytical basis for studying a broader class of nonlocal integral operators and may be useful in the analysis of fractional integral and differential problems exhibiting memory and multiscale effects.

Keywords: Sonin kernel, Sonin pair, generalized fractional integral operator, Morrey spaces, boundedness, integral inequalities, nonlocal operators

2020 Mathematics Subject Classification: 26A33, 42B35, 46E30

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Subword pattern statistics in Carlitz and palindromic compositions

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Abstract: A composition $\pi = \pi_1\pi_2\cdots\pi_m$ of a positive integer n is an ordered collection of one or more positive integers whose sum is n. The number of summands, namely m, is called the number of parts of π. A composition with distinct adjacent parts is called a *Carlitz composition*, while a composition that reads the same forwards and backwards is called a *palindromic composition*. We study pattern statistics in these classes of compositions by enumerating them with respect to the number of occurrences of a consecutive subword pattern τ and the sum of the first entries in these occurrences. Using multivariate generating functions and linear algebraic methods, we obtain formulas that simultaneously encode these statistics. In particular, we derive explicit generating functions for several short patterns in Carlitz compositions, palindromic compositions, Carlitz palindromic compositions, and their k-Carlitz extensions. These results provide a unified framework for studying refined pattern enumeration in restricted classes of compositions.

Keywords: Subwords, generating functions, Carlitz compositions, palindromic compositions, Cramer's method.

2020 Mathematics Subject Classification: 05A15, 15A06, 15A15

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Existence and uniqueness of an inverse source problem for a degenerate mixed-type equation with a variable coefficient

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Abstract: We investigate an inverse source problem for the degenerate mixed-type partial differential equation

$$u_{yy} + (\operatorname{sgn} y)|y|^m u_{xx} = f(x), \qquad m > 0, \tag{1}$$

in the rectangular domain $\Omega = \{(x, y) : 0 < x < 1, -\alpha < y < \beta\}$, where both the solution $u = u(x, y)$ and the source term $f = f(x)$ are unknown. The problem is supplemented by homogeneous Dirichlet conditions in the x-direction, prescribed boundary data at $y = -\alpha$ and $y = \beta$, and an additional condition for u_y at $y = -\alpha$. Related inverse problems for mixed-type equations were studied in [1, 2].

Applying the method of separation of variables and the Fourier method, we reduce the inverse problem to a sequence of ordinary differential equations whose solutions are expressed in terms of Bessel and modified Bessel functions. Explicit reconstruction formulas for the Fourier coefficients of the unknown source and the corresponding solution components are derived. Sufficient conditions ensuring the non-vanishing of the characteristic denominators are established. Under these conditions, the existence and uniqueness of a classical solution are proved. Moreover, uniform estimates for the Fourier coefficients and the solution components are obtained, which imply the absolute and uniform convergence of the corresponding Fourier series together with their required derivatives.

Keywords: inverse source problem, degenerate mixed-type equation, Fourier method, Bessel functions, characteristic denominator, existence and uniqueness.

2020 Mathematics Subject Classification: 35M10, 35R30, 35A01.

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On the Fourier transform of functions from the mixed-metric Lorentz space $L_{\bar{2},\bar{r}}$

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Abstract: Abstract: This extended abstract is devoted to Bochkarev-type inequalities for the Fourier transform and to sharp logarithmic estimates in Lorentz spaces, with particular emphasis on mixed-metric Lorentz spaces on $[0,1]^2$ and on $\mathbb{R}^2$.

1. Mixed-metric Lorentz space Definition 1.

Let $\bar{p}=(p_1,p_2)$ and $\bar{r}=(r_1,r_2)$ satisfy $0<p_i\le\infty$ and $0<r_i\le\infty$. The mixed-metric Lorentz space $L_{\bar{p},\bar{r}}\big([0,1]^2\big)$ is defined as the set of all measurable functions f on $[0,1]^2$ for which the quantity

$$\|f\|_{L_{\bar{p},\bar{r}}}=\Big\|\ \|f\|_{L_{p_1,r_1}([0,1])}\ \Big\|_{L_{p_2,r_2}([0,1])}$$

is finite. More explicitly, for $0<\bar{r}<\infty$,

$$\|f\|_{L_{\bar{p},\bar{r}}}=\left(\int_0^1\left[t_2^{1/p_2}\left(\int_0^1\Big(t_1^{1/p_1}f^{*1}(t_1,\cdot)\Big)^{r_1}\frac{dt_1}{t_1}\right)^{1/r_1}\right]^{r_2}\frac{dt_2}{t_2}\right)^{1/r_2},$$

and for $\bar{r}=\infty$,

$$\|f\|_{L_{\bar{p},\infty}}=\sup_{t_1,t_2}t_1^{1/p_1}\,t_2^{1/p_2}\,f^{*1*2}(t_1,t_2),$$

where f^{*1} and f^{*1*2} denote successive non-increasing rearrangements with respect to the variables.

2. Fourier coefficients in $L_{2,\bar{r}}\big([0,1]^2\big)$

Theorem 1. Let

$$\Phi_{m_1,m_2}(x_1,x_2)=\varphi_{m_1}(x_1)\,\psi_{m_2}(x_2),\qquad m_1,m_2\in\mathbb{N},$$

be an orthonormal system on $[0,1]^2$ which is uniformly bounded in L_∞, i.e., $\|\varphi_m\|_\infty\le M_1$ and $\|\psi_m\|_\infty\le M_2$. Then for every $f\in L_{2,\bar{r}}\big([0,1]^2\big)$ with $2<r_1,r_2<\infty$ the following inequality holds:

$$\sup_{\substack{|A_1|\ge 8\\ A_1\subset\mathbb{N}}}\sup_{\substack{|A_2|\ge 8\\ A_2\subset\mathbb{N}}}\frac{1}{|A_1|^{1/2}\,|A_2|^{1/2}\,(\log_2(|A_1|+1))^{\frac12-\frac{1}{r_1}}\,(\log_2(|A_2|+1))^{\frac12-\frac{1}{r_2}}}\sum_{k_1\in A_1}\sum_{k_2\in A_2}\big|\widehat{f}(k_1,k_2)\big|$$

$$\le 9M\,\|f\|_{L_{2,\bar{r}}},$$

where $M=(M_1+1)(M_2+1)$ and $\widehat{f}(k_1,k_2)$ are the Fourier coefficients of f with respect to the system $\{\Phi_{m_1,m_2}\}$.

3. Bochkarev-type inequality for the Fourier transform in $L_{2,r}(\mathbb{R})$

Theorem 2. Let

$$\mathcal{R}_{\mathbb{N}}=\left\{A=\bigcup_{i=1}^{N}A_i:\ A_i\text{ are intervals in }\mathbb{R}\right\}.$$

Then for every $f\in L_{2,r}(\mathbb{R})$ with $2<r<\infty$ one has

$$\sup_{N\ge 8}\sup_{A\subset\langle\mathbb{N}\rangle}\frac{1}{|A|^{1/2}\left(\log_2(1+N)\right)^{\frac12-\frac{1}{r}}}\left|\int_A\widehat{f}(\xi)\,d\xi\right|\le 23\,\|f\|_{L_{2,r}}.$$

4. Two-dimensional Fourier transform

Definition 2. Let $f \in L_1(\mathbb{R}^2)$. The two-dimensional Fourier transform of f is defined by

$$\widehat{f}(\xi_1, \xi_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_1, x_2)\, e^{\,2\pi i(x_1\xi_1 + x_2\xi_2)}\, dx_1\, dx_2.$$

5. Auxiliary lemma

Lemma 1. Let $\frac{4}{3} < q_1, q_2 < 2$ and $f \in L_{\bar{q},\bar{2}}(\mathbb{R}^2)$. For any measurable sets A_1 and A_2 of finite measure in $\mathbb{R}^N$ the following inequality holds:

$$\sup_{A_1 \subset \mathbb{R}^N} \sup_{A_2 \subset \mathbb{R}^N} \frac{1}{|A_1|^{1/q_1}\, |A_2|^{1/q_2}} \int_{A_1} \int_{A_2} \left|\widehat{f}(\xi_1, \xi_2)\right| d\xi_1\, d\xi_2$$
$$\leq C \left(\frac{q_1}{2(q_1-1)}\right)^{\left(\frac{1}{q_1}-\frac{1}{2}\right)} \left(\frac{q_2}{2(q_2-1)}\right)^{\left(\frac{1}{q_2}-\frac{1}{2}\right)} \|f\|_{L_{\bar{q},\bar{2}}}.$$

6. Main theorem for the two-dimensional Fourier transform

Theorem 3 (Main theorem) Let $2 < r_1, r_2 < \infty$. Then for every $f \in L_{2,\bar{r}}(\mathbb{R}^2)$ one has

$$\sup_{\substack{|A_1|\geq 8 \\ A_1 \subset \mathbb{N}}} \sup_{\substack{|A_2|\geq 8 \\ A_2 \subset \mathbb{N}}} \frac{1}{|A_1|^{1/2}\, |A_2|^{1/2}\, (\log_2(|A_1|+1))^{\frac{1}{2}-\frac{1}{r_1}}\, (\log_2(|A_2|+1))^{\frac{1}{2}-\frac{1}{r_2}}} \int_{A_1} \int_{A_2} \left|\widehat{f}(\xi_1, \xi_2)\right| d\xi_1\, d\xi_2$$
$$\leq \|f\|_{L_{2,\bar{r}}}.$$

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Keywords: Lorentz spaces, mixed metric, Fourier transform, Bochkarev-type inequalities.

A variable space grid explicit finite difference method for nonlocal boundary value problems with deviated arguments

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Abstract: This work presents a variable space grid (VSG) approach based on the explicit finite difference method for solving a nonlocal boundary value problem with deviated arguments.

The proposed numerical scheme is developed and validated through a series of numerical experiments. The computed results are compared with the corresponding exact solution and exhibit excellent agreement, demonstrating the accuracy and reliability of the proposed method. Furthermore, the convergence of the numerical solution toward the exact solution is verified as the mesh is refined. In addition, the stability of the proposed scheme is analyzed using the von Neumann stability criterion.

Keywords: Numerical solution, Non local partial differential equation, Deviated boundary condition, Variable space grid method

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On spectral properties of one discontinuous second-order differential operator with a spectral parameter in discontinuity conditions

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Abstract: Consider a spectral problem with a point of discontinuity

$$l(y) = -y''(x) = \lambda y(x), \qquad x \in \left(0, \frac{1}{3}\right) \cup \left(\frac{1}{3}, 1\right), \tag{1}$$

$$\begin{cases} \alpha_1 y'(0) + \alpha_0 y(0) = 0, \\ \beta_1 y'(1) + \beta_0 y(1) = 0, \\ y\left(\frac{1}{3} - 0\right) = y\left(\frac{1}{3} + 0\right), \\ y'\left(\frac{1}{3} - 0\right) - y'\left(\frac{1}{3} + 0\right) = \lambda m y\left(\frac{1}{3}\right). \end{cases} \tag{2}$$

where λ is a spectral parameter, $|\alpha_1| + |\alpha_0| \neq 0$, $|\beta_1| + |\beta_0| \neq 0$, $m \neq 0$ are a complex numbers. Such type of spectral problems were investigated in the papers [1], when instead of the boundary conditions $\alpha_1 y'(0) + \alpha_0 y(0) = 0$ and $\beta_1 y'(1) + \beta_0 y(1) = 0$ were taken $y(0) = y(1) = 0$, and in the work [2] were taken $y'(0) = y'(1) = 0$ and theorems on completeness and minimality were proved. We will prove theorems on completeness, minimality, basicity of eigenfunctions of the spectral problem in $L_p[0,1] \bigoplus C$ and $L_p[0,1]$ Lebesgue and $M^{p,\alpha}[0,1] \bigoplus C$ and $M^{p,\alpha}[0,1]$ Morrey type spaces.

This work was supported by the Azerbaijan Science Foundation-Grant No AEF-MGC-2024-2(50)-16/02/1-M-02.

Keywords: discontinuous differential operator, spectral parameter, eigenvalues, eigenfunctions, asymptotic formulas

2020 Mathematics Subject Classification: 34L10, 34L20, 34B24

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A stable second order difference scheme for numerical solution of the source identification problem for telegraph-parabolic equations

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Abstract: We construct a second order of accuracy difference scheme for the approximate solution of a source identification problem for telegraph-parabolic equations with an unknown spacewise dependent source term

$$u''(t) + \alpha u'(t) + Au(t) = p + f(t),\ 0 < t < 1, \tag{1}$$

$$u'(t) + Au(t) = p + g(t),\ -1 < t < 0, \tag{2}$$

$$u(0+) = u(0-),\ u'(0+) = u'(0-),\ u(-1) = \varphi,\ u(\lambda) = \psi,\ -1 < \lambda \leq 1. \tag{3}$$

The problem is considered in a Hilbert space H with a self-adjoint positive definite operator A satisfying $A \geq \delta I$, where $\delta > \frac{\alpha^2}{4} > 0$ (see [1]). A first-order of accuracy difference scheme for this problem was studied in [2].

The main goal of this work is to establish the unique solvability of the constructed second order of accuracy difference scheme and derive stability estimates for its solution. As applications of the abstract results, we obtain stability estimates for the solutions of second order of accuracy difference schemes for the approximate solutions of two source identification problems for telegraph-parabolic equations. We also discuss numerical procedures for implementing these difference schemes and provide an error analysis based on numerical results for simple test problems.

Keywords: Source identification problem, telegraph-parabolic equation, stable difference scheme

2020 Mathematics Subject Classification: 35M10, 35R30, 65M06, 65M32

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Mathematical methods for processing geological data and forecasting prospective ore-bearing areas

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Abstract: In recent years, machine learning methods have been increasingly used in Earth sciences due to their ability to model complex, multidimensional, and nonlinear patterns [1]. Algorithms such as Multilayer Perceptron (MLP) and Extreme Gradient Boosting (XGBoost) demonstrate high efficiency in forecasting and classification tasks. These methods allow for the effective integration of various data types, including geochemical, geological, and remote sensing data, and improve the accuracy of detecting subtle and hidden anomalies.

However, machine learning models often require large datasets and rigorous preprocessing; furthermore, they may be less interpretable from a geological perspective compared to classical geostatistical approaches. Machine learning is successfully applied across various Earth science disciplines, including hydrogeophysics, seismology, and seismic exploration. These studies demonstrate the efficacy of data-driven approaches in identifying significant patterns within complex geological and geophysical datasets. Thus, there is a growing need for a comprehensive comparison between classical interpolation methods and modern machine learning approaches in the context of geochemical anomaly detection. A thorough understanding of the strengths and limitations of each method is essential for selecting the appropriate tools for mineral exploration.

This paper investigates the mathematical aspects of machine learning methods used to detect and predict geochemical anomalies in ore-bearing regions. The research evaluates their effectiveness in terms of accuracy, sensitivity to local changes, and suitability for geological interpretation, while also exploring the potential benefits of combining these approaches.

Keywords:machine learning, geochemical anomalies, mineral exploration, geological data analysis, geospatial data classification, prediction

2020 Mathematics Subject Classification: 68T05, 68T07, 68T10 **References:**

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Maximal regularity conditions of the solution of a second-order differential equation

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Abstract: The following second-order singular differential equation is considered in this submission:

$$L_0 y = -s(x)\left(\rho(x)y'\right)' + r(x)\,y' + q(x)y = f(x), \tag{1}$$

where $x \in \mathbb{R} = (-\infty, +\infty)$, s, r, ρ are smooth functions, and q is continuous, $f \in L_2(\mathbb{R})$. By L we denote the closure in $L_2(\mathbb{R})$ of the operator

$$L_0 y = -s(x)\left(\rho(x)y'\right)' + r(x)y' + q(x)y,$$

defined on the set $C_0^{(2)}(\mathbb{R})$ of twice continuously differentiable functions with compact support. An element $y \in D(L)$ which satisfies the equality $Ly = f$, we call the solution of equation (1).

The singularity of the second-order differential equation (1) means that it is given on an infinite interval and its coefficients are unbounded functions. For equation (1) we consider the following issues:

a) the existence and uniqueness of the solution y,

b) the fulfillment of the following inequality for the solution y:

$$\left\|s\left(\rho y'\right)'\right\|_2 + \left\|ry'\right\|_2 + \left\|(1+|q|)y\right\|_2 \leqslant C\left\|f\right\|_2. \tag{2}$$

Also, we give one application of the estimate (2).

Equation (1) in the case of $s = \rho = 1$ and $r = 0$ is a Sturm-Liouville equation and it has been widely studied. The problem of obtaining inequality (2) for the Sturm-Liouville equation was investigated in the works of B. Everitt and W. Geertz, M. Otelbaev, K.Kh. Boymatov and others. A natural extension of the studies above is to study problems a) and b) for equation (1) in the case when the coefficients s and ρ may be unbounded functions. It is also important to describe the simultaneous influence of all the coefficients s, ρ, r, q on the issues considered, while retaining, as far as possible, their independence from each other.

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Keywords: differential equation, generalized solution, closed operator, existence of solution, uniqueness, maximal regularity

2020 Mathematics Subject Classification: 34A30, 34B40, 34C11

On the asymptotic and uniform difference schemes for hyperbolic perturbation problems

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Abstract: The abstract nonlocal boundary value perturbation problem

$$\begin{cases} \varepsilon^2 u^{''}(t) + Au(t) = f(t), 0 < t < T, \\ u(0) = \alpha u(T) + \varphi, \\ u^{'}(0) = \beta u^{'}(T) + \psi \end{cases}$$

for hyperbolic equations with the self-adjoint positive definite operator A in a Hilbert space H and an arbitrary small $\varepsilon \in (0, \infty)$ parameter multiplying the highest derivative term is considered. The asymptotic solution to the problem is presented. A uniform high-order two-step difference scheme corresponding to this problem is obtained. This work covers the general perturbation theory of uniform difference schemes on hyperbolic partial differential equations. The theoretical results are verified by the numerical implementations using Matlab. The error results of the numerical experiments are presented.

Keywords: Hyperbolic perturbation problems, nonlocal conditions, asymptotic formulas, uniform difference schemes

2020 Mathematics Subject Classification: 47A55, 34E10, 39A30, 35G15

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On a family of spectral problems for fourth-order ODE

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Abstract: We introduce

Definition 1. Denote by $V_1(0,l)$ and $V_2(0,l)$ the Hilbert spaces endowed with the inner products and the norms

$$V_1(0,l): \quad \begin{cases} (u'(x),v'(x))_{L^2(0,l)} \ \forall\, u,v \in \overset{\circ}{W}{}_2^1(0,l), \\ \|u\|_{V_1(0,l)} \equiv \left[(u'(x),u'(x))_{L^2(0,l)}\right]^{1/2}, \end{cases} \tag{1}$$

$$V_2(0,l): \quad \begin{cases} ((u,v)) = (u''(x),v''(x))_{L^2(0,l)} \ \ \forall\, u,v \in \overset{\circ}{W}{}_2^2(0,l) \\ \|u\|_{V_2(0,l)} \equiv \left[((u(x),u(x)))\right]^{1/2}. \end{cases} \tag{2}$$

We consider the following family of spectral problems:

$$\begin{cases} X^{IV}(x) + \mu X(x) = -\lambda X''(x), \quad x \in (0,l), \\ X(0) = X(l) = X'(0) = X'(l) = 0, \ \ \mu \in \mathbb{R}^1, \end{cases} \tag{3}$$

for which the following theorem holds true

Theorem 1. For each fixed value of the parameter $\mu \in \mathbb{R}^1$, the spectral problem (3) has a countable spectrum $\{\lambda_n,\ n \in \mathbb{N}\}$ with a single limit point $\lambda = +\infty$. The corresponding system of eigenfunctions $\{X_n(x),\ n \in \mathbb{N}\} \subset V_2(0,l)$ (2) forms an orthogonal basis in the space $V_1(0,l)$ (1).

Throughout this note we mainly use techniques from our works [1–7]. This theorem is used to construct a basis in the 3-D space of solenoidal functions.

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Keywords: Spectral problem, eigenvalues, eigenfunctions, Stokes operator

2020 Mathematics Subject Classification: 34L05, 34L10, 34M46

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On a method for constructing the Green function of the Dirichlet problem

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Abstract: Let $\Omega \subset \mathbb{R}^n$ be a bounded simply connected domain with smooth boundary $\partial\Omega$.

Dirichlet problem. To find the regular solution of the Laplace equation

$$-\Delta u = f(x), \qquad x \in \Omega, \tag{1}$$

satisfying the boundary condition

$$u\big|_{\partial\Omega} = 0. \tag{2}$$

The function $G(x, y)$ is called the Green's function of the Dirichlet problem if

$$u(x) = \int_\Omega G(x, y) f(y)\, dy, \tag{3}$$

where

$$-\Delta_x G(x, y) = \delta(x - y), \quad G(x, y)\big|_{x\in\partial\Omega} = 0. \tag{4}$$

Theorem 2.1. *The Green function of the Dirichlet problem in the domain of Ω is given by the formula*

$$G(x, y) = \chi_- G^-(x, y) - \chi_+ G^+(x, y), \tag{5}$$

where

$$G^-(x, y) = \varepsilon(x, y) + \int_{\partial\Omega} \frac{\partial}{\partial n^-_\xi} \varepsilon(x, \xi)\varepsilon(\xi, y), \quad x \in \Omega^- = \Omega, \tag{6}$$

$$G^+(x, y) = \int_{\partial\Omega} \frac{\partial}{\partial n^+_\xi} \varepsilon(x, \xi)\varepsilon(\xi, y), \quad x \in \Omega^+ = R^n \backslash \Omega, \tag{7}$$

here $\frac{\partial}{\partial n^-_\xi}$ is the derivative on the external normal to $\partial\Omega^-$, and $\frac{\partial}{\partial n^+_\xi}$ is the derivative on the normal to $\partial\Omega^+$,

$$\chi_- = \begin{cases} 1 & x \in \overline{\Omega^-} \\ 0 & x \in \Omega^+ \end{cases}; \qquad \chi_+ = \begin{cases} 1 & x \in \overline{\Omega^+} \\ 0 & x \in \Omega^- \end{cases} \tag{8}$$

Keywords: Green function, Laplace equation, Dirichlet problem.

2020 Mathematics Subject Classification: 35J05, 35J08, 35J25

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On noncommutative version of the corona theorem

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Abstract: The corona theorem is a result about the spectrum of the bounded holomorphic functions on the open unit disc $\mathbb{D}$, conjectured by Kakutani [1] and proved by L. Carleson [2]. Elementary formulation of corona theorem is that if $f_1, ..., f_n$ is a finite set of functions in $H^(\mathbb{D})$ satisfying $\sum_{j=1}^{n} |f_j(z)| \geq c > 0, \qquad z \in \mathbb{D}$, then there are functions $\{g_j\}_{j=1}^{n}$ in $H^(\mathbb{D})$ with $\sum_{j=1}^{n} f_j(z)g_j(z) = 1, \qquad z \in \mathbb{D}$.

In 1968 Fuhrmann [3] extended Carleson's corona theorem to the finite matrix case. In 1980 Rosenblum [4] and Tolokonnikov [5] proved the corona theorem for infinitely many generators $n =$. This was further generalized to the one-sided infinite matrix setting by Vasyunin in 1981 (see [6]). Finally Treil [7] showed in 1988 that the generalizations stop there by producing a counterexample to the two-sided infinite matrix case. Trent and Zhang [8] extended the matricial corona theorem of Fuhrmann and Vasyunin to more general algebras of functions which satisfy a corona theorem. In [9], Trent and Zhang proved a similar extension of the matricial corona theorem of M. Andersson [11] to the same algebras of functions satisfying a corona theorem.

This report is devoted to a noncommutative version of the corona theorem for finite subdiagonal subalgebras in Arveson's sense and its applications to noncommutative Hardy spaces H^p associated with a finite subdiagonal subalgebra.

Keywords: corona theorem, subdiagonal subalgebra, noncommutative Hardy H^p-spac, finite von Neumann algebra.

2020 Mathematics Subject Classification: Primary 46L51, 47L07; Secondary 47L30.

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Energy method for difference schemes of coupled Klein–Gordon equations with rational terms

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Abstract: In this paper, we study the initial value problem for a coupled system of nonlinear Klein–Gordon equations with rational terms under Neumann boundary conditions. We establish the existence and uniqueness of solution for the difference scheme and prove the associated energy conservation law. A difference scheme of second order accuracy is presented for the approximate solution of this problem. Numerical results for the one-dimensional coupled nonlinear hyperbolic equation are presented. Finally, a numerical example is given to confirm the theoretical analysis.

Keywords: Nonlinear hyperbolic equations, coupled Klein–Gordon system, difference scheme, stability, Hilbert space.
2020 Mathematics Subject Classification: 35L70, 35G50, 35A01, 35A02, 65M06, 65M12.
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A note on interpolation spaces generated by the second order differential operator with dependent coefficient and Samarskii-Ionkin conditions

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Abstract: In this study, we investigate the operator

$$A^x u = -(a(x)u_x)_x + \sigma u(x),\ \sigma \geqslant 1,\ 0 < x < 1,$$

where domain of A^x is

$$D(A^x) = \{u \in C^2[0,1] : u(0) = u(1),\ u'(1) + \mu u(1) = 0\},$$

$\mu > 0$, $a(0) = a(1)$, $a(x) \geqslant \delta_0 > 0$ is continuously differentiable function defined on $[0,1]$.

We construct the Green function of A^x. We also obtain the estimates for the Green function. We prove the positivity ([1]) of differential operator A^x in $C[0,1]$. Furthermore, we investigate the structure of the interpolation spaces $E_\alpha(C[0,1], A^x)$. In addition, we prove that for each $\alpha \in (0, 1/2)$ the norms of the interpolation space $E_\alpha(C[0,1], A^x)$ and the Hölder space $\mathring{C}^{2\alpha}[0,1]$ are equivalent. Hence, we also prove the positivity of A^x in $\mathring{C}^{2\alpha}[0,1]$.

To apply the our theoretical results, we consider the non-local boundary value problems of parabolic type

$$(1)\quad \begin{cases} v_t(x,t) - (a(x)v_x)_x + \sigma v(t,x) = \varphi(x,t),\ \sigma \geqslant 1,\ 0 \leqslant x \leqslant 1,\ 0 < t < T, \\ v(x,0) = \psi(x),\ 0 \leqslant x \leqslant 1, \\ v(0,t) = v(1,t),\ v_x(1,t) + \mu v(1,t) = 0,\ 0 < t < T, \end{cases}$$

and

$$(2)\quad \begin{cases} v_t(x,t) - (a(x)v_x)_x + \sigma v(t,x) = \varphi(x,t),\ \sigma \geqslant 1,\ 0 \leqslant x \leqslant 1,\ 0 < t < T, \\ v(x,0) = v(x,T),\ 0 \leqslant x \leqslant 1, \\ v(0,t) = v(1,t),\ v_x(1,t) + \mu v(1,t) = 0,\ 0 < t < T, \end{cases}$$

where μ is a fixed positive number, φ and ψ are sufficiently smooth functions.

We prove noval coercive inequalities for solution of nonlocal boundary value problems (1) and (2) of parabolic type.

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On a coupled system of semilinear time-fractional Schrödinger equations

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Abstract: This paper investigates a coupled system of time-fractional semilinear Schrödinger equations with disordered coefficients within a general abstract operator framework. This setting naturally accommodates variable-coefficient and nonstandard spatial operators, thereby extending the applicability of fractional Schrödinger models to heterogeneous media and interacting multi-component systems. Existence and uniqueness of mild solutions are established using operator-theoretic techniques under suitable assumptions. A time discretization scheme is constructed and analyzed within the same abstract framework, applicable to multi-dimensional problems. Numerical experiments in one- and two-dimensional configurations, involving variable-coefficient operators and various nonlinear interactions, confirm the theoretical results and demonstrate the robustness of the proposed method.

Keywords: Time-fractional Schrödinger equations, mild solutions, operator framework, semilinear systems, numerical discretization

2020 Mathematics Subject Classification: 35Q55, 65M12, 35R11

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Stability analysis for advection-diffusion equations and their numerical approximation

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Abstract: The advection-diffusion equation provides a fundamental mathematical framework for describing essential transport processes, including natural gas transport in pipelines and environmental fluid flow [1]. In this study, stability estimates for the advectionâ"diffusion equation are derived [2,3]. Rothe and Crank-Nicolson finite difference schemes are constructed [4,5], and their stability properties are established through rigorous theoretical analysis. The results are subsequently extended to one-dimensional advection-diffusion with nonlocal boundary conditions and involution, multidimensional advection-diffusion initial-boundary value problems with Dirichlet and Neumann boundary conditions. The derived stability estimates do not impose restrictions on the ratio of the temporal and spatial step sizes, thereby providing robust performance over a wide range of discretization parameters [6, 7]. Numerical experiments are conducted for two one-dimensional advection-diffusion test problems with Dirichlet and Neumann boundary conditions [8, 9], and the numerical solutions are compared with the exact solutions, with errors evaluated in the L_2 and C norms. The results confirm the theoretical convergence rates, with the Rothe scheme exhibiting first-order accuracy and the Crank-Nicolson scheme exhibiting second-order accuracy.

Keywords: Advection-diffusion equation, Stability estimates, Difference schemes, Rothe scheme, Crank-Nicolson scheme, Initial-boundary value problems

2020 Mathematics Subject Classification: 35K20, 65M06, 65M12 **References:**

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Stability threshold for a delayed discrete-time epidemic model with periodic transmission

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Abstract: We study a nonlinear delayed discrete-time epidemic model with periodic transmission and a saturation effect. Such models are appropriate when epidemic data are recorded at fixed intervals and transmission varies seasonally [1,2]. The model is

$$I_{n+1} = \frac{\alpha_n I_n}{1+\beta I_{n-m}}, \qquad \alpha_n = \alpha\left(1+\varepsilon\cos\frac{2\pi n}{T}\right), \tag{1}$$

where I_n is the infectious proportion, $\alpha > 0$ is the mean transmission rate, $\varepsilon \in (0,1)$ is the seasonal amplitude, T is the period, $\beta > 0$ is the saturation parameter, and $m \geq 0$ is the delay. The disease-free solution is $I_n = 0$. Set $\mathcal{R}_T = \prod_{n=0}^{T-1} \alpha_n$.

Theorem. The disease-free solution of (1) is locally asymptotically stable if $\mathcal{R}_T < 1$ and unstable if $\mathcal{R}_T > 1$.

Proof. Let $U_n = (u_n, \ldots, u_{n-m})^\top$. Since $\alpha_n u_n/(1+\beta u_{n-m}) = \alpha_n u_n - \alpha_n \beta u_n u_{n-m} + O(\|U_n\|^3)$, the delay and saturation terms are absent from the linear part, which is $u_{n+1} = \alpha_n u_n$. In the $(m+1)$-dimensional delayed state space, $U_{n+1} = A_n U_n$, where A_n is the lower triangular companion matrix with diagonal $(\alpha_n, 0, \ldots, 0)$. Hence $M = A_{T-1}\cdots A_0$ has diagonal $(\mathcal{R}_T, 0, \ldots, 0)$, so its Floquet multipliers are $\mathcal{R}_T$ and m zeros. Therefore $\mathcal{R}_T < 1$ gives local asymptotic stability, $\mathcal{R}_T > 1$ gives instability, and $\mathcal{R}_T = 1$ is critical. The delay and saturation parameters do not enter this linear threshold, although they affect the nonlinear dynamics away from zero.

Keywords: discrete-time epidemic model, seasonal transmission, time delay, saturation effect, disease-free solution, Floquet multiplier

2020 Mathematics Subject Classification: 39A30, 39A60, 92D30

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Spectral properties and global dynamics of interior fixed points in four-dimensional tournament Lotka–Volterra operators

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Abstract: We study the discrete Lotka–Volterra operator

$$(Vx)_i = x_i \left(1 + \sum_{j=1}^{5} a_{ij} x_j\right), \qquad i = 1, \ldots, 5,$$

on the simplex S^4, where $A = (a_{ij})$ is a real skew-symmetric matrix. The sign pattern of A determines an associated tournament and the corresponding fixed-point card [1]. We focus on the class of tournaments containing five cyclic triples.

A sufficient condition for the existence of an interior fixed point is obtained in terms of the principal Pfaffians $\theta_1, \ldots, \theta_5$. If they have the same nonzero sign, then

$$x^* = \frac{(\theta_1, \ldots, \theta_5)}{\theta_1 + \cdots + \theta_5} \in \operatorname{ri} S^4$$

satisfies $Ax^* = 0$ and is an interior fixed point.

The spectrum of the Jacobian at x^* has the form

$$\sigma(DV(x^*)) = \{1,\ 1 \pm i\omega_1,\ 1 \pm i\omega_2\}.$$

For $\omega_1\omega_2 \neq 0$, $|1 \pm i\omega_k| > 1, \qquad k = 1, 2$, so the interior fixed point is non-hyperbolic and is not a local attractor. Lyapunov-type functions are then used to exclude convergence of generic interior trajectories to x^* and to relate their asymptotic behavior to the boundary dynamics.

Keywords: Lotka–Volterra operator, tournament, cyclic triples, Pfaffians, interior fixed point, spectrum, global dynamics

2020 Mathematics Subject Classification: 37B25, 37C25, 37C35 **References:**

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Directed transition geometry in an eight-signature Lotka–Volterra system

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Abstract: We study the geometric structure and directed transitions of an eight-signature discrete Lotka–Volterra system on the simplex $S^3 = \{(S,E,I,R) \in \mathbb{R}^4 : S,E,I,R \geq 0,\ S+E+I+R=1\}$. The system is given by

$$(1) \qquad \begin{cases} S^{(n+1)} = S^{(n)}(1 - bE^{(n)} - cR^{(n)}), \\ E^{(n+1)} = E^{(n)}(1 + bS^{(n)} - dI^{(n)} + fR^{(n)}), \\ I^{(n+1)} = I^{(n)}(1 + dE^{(n)} - eR^{(n)}), \\ R^{(n+1)} = R^{(n)}(1 + cS^{(n)} + eI^{(n)} - fE^{(n)}), \end{cases} \qquad b,c,d,e,f > 0.$$

Writing (1) as $X_i^{(n+1)} = X_i^{(n)}(1+\Phi_i(X^{(n)}))$, we define the signature $\delta(X) = (\operatorname{sign}\Phi_1, \ldots, \operatorname{sign}\Phi_4)$. The sign conditions generated by $\Phi_1, \ldots, \Phi_4$ define eight nonempty regions $F_i = \{X \in S^3 : \delta(X) = \delta_i\}$, $i = 1, \ldots, 8$. Since $X_i^{(n+1)} - X_i^{(n)} = X_i^{(n)}\Phi_i(X^{(n)})$, their common faces determine adjacency and the signs of Φ_i determine transition directions. Thus, the main result is a directed adjacency graph describing admissible trajectory routes. Following the epidemiological interpretation of discrete Lotka–Volterra systems in [1], system (1) admits an extended SEIR-type interpretation with the principal interaction chain $S \to E \to I \to R$ and additional immunization and reinfection interactions.

Keywords: discrete Lotka–Volterra operator, signature, adjacency, directed transition, SEIR-type model.

2020 Mathematics Subject Classification: 37B25, 37C25, 37C27

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Solvability of a system of hyperbolic equations with nonlocal conditions

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Abstract: We consider the following nonlocal boundary value problem with two-point and integral boundary conditions for a system of hyperbolic equations in the domain $Q = [0, T] \times [0, l]$:

$$z_{tx} = f(t, x, z(t, x)), \tag{1}$$

$$A_1 z(0, x) + \int_0^T n(t) z(t, x)\, dt + B_1 z(T, x) = \varphi(x), \quad x \in [0, l], \tag{2}$$

$$A_2 z(t, 0) + \int_0^l m(x) z(t, x)\, dx + B_2 z(t, l) = \psi(t), \quad t \in [0, T] \tag{3}$$

where $z(t, x) = z_1(t, x), z_2(t, x), \ldots, z_n(t, x)$ is the unknown n-dimensional vector-function. $f : Q \times \mathbb{R}^n \to \mathbb{R}^n$, $\varphi(x)$, $\psi(t)$ are differentiable n-dimensional vector functions defined on the segments $[0, T]$, $[0, l]$. The functions $n(t)$, $m(x)$ are $n \times n$ matrix-valued functions, $A_1, A_2, B_1, B_2 \in \mathbb{R}^{n \times n}$ are given matrices with $A = A_1 + B_1 + \int_0^T n(t)\, dt$, $\det A \neq 0$ and $B = A_2 + B_2 + \int_0^l m(x)\, dx$, $\det B \neq 0$.

By applying the Banach's contraction mapping principle, sufficient conditions for the existence and uniqueness of a solution to the boundary value problem (1)–(3) are obtained.

Similar problems have been considered in [1].

Keywords: non-local boundary value problems, integral and two-point boundary conditions, Goursat-Darboux problem, existence and uniqueness.

2020 Mathematics Subject Classification: 35G35, 35G46, 35L53, 35L57

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On the numerical solution of an inverse source problem by model order reduction

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Abstract: In this study, an inverse time-dependent source problem for a linear parabolic equation arising in numerous practical applications involving diffusion and transport phenomena is considered. Typical examples of such problems aim to reconstruct the unknown time-dependent heat source in heat conduction processes, the temporally varying pollutant emissions in environmental monitoring, the atmospheric emission sources, *etc.* by using some direct observations. The derivation of numerical solutions of such a problem in high-dimensional spaces leads a huge computational cost. Taking into account that it may be necessary to obtain numerical solutions several times in some cases, it is a crucial task to reduce the computational cost, while preserving the accuracy. Motivated by this, for the numerical solution of the considered problem, we propose a model order reduction method by proper orthogonal decomposition [1,2]. In the method, finite difference method is implemented. Particularly, the time discretization is carried out by the implicit Euler's difference scheme. From the theoretical point of view the convergence rate of the method is deduced. The theoretical results and efficiency of the proposed method are illustrated by testing the method on a typical model problem.

Keywords: Inverse source problem, model order reduction, proper orthogonal decomposition

2020 Mathematics Subject Classification: 65J22, 65M32, 65N21, 65K99

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Comparison principle for a nonlocal fractional diffusion problem

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Abstract: In this paper, we study an initial–boundary value problem for a nonlinear time–space fractional diffusion equation in a smooth bounded domain $\Omega \subset \mathbb{R}^n$. The equation involves the Caputo fractional derivative $\mathcal{D}^{\alpha}_{0|t}$ of order $\alpha \in (0,1)$ (see [1]), the fractional p-Laplacian $(-\Delta)^s_p$, and the nonlinear reaction terms $\gamma|u|^{m-1}u + \mu|u|^{q-2}u$, where $s \in (0,1)$, $p \geq 2$, $m > 0$, $q \geq 1$, and $\gamma, \mu \in \mathbb{R}$.

The problem is equipped with the initial condition $u(x,0) = u_0(x)$ and the nonlocal Neumann-type boundary condition $\mathcal{N}^s_p u = 0$ in $\mathbb{R}^n \setminus \overline{\Omega}$, where $\mathcal{N}^s_p$ is defined by a nonlocal integral flux operator.

Using elementary algebraic inequalities, we establish a comparison principle for different ranges of the parameters γ, μ, m, and q. Based on this principle, we classify the finite-time blow-up of solutions and the existence of global weak solutions.

We also note that the corresponding problem with a Dirichlet boundary condition was studied in [2].

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Keywords: parabolic equations, nonlocal operator, polynomial nonlinearity comparison principle.

2010 Mathematics Subject Classification: 35R11, 35A01, 35B51, 35K55 **References:**

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On an inverse problem for the Poisson equation with an even source term

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Abstract: An inverse problem for the Poisson equation in the unit disk with an even source term depending only on the angular variable is considered. The problem consists in determining both the solution of the equation and the unknown source term from the Dirichlet boundary condition and an additional Neumann boundary condition. The proposed method is based on the decomposition of the solution into even and odd parts with respect to the angular variable, reducing the original inverse problem to the sequential solution of two auxiliary problems, which are solved by the Fourier method. Theorems on the existence and uniqueness of generalized and classical solutions are established.

Problem formulation. Consider the unit disk

$$\Omega = \{(r,\theta) : 0 < r < 1,\ 0 \le \theta \le 2\pi\}, \Gamma = \{(r,\theta) : r = 1,\ 0 \le \theta \le 2\pi\}.$$

Determine a pair of functions $u(r,\theta)$ and $f(\theta)$ satisfying the Poisson equation

$$-\Delta u(r,\theta) = f(\theta), \qquad (r,\theta) \in \Omega. \tag{1}$$

The Dirichlet boundary condition is specified on the boundary of the disk:

$$u(1,\theta) = \tau(\theta), \qquad 0 \le \theta \le 2\pi. \tag{2}$$

As an additional condition, the Neumann boundary condition is imposed on the upper semicircle:

$$\frac{\partial u}{\partial r}(1,\theta) = \nu(\theta), \qquad 0 \le \theta \le \pi. \tag{3}$$

The function $\tau(\theta)$ is assumed to be 2π–periodic. Moreover,

$$f(\theta) = f(2\pi - \theta), \qquad 0 \le \theta \le 2\pi.$$

Theorem 2.1. *Let*

$$\tau \in C^{3+\varepsilon}(\Gamma), \qquad \nu \in C^{2+\varepsilon}[0,\pi], \qquad 0 < \varepsilon \le 1$$

and let the conditions

$$\nu'(0) = \nu'(\pi) = 0$$

hold. Then problem (1)-(3) has a unique classical solution satisfying

$$u \in C^{2+\varepsilon}(\overline{\Omega}), \qquad f \in C^{\varepsilon}(\Gamma).$$

Theorem 2.2. *Let*

$$\tau \in W_2^2(\Gamma), \qquad \nu \in W_2^1[0,\pi].$$

Then problem (1)-(3) has a unique generalized solution satisfying

$$u \in W_2^2(\Omega), \qquad f \in L_2(\Gamma).$$

Keywords: Poisson equation, inverse problem, Sobolev spaces, Hölder spaces

2020 Mathematics Subject Classification: 35J05, 31A25

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Unconditional basis property of root functions of the operator of multiple differentiation on a segment with a spectral parameter in one boundary condition

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Abstract: Let us consider a spectral problem with a linear occurrence of a spectral parameter in one of the boundary conditions:

$$l(y) \equiv -y'(x) = \lambda y(x), \quad 0 < x < 1, \tag{4}$$

$$y'(0) = 0, \quad y'(1) = \lambda \left(y(0) + y(1)\right). \tag{5}$$

The characteristic determinant of the problem (1)-(2) has the form

$$\Delta(\lambda) = \sqrt{\lambda}\left(1 + \cos\sqrt{\lambda}\right) + \sin\sqrt{\lambda} = \cos\frac{\sqrt{\lambda}}{2}\left(\sqrt{\lambda}\cos\frac{\sqrt{\lambda}}{2} + \sin\frac{\sqrt{\lambda}}{2}\right).$$

Solving the equation $\Delta(\lambda) = 0$, we have two series of eigenvalues

$$\lambda_k^{(1)} = ((2k+1)\pi)^2; \;\; \lambda_k^{(2)} = ((2k+1)\pi + \delta_k)^2; \;\; \delta_k = O\left(\frac{1}{k}\right).$$

$\lambda_k^{(1)}$, $\lambda_k^{(2)}$ are simple (one-time) eigenvalues of problem (1)-(2). The system of eigenfunctions of problem (1)-(2) has next form

$$y_k^{(1)} = \cos\left((2k+1)\pi\right)x; \quad y_k^{(2)} = \cos\left((2k+1)\pi + \delta_k\right)x; \tag{6}$$

$$k = 0, 1, 2, 3, \ldots, \quad \delta_k = O\left(\frac{1}{k}\right).$$

Due to the asymptotics of δ_k, when one element is removed from this system, we obtain a complete and minimal system in $L_2(0,1)$. This system does not form an unconditional basis, since for sufficiently large $k \gg 1$ two series of eigenfunctions $y_k^{(1)}$ and $y_k^{(2)}$ "stick together", although according to the results of A.A.Shkalikov [1], this system forms a Riesz basis with brackets in $L_2(0,1)$.

Let us introduce an auxiliary system of functions

$$v_k^{(1)}(x) = y_k^{(1)}(x); \quad v_k^{(2)}(x) = \frac{1}{\delta_k}\left(y_k^{(2)}(x) - y_k^{(1)}(x)\right), \quad k = 0, 2, 3, \ldots,$$

that is, the auxiliary system of functions has the form

$$\begin{gathered} v_k^{(1)}(x) = \cos(2k+1)\pi x; \\ v_k^{(2)} = -\frac{\sin(\delta_k x)}{\delta_k}\sin(2k+1)\pi x + \frac{\cos(\delta_k x) - 1}{\delta_k}\cos(2k+1)\pi x, \\ k = 0, 2, \ldots. \end{gathered} \tag{7}$$

Theorem 1. The system of eigenfunctions (3) of the spectral problem (1)-(2) does not form an unconditional basis in $L_2(0,1)$. The auxiliary system of functions (4) forms an unconditional basis in $L_2(0,1)$.

Remark. Problems of this type naturally arise when solving (by the method of separation of variables) initial boundary value problems for evolutionary equations (in particular, for the equation of heat conductivity), when the boundary condition includes derivatives of the solution with respect to the time variable:

$$u_t = u_{xx}, \quad 0 < x < 1, \quad t > 0,$$

$$u(x,0) = \varphi(x), \quad u_x(0,t) = 0, \quad u_x(1,t) = -\left(u_t(0,t) + u_t(1,t)\right).$$

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Keywords: differentiation operator, eigenfunctions, unconditional basis, boundary condition, spectral parameter

2020 Mathematics Subject Classification: 45A47

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Anisotropic Morrey spaces and the boundedness of the Riesz operator in them

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Abstract: In this paper, we define anisotropic Morrey-type spaces and consider the Riesz-type potential operator acting in them.

Let $k \in \mathbb{Z}$ and denote by G_k the set of all cubes in $\mathbb{R}^d$ of the form $[0, 2^k)^d + 2^k \bar{m}$, $\bar{m} \in \mathbb{Z}^d$.

Assume $\bar{k} = (k_1, ..., k_d)$, $\bar{n} = (n_1, ..., n_d)$, G_{k_i} be the set of cubes in R^{n_i}, $\mathbb{G}_{\bar{k}} = \{Q = Q_1 \times \cdots \times Q_d : \ Q_i \in G_{k_i}\}$. $\mathbb{R}^{\bar{n}} = \mathbb{R}^{n_1} \times \cdots \times \mathbb{R}^{n_d}$, $\bar{x} = (x_1, ..., x_d)$, $x_i \in R^{n_i}$, $i = \overline{1,d}$

Let $\bar{p} = (p_1, ..., p_d)$, $\bar{q} = (q_1, ..., q_d)$, $\bar{\lambda} = (\lambda_1, ..., \lambda_d)$ such that $0 < p_i \leq \infty$, $0 < q_i \leq \infty$, $0 < \lambda_i < \infty$. The anisotropic Morrey-type spaces $M^{\bar{\lambda}}_{\bar{p}}$ are defined as the set of all Lebesgue measurable functions $f \in L^{loc}_{\bar{p}}(\mathbb{R}^{\bar{n}})$ for which $\|f\|_{M^{\bar{\lambda}}_{\bar{p}}} = \sup_{\bar{k} \in \mathbb{Z}^d} \left(2^{-\langle \bar{k}, \bar{\lambda} \rangle} \sup_{Q \in G_{\bar{k}}} \|f\|_{L_{\bar{p}}(Q)}\right) < \infty$.

Let $\bar{\delta} = (\delta_1, \ldots, \delta_d)$, $\bar{n} = (n_1, ..., n_d)$, $\delta_i < n_i$, $i = \overline{1,d}$. Consider the Riesz-type potential operator $I_{\bar{\delta}} f(\bar{x}) = \int_{\mathbb{R}^{|\bar{n}|}} \frac{f(\bar{y})}{|\bar{x} - \bar{y}|^{\bar{n} - \bar{\delta}}} d\bar{y}$. The following theorems holds. Let $\bar{p} = (p_1, \ldots, p_d)$, $\bar{q} = (q_1, \ldots, q_d)$, $\bar{\lambda} = (\lambda_1, \ldots, \lambda_d)$, $\bar{\delta} = (\delta_1, \ldots, \delta_d)$, $\bar{\gamma} = (\gamma_1, \ldots, \gamma_d)$, $\bar{n} = (n_1, \ldots, n_d)$.

Theorem 1. Let $1 < p_i < q_i < \infty$, $0 \leq \lambda_i < \frac{n_i}{q_i}$, $\delta_i = \frac{n_i}{p_i} - \frac{n_i}{q_i}$, $i = \overline{1,d}$. Then $I_{\bar{\delta}}$ is bounded from $M^{\bar{\lambda}}_{\bar{p}}$ to $M^{\bar{\lambda}}_{\bar{q}}$.

Theorem 2. Let $1 < p_i < q_i < \infty$, $1 \leq \lambda_i < \frac{n_i}{p}$, $0 \leq \lambda_i < \frac{n_i}{q_i}$, $0 \leq \gamma_i < \frac{n_i}{p_i}$, $\lambda_i q_i = \gamma_i p_i$, $\delta_i = \lambda_i + \frac{n_i}{p_i} - \frac{n_i}{q_i}$, $i = \overline{1,d}$. Then $I_{\bar{\delta}}$ is bounded from $M^{\bar{\gamma}}_{\bar{p}}$ to $M^{\bar{\gamma}}_{\bar{q}}$.

Keywords: anisotropic Morrey space, Riesz-type potential operator

2020 Mathematics Subject Classification: 35Q79, 35K05, 35K20

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A new non-polynomial fractional hyperbolic spline approach with convergence study for fractional Fredholm integro-differential equations

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Abstract: Fractional Fredholm integro-differential equations arise in many mathematical models involving memory, hereditary effects, and nonlocal interactions. Their numerical treatment is challenging because fractional differential operators and Fredholm integral terms act simultaneously on the unknown solution. In this work, a new non-polynomial fractional hyperbolic spline method is proposed for the numerical solution of Caputo fractional Fredholm integro-differential equations. The spline combines fractional-power terms with modified hyperbolic basis functions and is constructed locally on a uniform partition. Six interpolation conditions are imposed on each subinterval to determine the spline coefficients, while a sequential Caputo fractional continuity condition is used to connect adjacent spline segments. The resulting relations lead to a compact matrix formulation for the numerical scheme. The local coefficient matrix is shown to be nonsingular for sufficiently small mesh sizes, which guarantees unique determination of the local spline coefficients. In addition, the reduced discrete formulation is shown to be well posed, and a convergence analysis establishes uniform convergence of the spline approximation as the mesh size tends to zero. Numerical examples with known exact solutions are used to assess the method. The computed maximum absolute and root mean square errors decrease consistently under mesh refinement, confirming the accuracy and reliability of the proposed approach. The results indicate that the fractional hyperbolic spline provides an effective framework for the numerical approximation of fractional Fredholm integro-differential equations.

Keywords: Non-polynomial fractional spline, hyperbolic spline, fractional Fredholm integro-differential equations, Caputo fractional derivative, numerical approximation, convergence analysis

2020 Mathematics Subject Classification: 45J05, 65R20, 41A15, 65D07, 26A33

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A numerical algorithm based on first and second order of accuracy difference schemes for third order delay partial differential equation with involution

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Abstract: In this paper, first and second order of accuracy difference schemes are developed for solving a third order delay partial differential equation with involution. Illustrative numerical results are presented to demonstrate the accuracy and effectiveness of the proposed schemes..

Throughout this note we mainly use techniques from our works [1].

Keywords: Keywords: Time delay, third order partial differential equations, first- and second-order difference schemes

2020 Mathematics Subject Classification: 35G10, 35L90, 58D25

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Pathwise Fourier continuation for Bermudan and American options: a sinh-accelerated NUFFT–RQMC framework

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Abstract: We develop a pathwise Fourier-continuation framework for Bermudan and American option pricing. The method combines randomized quasi-Monte Carlo path simulation, sinh-accelerated Fourier inversion, and type-3 nonuniform FFT evaluation so that, at each exercise date, continuation values are computed from the one-step characteristic function directly at the simulated states. This replaces the regression fit in least-squares Monte Carlo by a transform-based conditional expectation while retaining a simulation architecture. The framework is formulated for one-factor exponential models with explicit one-step characteristic functions; the continuously monitored barrier treatment considered here is specific to geometric Brownian motion, where a Brownian-bridge survival correction is applied in log space without changing the Fourier continuation step.

Numerical experiments recover standard GBM American-put benchmarks accurately, show competitive bias and RMSE relative to polynomial and Laguerre-basis LSMC on representative cases, and demonstrate the continuation machinery under a pure-jump CGMY model. For continuously monitored up-and-out puts under GBM, the barrier extension gives prices in close agreement with published benchmark values.

Keywords: American options; Bermudan options; Fourier continuation; nonuniform fast Fourier transform; randomized quasi-Monte Carlo; sinh acceleration; barrier options; CGMY model.

JEL classification: C63; G12; G13.

On the positivity of a second-order linear differential operator with constant coefficients in Banach spaces

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Abstract: Positivity of linear differential operators in Hilbert and Banach spaces is a fundamental concept in the analysis of partial differential equations (PDEs). In Hilbert spaces, positivity is defined using the notion of an inner product. This concept can be generalized to Banach spaces, where an inner product is not available. However, establishing positivity in Banach spaces is technically more challenging than in Hilbert spaces [1].

The main goal of this work is to establish sufficient conditions for the positivity of the second-order linear differential operator

$$Av = -\alpha v''(x) + \beta v'(x) + \delta v(x), \quad 0 < x < 1, \tag{1}$$

where α, β, and δ are positive constants, with domain

$$\mathcal{D}(A) = \left\{ v \ : \ v \in C^{(2)}(0,1); \ v(0) = v(1) = 0 \right\}. \tag{2}$$

The positivity is considered in abstract Banach spaces. The obtained result can be applied to the theory and analysis of various initial-boundary value problems for parabolic PDEs.

Keywords: Linear differential operator, positive operator, Banach spaces

2020 Mathematics Subject Classification: 47B01, 47B65

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On the new eighthrder-order of approximation difference schemes for fourth-order differential equations

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Abstract: Two types compact four-step difference schemes of eighth-order of approximation for the numerical solutions of local and nonlocal boundary value problems for fourth-order differential equations with dependent coefficients are studied. These these difference schemes are generated by a new Taylor decomposition on five points [1]. The theoretical statements for the solution of these difference schemes are supported by the results of numerical experiments. The main goal of this work is to establish sufficient conditions for the positivity of the second-order linear differential operator.

Keywords: Taylor's decomposition on five points, local and nonlocal problems, difference schemes, approximation, numerical experiments.

2020 Mathematics Subject Classification: 34B10, 35K10, 49K40

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Method of generalized Jacobi elliptic functions for solving some classes of nonlinear elliptic systems of second-order equations on the plane

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Abstract: In this paper, generalized Jacobi elliptic functions $dn\,\omega(z)-$ delta-amplitude are applied to the construction of particular solutions of certain classes of elliptic systems of partial differential equations of second order on the complex plane C with Cauchy–Riemann operators $2\partial_{\overline{z}} = \partial_x + i\partial_y$, $2\partial_z = \partial_x - i\partial_y$, and the Bitsadze operator $4\partial^2_{z\,\overline{z}} = 4\partial_{\overline{z}}(\partial_{\overline{z}}) = \partial^2_{xx} - \partial^2_{yy} + 2i\partial^2_{xy}$, where the variables $z = x + iy$, $\overline{z} = x - iy$ are independent.

The function $\omega(z)$ is a quasi-periodic homeomorphism of the Beltrami equation

$$\omega_{\overline{z}} - q\omega_z = 0, \quad |\,q\,| \neq 1,$$

satisfying the condition

$$\omega(0) = 0, \quad \omega(z + h_j) = \omega(z) + \tau_j, \quad j = 1, 2,$$

where $\mathrm{Im}(h_2/h_1) \neq 0$, $\mathrm{Im}(\tau_2/\tau_1) \neq 0$.

The function $\omega(z)$ quasiconformally maps any parallelogram Ω with vertices z_0, $z_0 + h_1$, $z_0 + h_1 + h_2$, $z_0 + h_2$ onto a quadrilateral Ω' with vertices $\omega(z_0)$, $\omega(z_0) + \tau_1$, $\omega(z_0) + \tau_1 + \tau_2$, $\omega(z_0) + \tau_2$, and for $|\,q\,| < 1$ the orientation of the boundaries of Ω and Ω' is preserved.

Keywords: solution of equations, modulus of elliptic function, Jacobi elliptic functions, analytic function.

2020 Mathematics Subject Classification: Primary 35A09; Secondary 35B36

Numerical solution of the inverse coefficient problem for an elliptic equation

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Abstract: This report is devoted to the inverse recovery problem of the coefficient $c(y)$ and solving $u(x,y)$ from the equation

$$-\Delta u(x,y) + c(y)u(x,y) = f(x,y), \quad (x,y) \in \Omega \subset R^2, \tag{1}$$

with Robin's condition

$$\frac{\partial u}{\partial n}(x,y) = -q(x,y)u(x,y) + g(x,y) \tag{2}$$

and additional data

$$u\rfloor_{\partial\Omega} = U(x,y), \quad (x,y) \in \partial\Omega \tag{3}$$

The uniqueness theorem of the solution of the inverse problem is proved. A finite-difference scheme and an iterative gradient algorithm based on the sequential solution of direct and adjoint problems are constructed. The results of numerical experiments at different noise levels in the input data are presented. The numerical algorithm is based on the approach described in [1].

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Keywords: coefficient inverse problem, Poisson equation, Robin boundary condition, finite-difference method, adjoint problem, gradient method.

2020 Mathematics Subject Classification: 35R30, 65N21

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Unique determination of the time-dependent source factor and the order of a fractional derivative with an unknown initial value

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Abstract: This paper considers the inverse problem of simultaneously recovering the fractional order $\alpha \in (0,1)$ and the time-dependent source factor $p(t)$ in a boundary-value problem for the subdiffusion equation with a Dirichlet Laplacian. The overdetermination condition is defined by the scalar product $(u(t),\psi)_X$ for $0 < t < T$, where $\psi \in D(A)$ is an arbitrary fixed element.

Uniqueness is established independently of the specific form of the initial function.

On a class of a loaded overdetermined system of second-order differential equations with regular coefficients

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Abstract: We consider a system of two loaded differential equations: one linear second-order hyperbolic equation and one first-order differential equation with regular coefficients. Under certain conditions on the coefficients of these equations, we obtain a representation of the variety of solutions via two arbitrary constants and prove the unique solvability of the initial value problem.

Key words: overdetermined system, solution manifold, parallelepiped, weak singular coefficient.

An approximation operator for the Haar wavelet spaces

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Abstract: This report is devoted to constructing a concrete approximation operator for multivariate functions. The functions can be written as the Fourier–Haar series and their Fourier–Haar coefficients decay rapidly. We call a set of such functions a Haar wavelet space. We show that the classical Besov and Sobolev spaces with a dominant mixed derivative is continuously embedded into the Haar wavelet space. Thus, the approximation operator can be applied to Besov and Sobolev spaces.

We also investigate the convergence rate of the approximation operator for the Haar wavelet spaces. Due to the embeddings of the Besov and Sobolev spaces with a dominant mixed derivative we also receive the convergence rate for these spaces.

Keywords: multi-dimension, function approximation, Besov space, Sobolev space.

2020 Mathematics Subject Classification: 41A25, 41A63

The BD property of a class of high-order singular difference operators

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Abstract: This report is devoted to the discreteness of the spectrum and its boundedness from below for a high-order one term difference operator with arbitrary positive coefficients.

Let $n \in \mathbb{N}$. Let $\rho = \{\rho_k\}_{k=0}^{\infty}$, $v = \{v_k\}_{k=0}^{\infty}$ be positive real number sequences.

Let us consider the symmetric difference expression

$$ly_{k+n} = (-1)^n \frac{\bar{\Delta}^n(\rho_k \Delta^n y_k)}{v_k},$$

where $\Delta y_k = y_{k+1} - y_k$, $\bar{\Delta} y_k = y_k - y_{k-1}$, $\Delta^n y_k = \Delta(\Delta^{n-1} y_k)$, $\Delta^0 y_k \equiv y_k$, $k \in \mathbb{N}$ and similarly for $\bar{\Delta}^n y_k$.

Let $N \in \mathbb{N}$, and E_N be the set of sequences of real numbers $y = \{y_k\}_{k=N}^{\infty}$ such that $y_N = y_{N+1} = \cdots = y_{N+n-1} = 0$. Let $\mathring{E}_N \subset E_N$ and for any $y \in \mathring{E}_N$ there exists a number $M(y) \in \mathbb{N}$ such that $y_k = 0$ for all $k \geq M(y)$.

Let $\mathbb{N}(N) = \{k \in \mathbb{N} : k \geq N\}$ and $l_{2,v}$ be the Hilbert space of sequences $f = \{f_k\}_{k=0}^{\infty}$ with an inner product $(f,g)_{2,v} = \sum_{i=0}^{\infty} f_i g_i v_i$, $f, g \in l_{2,v}$, and with norm $\|f\|_{2,v} = \left(\sum_{i=N}^{\infty} v_i |f_i|^2\right)^{\frac{1}{2}}$.

We determine the following operator L_{min} with domain $\mathcal{D}(L_{min}) = \mathring{E}_N$

$$L_{min} y_{k+n} = l y_{k+n} = (-1)^n \frac{\Delta^n(\rho_k \Delta^n y_k)}{v_k}, \quad \forall y \in \mathcal{D}(L_{min}),$$

where outer $\Delta^n \equiv \bar{\Delta}^n$.

The operator L_{min} is symmetric in the space $l_{2,v}$. Denote its closure in $l_{2,v}$ by $\bar{L}_{min}$. It is known that all self-adjoint extensions L of the operator $\bar{L}_{min}$ have similar spectral properties.

Here, we investigate the discreteness of the spectrum of the operator L and its boundedness from below. According to the terminology introduced in [1], this property is called the BD property.

Our main result reads as follows.

Theorem 1. The operator L has the BD property if and only if

$$\lim_{N \to \infty} \sup_{s \geq N} \sum_{j=N}^{s} \rho_j^{-1} \left[(s-j)^{(n-t)}\right]^2 \sum_{k=s}^{\infty} v_{k-(n-1)} \left[(k-s)^{(t-1)}\right]^2 = 0 \tag{1}$$

for all $t = 1, 2, \ldots, n$, where $k^{(i)} = k(k-1)\cdots(k-i+1)$.

Keywords: difference operator, spectral properties, BD property, roundabout theorem, variational method.

2020 Mathematics Subject Classification: 39A70, 34C10, 26D15

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Stability analysis of multipoint nonlocal problems for third-order partial differential equations

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Abstract: This paper investigates a finite multipoint nonlocal boundary value problem for a third-order differential equation in a Hilbert space. Assuming that the governing operator is self-adjoint and positive definite, we derive an explicit condition on the multipoint coefficients that guarantees the bounded invertibility of the operator associated with the nonlocal conditions. This result allows the solution to be represented by means of cosine and sine operator functions. Using this representation, we prove the existence and uniqueness of the solution and derive stability estimates with respect to the given data. The proposed formulation extends earlier single-point nonlocal problems to a finite multipoint setting, allowing several temporal evaluation points and their corresponding coefficients. The abstract results are then applied to one-dimensional and multidimensional third-order partial differential equations subject to periodic, Dirichlet, and conormal boundary conditions. These applications demonstrate the direct applicability of the operator-theoretic results to several classes of evolution problems.

Keywords: Third-order partial differential equations, multipoint nonlocal boundary conditions, stability estimates, cosine and sine operator functions, operator methods, Hilbert spaces.

2020 Mathematics Subject Classification: 34B10, 335G15, 47D09.

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Two-sided estimates of the norm of a elliptic-hyperbolic type operator with Parabolic Degeneration

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Abstract: The aim of the work is to construct a solvability theory (invertibility, separability, resolvent compactness) and to obtain sharp two-sided estimates for the s-numbers and eigenvalues of the operator $Lu = k(y)u_x x - u_y y + b(y)u_x + q(y)u$ on the cylinder $\omega = \|x\| \leq \pi, y \in R$ with periodic conditions in x, where $k(y)$ is an arbitrary piecewise continuous (including oscillating) function and $b(y)$, $q(y)$ are unbounded as $|y| \to \infty$.

The continuous invertibility of $L + \mu I$ is proved, and its resolvent is represented as a series over the resolvents of l_n. Under slow-variation conditions on b, q and the relation $q \leq C_0 b^2$, separability of the operator is established. A compactness criterion for the resolvent is obtained. A two-sided estimate for the counting function $N(\lambda)$ of singular numbers is derived in terms of the level-set measures of $Q_n(y)$, $K_n(y)$. For a model example ($k(y) = sin(e^{10|y|}), b(y) = q(y) = |y| + 1$), explicit power-type two-sided s-number estimates are obtained, and, via Weyl's inequality, eigenvalue estimates.

Throughout this note we mainly use techniques from our works [1].

Keywords: Invertibility, separability, compactness of the resolvent, eigenvalues, s-numbers, two-sided estimates

2020 Mathematics Subject Classification: 34B24, 34L05, 35M10, 47A10

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Two-weight estimates for integral operators with power-logarithmic singularities, $p > q$

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Abstract: Let $I = (0, \infty)$ and the functions v, u are positive almost everywhere, locally integrable on the interval I. Let $1 < p, q < \infty$ and $p' = \frac{p}{p-1}$. Let us denote by $L_{q,v} \equiv L_q(v, I)$ the set of functions f measurable on I for which $\|f\|_{q,v} = \left(\int\limits_0^\infty |f(x)|^q v(x)dx\right)^{\frac{1}{q}} < \infty$. Let W is a positive, strictly increasing and locally absolutely continuous function on the interval I. Assume $\frac{dW(x)}{dx} = w(x)$, for almost all $x \in I$. We consider the operator $T_{\alpha,\beta}$, where $\alpha > 0$, $\beta \geq 0$:

$$T_{\alpha,\beta}f(x) = \int\limits_0^x \frac{\left(ln\frac{W(x)}{W(x)-W(s)}\right)^\beta}{(W(x)-W(s))^{1-\alpha}} u(s)f(s)dW(s), \quad x \in I.$$

Theorem. Let $0 < \alpha < 1$, $0 < q < p < \infty$, $p > \frac{1}{\alpha}$ and $\beta \geq 0$. Let the function u be nonincreasing on I. Then the operator $T_{\alpha,\beta}$ is bounded from $L_{p,w}$ to $L_{q,v}$ if and only if

$$B_{\alpha,\beta} = \left(\int\limits_0^\infty \left(\int\limits_0^z W^{p'\beta}(s)u^{p'}(s)w(s)ds\right)^{\frac{q(p-1)}{p-q}} \times \right.$$

$$\left. \times \left(\int\limits_z^\infty v(x)W^{q(\alpha-\beta-1)}(x)dx\right)^{\frac{q}{p-q}} W^{q(\alpha-\beta-1)}(z)v(z)dz\right)^{\frac{p-q}{pq}} < \infty,$$

and $\|T_{\alpha,\beta}\| \approx B_{\alpha,\beta}$, where $\|T_{\alpha,\beta}\|$ is the norm of the operator $T_{\alpha,\beta}$ from $L_{p,w}$ to $L_{q,v}$.

Keywords: Boundedness; Weight function; Logarithmic singularity

2020 Mathematics Subject Classification: 26A33; 26D10; 47G10

An interpolation theorem for Lorentz spaces

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Abstract: In this work we investigate a real interpolation method with parameter functions. In particular, an interpolation theorem for Lorentz spaces is obtained.

Let $0 < \bar{q} = (q_1, q_2) \leq \infty$, and $\overline{\varphi} = (\varphi_1, \varphi_2)$, where $\varphi_i > 0$, $i = 1, 2$. The anisotropic Lorentz space $\Lambda^{\bar{q}}(\bar{\varphi})$ consists of all measurable functions f such that the quantity

$$\|f\|_{\Lambda^{\bar{q}}(\bar{\varphi})} := \left(\int_0^{+\infty} \left(\int_0^{+\infty} (f^{*_1 *_2}(t_1, t_2) \varphi_1(t_1) \varphi_2(t_2))^{q_1} \, \frac{dt_1}{t_1} \right)^{\frac{q_2}{q_1}} \frac{dt_2}{t_2} \right)^{\frac{1}{q_2}}$$

is finite. Here $f^{*_1 *_2}$ is the repeated nonincreasing rearrangement of the function f‘ (see [1]).

Theorem 2.3. *Let* $0 < \bar{p}_0 < \bar{p}_1 < \infty$, $0 < \bar{q}_0 < \bar{q}_1 < \infty$, $0 < \bar{\tau} \leq \infty$, $\gamma_i = \frac{1}{p_i^0} - \frac{1}{p_i^1}$, $\beta_i = \frac{1}{q_i^0} - \frac{1}{q_i^1}$, $i = 1, 2$ *and* $\bar{\varphi} = (\varphi_1, \varphi_2)$, $\varphi_i > 0$, $\varphi_i \in C$, $i = 1, 2$,
$E = \{\bar{\varepsilon} = (\varepsilon_1, \varepsilon_2) : \ \varepsilon_1 = 0, 1, \ \varepsilon_2 = 0, 1\}$. *If* T *is a quasilinear operator such that*

$$\|Tf\|_{L_{\bar{q}_\varepsilon, \infty}} \leq M_\varepsilon \|f\|_{L_{\bar{p}_\varepsilon}}, \quad \varepsilon \in E \ \textit{with norm} \ M_\varepsilon.$$

Then the operator T *is bounded from* $\Lambda^{\bar{q}}(\overline{\psi}_1)$ *to* $\Lambda^{\bar{q}}(\overline{\psi}_2)$, *more precisely, the inequality*

$$\|Tf\|_{\Lambda^{\bar{\tau}}(\bar{\psi}_2)} \leq c_1 \max_{\varepsilon \in E} M_\varepsilon \|f\|_{\Lambda^{\bar{\tau}}(\bar{\psi}_1)}$$

holds, where

$$\overline{\psi}_1(t_1, t_2) = \left(\frac{t_1^{\frac{1}{p_1^0}}}{\varphi_1(t_1^{\gamma_1})}, \ \frac{t_2^{\frac{1}{p_2^0}}}{\varphi_2(t_2^{\gamma_2})} \right) \quad \textit{and} \quad \overline{\psi}_2(t_1, t_2) = \left(\frac{t_1^{\frac{1}{q_1^0}}}{\varphi_1(t_1^{\beta_1})}, \ \frac{t_2^{\frac{1}{q_2^0}}}{\varphi_2(t_2^{\beta_2})} \right).$$

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Keywords: Lorentz spaces, interpolation methods, interpolation theorem

2020 Mathematics Subject Classification: 46B70, 46E30

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Keywords: Lebesgue and Lorentz spaces, interpolation methods, interpolation theorem

2010 Mathematics Subject Classification: 46B70, 46E30

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Data-driven modeling and low-rank approximation of the dielectric response of PVC prepared under electric fields

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Abstract: Permittivity was recalculated from 34,848 capacitance–loss pairs for poly(vinyl chloride) samples cast under DC voltages [1]. The data span two configurations, 11 voltages, 33 temperatures and 48 frequencies. Screening excluded frequencies below 250 Hz, nonphysical entries, 400 K and the heating-sensitive 5 kV condition, leaving 23,590 primary points.

For $f \geq 250$ Hz and 0–4.5 kV, the corrected Akaike information criterion (AICc) [2] preferred linear over quadratic field models in 2,366 of 2,368 cells (99.92%). The median $R^2 = 0.26$ indicates scattered data: quadratic curvature was rarely justified.

In 80 spectra (330, 350, 370 and 390 K), Havriliak–Negami, Cole–Cole and Cole–Davidson [3] were selected in 43, 20 and 17 cases; Debye was not selected. With eight 5 kV sensitivity spectra included, the totals were 47, 23, 18 and 0.

Singular value decomposition [4] of temperature–frequency field-sensitivity matrices retained 99.33% and 99.65% of the variance with two modes for the base and inner-wall configurations. Repeated ten-fold entry-wise cross-validation gave median relative prediction errors of 7.86% and 4.83%, respectively. Two modes captured most of the variation in field sensitivity.

Keywords: Dielectric spectroscopy, model selection, low-rank approximation, singular value decomposition, poly(vinyl chloride).

2020 Mathematics Subject Classification: 65F55, 62P35

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On singular solutions of one boundary value problem with an internal condition for the Lavrent'ev-Bitsadze equation

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Abstract: In the finite domain $\Omega \subset R^2$, bounded at $y > 0$ by the arc of the circle $\sigma_\delta = \left\{(x,y):\ \left(x-\frac{1}{2}\right)^2 + (y+\delta)^2 = \frac{1}{4}+\delta^2,\ y>0\right\}$, and at $y<0$ - by characteristics $AC:\ x+y=0$ and $BC:\ x-y=1$, the following problem for the homogeneous Lavrent'ev-Bitsadze equation is considered

$$sgny\, u_{xx} + u_{yy} = 0, \tag{1}$$

$$u|_{\sigma_\delta} = 0, \tag{2}$$

$$u(x,0) = \frac{1-\alpha}{1+\alpha}\int\limits_0^x u_y(t,0)dt, \tag{3}$$

$$u(x,+0) = u(x,-0), \quad u_y(x,+0) = u_y(x,-0). \tag{4}$$

The following theorem is proved.

Theorem 1. There are an infinite number of singular solutions of the problem (1)-(4), defined by the formula

$$u_k(x,y) = \begin{cases} C\cdot Re\left(\left(\frac{1-\alpha}{1+\alpha}-i\right)\left(\frac{1-x+iy}{(1-x)^2+y^2}-1\right)^{\mu_k}\right), & y>0, \\ C\cdot\left(\frac{1}{1+\alpha}\left(\frac{x+y}{1-x-y}\right)^{\mu_k} - \frac{\alpha}{1+\alpha}\left(\frac{x-y}{1-x+y}\right)^{\mu_k}\right), & y<0, \end{cases} \tag{5}$$

where $C-const$, $\mu_k = \frac{\gamma_\alpha+\frac{\pi}{2}+\pi k}{\gamma_\delta}$, $k=0,1,2,\ldots$, $\gamma_\alpha = arcctg\,\frac{1-\alpha}{1+\alpha}$, $\gamma_\delta = arcctg\,2\delta$, $0<\gamma_\delta<\pi$, and the problem itself is ill-posed in the class of discontinuous functions.

This research was funded by the Science Committee of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant No. AP23488701).

Keywords: singular solution, boundary value problem, mixed type equation, internal condition, matching condition

2020 Mathematics Subject Classification: 35L80, 35M10, 35N30, 33C05

Stability of difference schemes for the space-dependent source identification problem in the two-dimensional neutron transport equation with periodic boundary conditions

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Abstract: This paper investigates the space-dependent source identification problem [1] for the two-dimensional neutron transport equation [2] with spatially variable coefficients and periodic boundary conditions. A first-order finite difference scheme [3] is constructed for the numerical approximation of the problem. The stability properties of the corresponding discrete transport operator are analyzed, and stability estimates for the difference solution are established in appropriate grid-function spaces. Using these estimates, the inverse problem of recovering an unknown space-dependent source term is studied. The results provide a theoretical foundation for the reliable numerical treatment of inverse source problems arising in neutron transport models.

Keywords: Space-dependent source identification problem, two-dimensional neutron transport equation, finite difference scheme, inverse source problem, stability estimates, periodic boundary conditions, spatially variable coefficients.

2020 Mathematics Subject Classification: 35R30, 82D75, 65M32.

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Efficient computation of graph polynomials on cographs

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Abstract: Let $G = (V, E)$ be a simple graph. It is called a cograph if it can be formed by recursively applying join (∇) and union ($\cup$) operations. A graph polynomial is an invariant on graphs that is preserved under isomorphism. We develop an $O(N^2 \log N)$ algorithm to compute several graph polynomials for cographs. For their computation, we use the notion of graph operators:

Theorem 2.4. *The following graph polynomials can be computed for a cograph G using corresponding graph operators in $O(N^2 \log N)$ time complexity:*

(1) *The matching polynomial $\mu_G(t)$ with the operator $\phi_\mu = \exp(\partial_t^2/2)$,*
(2) *The path-cover polynomial $\pi_G(t)$ with the path-cover operator $\phi_\pi = \exp(t\partial_t^2)$,*
(3) *The chromatic polynomial $\chi_G(t)$ with the chromatic operator $\phi_\chi[t^n] = (t)_n$ defined on standard basis elements as the falling factorial,*
(4) *The polynomial of Pfaffian matchings $\theta_G(t)$ with the Pfaffian matchings operator $\phi_\theta = \exp(\partial_{t^2})$.*

Given a graph polynomial, it is possible to extract various useful information about the graph. For example, it is possible to obtain the number of perfect matchings, Hamiltonian cycles, and Hamiltonian paths, as well as the Pfaffians. Earlier, enumeration of Hamiltonian cycles in the complete multipartite graphs was considered by Horak and Tovarek [1], where they give a polynomial-time algorithm for fixed m.

Keywords: Cographs, graph polynomials, matchings, graph colorings, Hamiltonian paths, Hamiltonian cycles, determinant

2020 Mathematics Subject Classification: 05C31, 33C45, 05C38, 05C85

Acknowledgements. This research was funded by the Science Committee of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant No. BR 28713025).

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Stable difference schemes for time-dependent source identification in the Schrodinger equation

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Abstract: In this study, single-step absolute stable difference schemes for the approximate solutions to the time-dependent source identification problem for Schrödinger equations are investigated. The first and second order of accuracy implicit and r-modified Crank-Nicolson difference schemes are presented. The stability of these difference schemes are established. In applications, theorems on the stability inequalities for the numerical solutions of several types of source identification problems for Schrödinger equations are proved.

Keywords: Schrödinger equation, source identification problem, stability, Hilbert space.

2020 Mathematics Subject Classification: 35N05, 35N10, 35N20, 35N30.

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Inequalities for the Fourier transforms

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Abstract: We survey old and new estimates for the Fourier transforms in various function spaces. In particular, we discuss a solution to an old problem by Kolmogorov.

Keywords: Fourier transform/series

2020 Mathematics Subject Classification: 42

Riemann-Hilbert problem for a fourth-order elliptic equation in a quadrant

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Abstract: Let the domain D be bounded by a simple smooth contour Γ, and let $n(t) = n_1(t) + in_2(t)$ be the inward normal to Γ at the point t. It is assumed that the contour Γ belongs to the class $C^{3,\nu}$ with some $0 < \nu < 1$. The latter means that the function $n(t)$ with respect to the arc length parameter on Γ belongs to the class $C^{2,\nu}$.

Consider in this domain the elliptic equation of the fourth order

$$\frac{\partial^4 u}{\partial y^4} - a_1 \frac{\partial^4 u}{\partial x^4} - a_2 \frac{\partial^4 u}{\partial x^3 \partial y} - a_3 \frac{\partial^4 u}{\partial x^2 \partial y^2} - a_4 \frac{\partial^4 u}{\partial x \partial y^3} + \sum_{i+j \le 3} a_{ij} \frac{\partial^3 u}{\partial x^i \partial y^j} = f^0, \tag{1}$$

with coefficients $a_i \in \mathbb{R}$ and $a_{ij} \in C^{\mu}(\overline{D})$, $0 < \mu < \nu$, and the right-hand side $f^0 \in C^{\mu}(\overline{D})$. Its solution is sought in the class of functions $u \in C^4(D) \cap C^{3,\mu}(\overline{D})$. For equation (1), the boundary value problem is posed

$$u^+ = f_1, \quad \left(\frac{\partial^3 u}{\partial n^3}\right)^+ = f_2. \tag{2}$$

In the general case of 2l-th order equations of this type of problem from the Fredholm point of view were studied in the work [1], when the simply connected domain D is bounded by a simple smooth contour Γ. Weighted Holder spaces were defined and studied in detail in [2,3] for functions defined on arbitrary sets.

Theorem 2.5. *Theorem 1. Problem (1), (2) is Fredholm in the class $C^{3,\mu}(\overline{D}) \cap C^4(D)$ if and only if the matrix-function $G = C\,B$ is invertible. Under this condition, the index of the problem is given by the formula*

$$æ = -2\mathrm{Ind}\, G + 8. \tag{3}$$

The report will also discuss the issue of Fredholm solvability of problem (1), (3) when the region under consideration is a region with a non-compact piecewise smooth boundary, namely in a quadrant of the plane.

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Keywords: fourth-order elliptic equations, generalized Neumann problem, J-analytic functions, Riemann-Hilbert problem.

2010 Mathematics Subject Classification: 35J25, 35J40, 35J45

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On finite lattices generating non-finitely based non-standard quasivarieties

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Abstract: There are two closely related problems in lattice theory: which finite lattices generate finitely based quasivarieties and which finite lattices generate standard quasivarieties. The aim of this talk is to contribute to the study of both problems.

We present sufficient conditions under which a quasivariety generated by a finite lattice is neither finitely axiomatizable nor standard. Based on these conditions, we construct and discuss several examples of finite lattices that generate quasivarieties with these properties.

Keywords: modular lattice, quasivariety, finite basis problem, standardness problem, profinite algebra, topological algebra

2020 Mathematics Subject Classification: 06C99, 08C15.

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Formula-definable classes of models

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Abstract: This report outlines one of the research areas pursued at the Department of Algebra and Geometry of L.N. Gumilyov Eurasian National University: the continued investigation into the classification of classes of models, including the study of the properties of formula-definable classes of models.

One of the methodological aspects of model theory is the classification of classes of models based on properties expressible by sentences of a language that are true in all models within the class. A set Φ of sentences of a language L (with a given signature) defines a class of models K: a model A belongs to K if and only if A is a model for every sentence in the set Φ. The set of sentences Φ is called the theory of this class of models, and the class itself is termed an axiomatizable class; in other words, the set of sentences Φ "recognizes" which models belong to the class K.

Definition 2.6 ([1])**.** *A class of models K is called a formula-definable class of models if there exists a model A such that, for any model B, $B \in K$ if and only if* $\mathrm{Th}(B) \cdot \mathrm{Th}(A) = \mathrm{Th}(A)$. *In this case, the model Aâ"and sometimes the elementary type* $\mathrm{Th}(A)$ *of the model Aâ"is called a definer of the class K. If the definer of a formula-definable class K is an idempotent model, then the class K is called idempotently formula-definable.*

In other words, model Aâ"specifically its elementary typeâ"identifies which models belong to the formula-definable class K, effectively doing so through the operation of the Cartesian product of elementary types. This property of a formula-definable class of models essentially reveals the algebraic nature of such a class.

It has been proven that a formula-definable class of models is an axiomatizable class of models. Various classifications of formula-definable classes of models are considered.

Keywords: formula-definable class, axiomatizable class, model theory, elementary type, definer, idempotent model

2020 Mathematics Subject Classification: 03C52, 03C95, 03C07

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Boas type inequalities

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Abstract: Boas' conjecture (1967) identifies, for an even monotone function on the line, exactly which power-weighted integrability of the function is equivalent to a given power-weighted integrability of its Fourier transform. It was proved by Sagher for monotone functions and extended by Liflyand and Tikhonov to general (not necessarily monotone) functions of bounded variation satisfying a general-monotonicity condition; its extension to radial functions on ^{n} is due to Gorbachev, Liflyand and Tikhonov. We study the natural substitute for "non-radial" in this circle of results: functions on $^{d}_{0}$ whose bounded variation is understood through mixed differences, in the sense of Vitali and Hardy, and whose general monotonicity is imposed coordinate-wise, so that the resulting inequalities carry product power weights. We formulate the corresponding Boas-type equivalence in ^{d} and prove it for all functions in the class, of either sign, non-product functions included: the two weighted norms are comparable, with constants depending only on the class, whenever the function-side norm is finite, and for integrable functions the finiteness of either side implies that of the other. The one-sided (Pitt) inequality is available from the author's coordinate-wise theory and is recalled here; the reverse inequality is obtained by bounding $|f|$ on small boxes by a signed average plus variation terms, relating the signed averages to the Fourier transform through a single Parseval identity, and absorbing the variation terms, which carry a small factor, by means of iterated one-dimensional Hardy inequalities – the range of exponents being exactly the product A_p range. The same argument gives the two-sided estimate for a class in which the variation is controlled by the primitive of f, the continuous analogue of a class recently introduced for double Fourier series. We also show that the two-sided statement, unlike the one-sided one, is specific to the classical general monotone class.

Keywords: Boas' conjecture, Fourier transform, general monotone functions, bounded variation in the sense of Vitali and Hardy; product weights, Pitt inequality

2020 Mathematics Subject Classification: 42B10, 26B30, 26D10, 42A38, 46E30

Direct and inverse problems for the distributed order diffusion equation

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Abstract: In this report, we consider the distributed order diffusion equation

$$\left(D_{0y}^{[\mu]} - \Delta_x\right) u(x,y) = f(x,y), \tag{1}$$

where $D_{0y}^{[\mu]}$ is the differentiation operator of distributed order associated with the Borel measure μ, defined as [1]

$$D_{0y}^{[\mu]} u(x,y) := \int_{\mathbb{R}} D_{0y}^{t}\, u(x,y)\, \mu\,(dt)\,;$$

Δ_x is the Laplace operator with respect to $x = (x_1, x_2, ..., x_n) \in \mathbb{R}^n$. In (1) it is assumed that $0 < \sup \operatorname{supp}(\mu) < 1$.

The report discusses the solvability of a class of direct and inverse problems for the equation (1). In particular, we find a representation for the fundamental solution of (1) in terms of the Wright function with distributed parameters [2], construct solutions of the initial value problem in an infinite layer, investigate the question of the uniqueness of solutions in the class of rapidly growing functions, and also investigate the inverse problem of recovering the measure for (1).

Keywords: initial value problem, distributed order diffusion equation, differentiation operator of distributed order, measure recovery, inverse problem

2020 Mathematics Subject Classification: 26A33, 35R11, 35R30

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On grand Lorentz sequence spaces

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Abstract: Nursultanov, Rafeiro and Suragan [2] introduced the grand Lorentz space $GL^{\theta}_{p,q}(\Omega)$, with aggrandisation of the index p. We introduce its discrete analogue (cf. [1]). Let $a^* = \{a^*_n\}_{n\in\mathbb{N}}$ be the non-increasing rearrangement of a sequence $a = \{a_k\}_{k\in\mathbb{Z}}$ and let $\ell_{p,q}$ be the Lorentz sequence space [3].

Definition 1. Let $\theta \in \mathbb{R}$, $0 < p, q < \infty$. The grand Lorentz sequence space $Gl^{\theta}_{p,q}$ is the set of sequences a with finite quasi-norm

$$\|a\|_{Gl^{\theta}_{p,q}} = \begin{cases} \sup\limits_{0<\varepsilon<1/p} \varepsilon^{\theta} \big(\sum_{n=1}^{\infty} (n^{\frac{1}{p}-\varepsilon} a^*_n)^q \frac{1}{n}\big)^{1/q}, & \theta \geq 0, \\ \inf\limits_{0<\varepsilon<1} \varepsilon^{\theta} \big(\sum_{n=1}^{\infty} (n^{\frac{1}{p}+\varepsilon} a^*_n)^q \frac{1}{n}\big)^{1/q}, & \theta < 0, \end{cases}$$

with the usual modification for $q = \infty$. Note that $Gl^0_{p,q} = \ell_{p,q}$.

Theorem 1. Let $0 < p < \infty$, $0 < q \leq \infty$. Then $Gl^{-\theta}_{p,q} \hookrightarrow \ell_{p,q} \hookrightarrow Gl^{\theta}_{p,q}$ for $\theta > 0$; $Gl^{\theta}_{p,q} \hookrightarrow Gl^{\theta_1}_{p,q}$ if $\theta \leq \theta_1$; and $Gl^{\theta}_{p,q} \hookrightarrow Gl^{\theta}_{p,q_1}$ if $q < q_1 \leq \infty$.

Theorem 2. Let $0 < p < \infty$ and $\theta > 0$. Then we have $\|a\|_{Gl^{\theta}_{p,\infty}} \asymp \sup_{n\in\mathbb{N}} n^{1/p} (\ln(n+1))^{-\theta} a^*_n$, i.e. $Gl^{\theta}_{p,\infty}$ coincides with the Lorentz–Karamata sequence space $\ell_{p,\infty,\theta}$.

Theorem 3. Let $1 < p < \infty$, $1 \leq q \leq \infty$, $\theta > 0$. Then $\sum_{k\in\mathbb{Z}} |a_k b_k| \lesssim \|a\|_{Gl^{\theta}_{p,q}} \|b\|_{Gl^{-\theta}_{p',q'}}$.

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Keywords: grand Lorentz space, sequence space, embedding, Hölder inequality

2020 Mathematics Subject Classification: 46A45, 46E30

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Some properties of Besov-Morrey type spaces

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Abstract: This report is Sobolev-Morrey and Besov-Morrey spaces. For Morrey spaces, multipliers are studied, and a theorem on multipliers from $M_p^\lambda(\mathbb{T})$ to $M_p^\lambda(\mathbb{T})$ for $1 < p < \infty, 0 \le \lambda < \frac{1}{p}$ is obtained. Based on this theorem, embeddings of $B_1^\alpha\left(M_p^\lambda\right) \hookrightarrow W^\alpha\left(M_p^\lambda\right) \hookrightarrow B_\infty^\alpha\left(M_p^\lambda\right)$ for $0 < p \le \infty$, $0 \le \lambda \le \frac{1}{p}$, $\alpha \in \mathbb{R}$ are derived. An interpolation theorem for Besov-Morrey spaces is proved.

In this work, the boundedness of the Fourier partial sum operator on Morrey spaces $M_p^\lambda(\mathbb{T})$ is established, providing a sufficient condition for sequences to act as bounded Fourier multipliers. Using these multiplier properties, continuous embeddings connecting Sobolev-Morrey spaces with limiting scales of Besov-Morrey spaces are established. Furthermore, real interpolation identities for Besov-Morrey spaces and Sobolev-Morrey spaces are derived.

Throughout this note we mainly use techniques from work [1].

Keywords: Morrey spaces, Besov-Morrey spaces, Sobolev-Morrey spaces, Fourier multipliers, Embedding theorems.

2020 Mathematics Subject Classification: 46E35, 42B35, 42B15, 46M35

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Universal approach to pointwise and integral estimates in cones of monotone functions

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Abstract: In the report, we present a unified framework for computing pointwise and integral estimates (majorants) in cones of monotone functions. Such cones are central in the analysis of rearrangements of generalized Riesz and Bessel potentials. Explicit formulas are derived, highlighting the method's universality and applicability across diverse functional settings.

Let f^* denote the decreasing rearrangement of a Lebesgue measurable function f. The symmetric rearrangement $f^{\#}$ is defined via the volume of the unit ball in $\mathbb{R}^n$. Rearrangement-invariant spaces (RIS) are considered, with norms monotone with respect to decreasing rearrangements. The results obtained are used to derive formulas, admit to evaluate pointwise and integral majorants of decreasing rearrangements for generalized Riesz and Bessel potentials [1].

Potential space $H_E^G \equiv H_E^G(\mathbb{R}^n)$ on $n-$ dimensional Euclidean space is introduced by using convolutions:

$$H_E^G(\mathbb{R}^n) = \{u = G * f : \quad f \in E(\mathbb{R}^n)\}$$

where $E(\mathbb{R}^n)$ - *is a rearrangement invariant space (shortly : RIS).* Convolution kernel G is *admissible*, if $G \in L_1(\mathbb{R}^n)+E'(\mathbb{R}^n)$. Here $E'(\mathbb{R}^n)$ means the associated RIS for RIS $E(\mathbb{R}^n)$.

We focus on the kernel

$$\psi(t) = \frac{1}{1+t}, \quad t \in (0,T),$$

and construct the cone $K_{\psi,E}(T)$ generated by functions from the space $\tilde{E}(0,T)$. The task is to compute pointwise and integral majorants for this cone.

If $\psi \in \Omega(T)$ and $f_\psi(t,\cdot) \in \tilde{E}'(0,T)$, then the following formulas hold:

$$\lambda_K(t) = \|f_\psi(t,\cdot)\|_{\tilde{E}'(0,T)}, \quad t \in (0,T),$$

$$\tilde{\lambda}_K(t) = \frac{1}{t}\int_0^t f_\psi(\xi,\cdot)\,d\xi\ \tilde{E}'(0,T), \quad t \in (0,T).$$

These explicit expressions demonstrate the universality of the method and extend previous approaches to cones of monotone functions.

Keywords: Monotone functions; Cones; Pointwise and integral estimates; Rearrangement-invariant spaces; Generalized Riesz potentials; Cones of decreasing rearrangements of potentials

2020 Mathematics Subject Classification: MSC 46E30

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A functional parameterization method for a nonlocal boundary value problem with involution

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Abstract: We investigate a one-dimensional nonlocal initial-boundary value problem for a parabolic equation with spatial involution. The equation contains the involution term $u_{xx}(l-x,t)$, while the boundary conditions include a nonlocal integral condition. To study the problem, Dzhumabaev's parameterization method [1] is modified by introducing a time-dependent functional parameter $\mu(t) = u(l,t)$, and the solution is represented in the form

$$u(x,t) = v(x,t) + \sigma(x)\mu(t),$$

where $\sigma(x)$ is an appropriately chosen shape function. This transformation reduces the original nonlocal problem to an auxiliary Dirichlet problem coupled with an equation for the functional parameter. Using Green's functions, an integral representation of the solution is obtained, which leads to a Volterra integral equation for $\mu(t)$. Under appropriate assumptions on the data and the resulting kernel, the existence and uniqueness of a classical solution are established. The proposed approach provides a constructive method for treating nonlocal parabolic problems with involution without requiring eigenfunction expansions.

Keywords: functional parameterization; nonlocal boundary value problem; involution; Green's function; Volterra integral equation; heat equation

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Time-fractional parabolic equation with Ionkin-Samarskii conditions

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Abstract: The present study is devoted to the investigation of the time-fractional parabolic equation with Ionkin-Samarskii conditions:

$$D_t^\alpha u(t,x) - (\sigma(x)u_x(t,x))_x + \zeta u(t,x) = f(t,x),\ t \in (0,T),\ x \in (0,L),$$

$$u(t,0) = 0,\ u_x(t,0) = u_x(t,L),\ t \in [0,T], \tag{1}$$

$$u(T,x) = 0,\ x \in [0,L],$$

where $D_t^\alpha = D_{T-}^\alpha$ denotes the right-sided Caputo type fractional derivative of order $\alpha \in (0,1)$. We assume that $\sigma(x)$ $(x \in (0,L))$ and $f(t,x)$ $(t \in (0,T),\ x \in (0,L))$ are given smooth functions such that $\sigma(L-x) = \sigma(x)$, $\sigma(x) \geq \sigma_0 > 0$ and $\zeta > 0$.

In this work, stable difference schemes are constructed for the approximate solution of the boundary value problem (1) [1–5]. First, the nonlocal problem is reduced to two local problems. Then, a modified Gaussian elimination method is applied to solve the first- and second-order accurate difference schemes corresponding to these local problems.

Keywords: time-fractional parabolic equations, Ionkin-Samarskii conditions, Caputo type fractional derivative, difference schemes, modified Gaussian elimination method, stability

2020 Mathematics Subject Classification: 35K20, 35R11, 65M06

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Very weak solutions for a wave equation driven by a mixed local–nonlocal operator with singular coefficients

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Abstract: In this paper, we consider the Cauchy problem for the wave equation

$$\begin{cases} \partial_{tt}u(t,x) + \mathcal{L}u(t,x) = 0, & (t,x) \in (0,T)\times\mathbb{R}^d, \\ u(0,x) = u_0(x), \qquad u_t(0,x) = u_1(x), & x \in \mathbb{R}^d, \end{cases} \tag{1}$$

where $\mathcal{L}$ is the mixed local–nonlocal operator

$$\mathcal{L}u = -\nabla\cdot\big(h(x)\nabla u\big) + (-\Delta)^{s/2}\Big(g(x)\,(-\Delta)^{s/2}u\Big) + m(x)u, \qquad 0<s<1.$$

Here u denotes a wave function, u_0 and u_1 are the initial datum. The coefficients h, g, m are assumed real-valued, with $h(x) \geq h_0 > 0$, $g(x) \geq g_0 > 0$, and $m(x) \geq m_0 > 0$ for a.e. $x \in \mathbb{R}^d$.

Here, the local term $-\Delta$ describes short-range random motion, while the fractional Laplacian $(-\Delta)^s$ accounts for long-range interactions and jump processes. In the whole space $\mathbb{R}^d$, the operator $(-\Delta)^s$ can be defined as follows $(-\Delta)^s u(x) = c_{d,s}\,\text{P.V.}\int_{\mathbb{R}^d}\frac{u(x)-u(y)}{|x-y|^{d+2s}}\,dy$, where P.V. stands for the principal value integral, and $c_{d,s}$ denotes a normalization constant.

We assume that the coefficients and the initial data are spatial distributions. In this setting, one faces the classical obstruction of Schwartz: in general, there is no consistent multiplication of distributions extending the pointwise product of smooth functions [1]. Therefore, the equation cannot be understood in the classical sense, and even standard weak formulations usually fail when products of distributional coefficients are involved. In the present work, we extend the very weak solution framework [2] to the wave equation governed by the mixed local–nonlocal operator (1), allowing the coefficients $h, g, m \in \mathcal{D}'(\mathbb{R}^d)$ to be spatial distributions and permitting the initial data to be distributional in space as well. Within this setting, we establish the existence, uniqueness, and consistency of very weak solutions.

Keywords: wave equation, mixed local-nonlocal operator, very weak solutions, singular coefficients, energy estimate

2020 Mathematics Subject Classification: 35D99, 35K67, 34A45

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Universal approach to pointwise and integral estimates in cones of monotone functions

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Potential space $H_E^G \equiv H_E^G(\mathbb{R}^n)$ on $n-$ dimensional Euclidean space is introduced by using convolutions: $H_E^G(\mathbb{R}^n) = \{u = G * f : \quad f \in E(\mathbb{R}^n)\}$, where $E(\mathbb{R}^n)$ - *is a rearrangement invariant space (shortly : RIS).* Convolution kernel G is *admissible*, if $G \in L_1(\mathbb{R}^n) + E'(\mathbb{R}^n)$. Here $E'(\mathbb{R}^n)$ means the associated RIS for RIS $E(\mathbb{R}^n)$.

We focus on the kernel $\psi(t) = \frac{1}{1+t}, \quad t \in (0,T)$, and construct the cone $K_{\psi,E}(T)$ generated by functions from the space $\tilde{E}(0,T)$. The task is to compute pointwise and integral majorants for this cone.

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$\tilde{\lambda}_K(t) = \frac{1}{t}\int_0^t f_\psi(\xi,\cdot)\,d\xi\,\tilde{E}'(0,T), \quad t \in (0,T).$

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